

Multimedia Retrieval

University of Basel Course Textbook

Multimedia Retrieval - 2026 - Uni Basel

Preface

This textbook accompanies the Multimedia Retrieval course at the University of Basel. The online edition remains the authoritative source for interactive material and updates.

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Overview

Timeline and Organization of the course

Theory: Please study the material in advance. During lessons, we will cover the essentials with demonstrations and discussions, but some details will be left for self-study to keep a good pace. Check the schedule and announcements in ADAM. As a general rule, aim to read about one chapter ahead.

Practice: During the third hour, we combine theory and hands-on exercises. We work with Python and software packages related to the retrieval topics we cover. This session is optional, but active participation can help you in the exams, as outlined on the next page

- University: Spiegelgasse 5, Seminarraum 05.002, no zoom available, no video uploads after lecture**
Zoom: see meeting link on Web / in ADAM, video uploads after lecture

Exams and how to prepare for them

Exams are scheduled in January / February 2026 (dates will be announced)

- each student will have a 30-minute slot
- slots will be assigned in December based on your preferences and available slots
- each exam covers three topics, with each topic allocated around 8-10 minutes
- each topic will include several questions of increasing difficulty, so be accurate and fast to earn maximum points

Prerequisites for Exams: all exercises are optional, but practical exercises can help to round-up exam grades

- you don't need to submit theoretical exercises; we'll provide and discuss solutions during the 3rd hour
- you can submit practical exercises but you won't receive grades; instead, you earn points for the exam
- you do not earn points for theoretical exercises
- points will be converted into an upgrade of your grade ranging from 0 to 0.3
- the upgrade is added to your oral exam grade for the final result (rounded to the nearest 0.5 grade step)
- Examples:
 - ▶ Student submits some practical exercises, earning an upgrade of 0.2. In the oral exam, the student doesn't perform well and receives a grade of 3.6. Final result: $3.6 + 0.2$ rounded up to 4.0 (pass)
 - ▶ Student earns an upgrade of 0.3 for very good submissions. In the oral exam, the student receives a grade of 5.5, due to some difficulty with challenging questions. Final result: $5.5 + 0.3$ rounded up to 6.0 (pass, excellent)
 - ▶ Student doesn't have time, earns no upgrade, but performs well in the oral exam, receiving a grade of 5.7 struggling only with the toughest questions. Final result: $5.7 + 0.0$ rounded down to 5.5 (pass, very good)
 - ▶ Student doesn't have time, earns no upgrade, and doesn't perform well in the oral exam, receiving a grade of 3.7 with struggles in many questions. Final result: $3.7 + 0.0$ rounded down to 3.5 (failed)

Submission deadlines:

- Theoretical exercises: no submissions to ADAM; self correction against distributed solution
- Practical exercises: submission to ADAM by 31st Dec (23:59); groups of 2 are possible, but you need to provide more / better results to earn points

The Long Road to Modern Search

The study of information retrieval and knowledge management is closely linked to the broader story of how societies create, store, and access knowledge. From early library catalogues to today's vast digital ecosystems, each technological advance has reshaped the ways in which people find and use information. Understanding this progression is essential for students of information science, computer science, and related disciplines because it highlights not only the technical breakthroughs but also the social and economic forces that drive innovation.

Before computers became common tools for research and communication, information retrieval was a slow, manual process. Scholars and librarians relied on carefully maintained card catalogues, subject indexes, and classification systems to locate relevant works. These methods were effective for the printed collections of the time, but they could not scale to meet the demands of a rapidly growing body of scientific literature and government documentation. By the mid-twentieth century, the need for more efficient retrieval methods had become urgent. Governments, universities, and industries all required faster ways to search large collections of technical documents, and this demand set the stage for the field's first major transformation.

The arrival of digital computing in the post-war era provided the tools to address these challenges. Early experiments with computerized catalogues and automated indexing showed that machines could, in principle, manage and search much larger volumes of information than human indexers. Researchers began to formalize concepts such as "recall" and "precision" which remain core metrics today. Yet the transition from theory to practical systems was gradual. Early machines were expensive, text collections were limited, and few documents were available in machine-readable form. Nevertheless, the foundations laid during this period would guide the next decades of development.

The history that follows traces how information retrieval evolved from these early experiments into the sophisticated, large-scale systems of the present. Each decade introduced new models—Boolean retrieval, vector spaces, probabilistic approaches, and eventually neural networks—that improved how systems interpret queries and rank results. At the same time, the volume of data expanded exponentially, requiring continual advances in storage, processing power, and algorithmic efficiency. By exploring these developments chronologically, we can see how the field adapted to shifting technological landscapes, from the first keyword indexes to today's transformer-based semantic search and retrieval-augmented generation. In the following, we provide the context needed to appreciate both the theoretical underpinnings and the practical achievements described in this course.

The 1960s: Foundations of Knowledge Management

- The 1960s saw the rise of computerized information retrieval, built on centuries of manual cataloging. Calvin Mooers had coined the term "information retrieval" in 1950, but serious research did not begin until the 1960s. That decade became a "boom time for information retrieval" because of widespread research and development.
- Important advances included H. P. Luhn's creation of KWIC (Key Word In Context) indexes and Calvin Mooers's development of edge-notched card systems. Western Reserve University's Searching Selector, built by Allen Kent, was one of the era's best-known machines. Most work, however, remained experimental, and there was little computerized retrieval because few texts were machine-readable.
- During the 1960s, researchers established core evaluation metrics for retrieval systems, defining recall and precision. IBM developed STAIRS, the Storage and Information Retrieval System, one of the first large-scale, general-purpose information retrieval systems for text datasets. At Cornell, Gerard Salton began developing SMART, the System for the Mechanical Analysis and Retrieval of Text, which became foundational to modern search technology.

Characteristics:

- Users: Researchers, government/industrial R&D groups
- Use Cases: Experimental text search, bibliographic retrieval, indexing scientific literature, cataloging documents

- Key Technologies: Boolean retrieval, KWIC (Key Word In Context) indexing, edge-notched cards, classification systems (Dewey Decimal)
- Retrieval model: Retriever-only, Retriever-Filter
- Limitations: Small machine-readable text collections, slow response times, complex Boolean queries required, limited scope, largely experimental, few operational systems

The 1970s: The Rise of Computational Models

- The 1970s marked a turning point when retrieval systems became practical. Widespread use of computer typesetting and word processing produced the machine-readable text needed for large-scale systems. By the end of the decade, most printed material passed through computer input stages, creating large repositories for retrieval systems.
- This decade saw the rise of classical retrieval models that would shape the field for years. Gerard Salton's vector space model represented documents and queries as vectors in a high-dimensional space, allowing their similarity to be measured mathematically. During the same period, Stephen E. Robertson, Karen Sparck Jones, and others developed a probabilistic retrieval framework, which laid the groundwork for probabilistic models. Both approaches were later combined into BM25, which still produces excellent retrieval results.
- Large-scale commercial systems like the Lockheed Dialog system came into operation in the early 1970s. The MEDLARS (Medical Literature Analysis and Retrieval System) became operational, providing computerized access to biomedical literature. These systems demonstrated the practical viability of automated information retrieval on a commercial scale.

Characteristics:

- Users: Researchers, Librarians & Government analysts
- Use Cases: Academic & biomedical literature search, Library catalog/book search, Bibliographic & reference retrieval
- Key Technologies: Boolean, vector space, probabilistic retrieval
- Retrieval model: Retriever-only, Retriever-Filter, Retriever-Ranker
- Limitations: Limited coverage, Expensive access (subscriptions, terminals, telecom), Slow response times, Complex queries requiring trained intermediaries, Often abstract/metadata-only (not full-text)

The 1980s: Probabilistic Models and Boolean Systems

- In the 1980s, retrieval models developed in the previous decade were refined and widely adopted. Boolean retrieval systems, derived from century-old indexing practices, became the dominant commercial approach. These systems let users combine search terms with logical operators (AND, OR, NOT), but they remained difficult for new users to operate effectively.
- The probabilistic retrieval approach continued to develop, notably advancing the Binary Independence Retrieval (BIR) model. Robertson formalized its theoretical basis, the Probability Ranking Principle, showing that ordering documents by their probability of relevance from highest to lowest yields the best retrieval performance.
- The Okapi information retrieval system was developed at London's City University during this decade; it later gave its name to the well known BM25 ranking function. Latent Semantic Indexing (LSI) appeared as a new method that uses singular value decomposition to reveal hidden relationships between terms and documents.

Characteristics:

- Users: Researchers, librarians, and commercial search system operators
- Use Cases: Academic literature retrieval, library book searches, bibliographic/reference lookup
- Key Technologies: Boolean retrieval, vector space model, probabilistic retrieval (e.g., Binary Independence Model)
- Retrieval model: Retriever-only, Retriever-Filter, Retriever-Ranker
- Limitations: Small-scale datasets, slow query response times, high learning curve for formulating queries, costly system implementation and operation

The 1990s: The Web Revolution and Search Engines

- The 1990s brought a major change with the arrival of the World Wide Web and the first web search engines. When the Web opened to the public in 1991, available information expanded rapidly, and new methods were needed to find it.
- Early search engines such as Archie, Gopher, and WebCrawler appeared to help users find their way through the new online world. Yahoo! Search became one of the first widely used web search services; it began as a directory-based system. Launched in 1995, AltaVista introduced full-text searching and indexed multimedia content.
- The most significant development occurred when Larry Page and Sergey Brin created PageRank at Stanford University in 1996. The algorithm changed web search by ranking pages according to the authority of links pointing to them instead of relying only on keyword matching.
- During this period, machine learning began to be used in information retrieval. Researchers explored neural networks, symbolic learning, and genetic algorithms for various IR tasks. Relevance feedback grew more sophisticated, enabling systems to learn from user interactions.

Characteristics:

- Users: General public, e-commerce consumers, professionals and business users
- Use Cases: web search, e-commerce search, business/market intelligence, academia
- Key Technologies: Vector Space Retrieval, Probabilistic Retrieval, Web Search Engines, Query Expansion, separation of retrieval & sort
- Retrieval model: Retriever-Ranker, (Retriever-Filter)
- Limitations: data & index explosion due to exponential growth, large retrieval and indexing costs, good recall values but poor perceived precision (i.e., document not relevant for the user), quality of data

The 2000s: Machine Learning and Ranking Algorithms

- In the 2000s, information retrieval systems began using advanced machine learning. Learning to Rank (LTR) became an important approach: supervised models were trained to sort search results by relevance. This shift moved ranking away from hand-designed features toward data-driven optimization.
- Based on a probabilistic framework from earlier decades, BM25 became a standard ranking function. It combines term frequency and inverse document frequency with document-length normalization, delivering strong performance across diverse collections.
- Deep learning reemerged after research largely stopped at the end of the 1990s. More powerful computers, especially GPUs, larger datasets, and better training methods brought neural networks back into use. Researchers developed deep belief networks (Hinton, 2006) and applied convolutional neural networks to vision tasks.
- Search engines continued to evolve their algorithms, and Google's PageRank became the basis of its dominance in web search. User behavior signals and personalization began to influence the ranking.

Characteristics:

- Users: Web search users, e-commerce consumers, data scientists & academics, social media users
- Use Cases: Web search personalization, e-commerce search & recommendation, emergent semantic search research
- Key Technologies: LTR, BM25, personalization via behavior logs
- Retrieval model: Retriever-Ranker (enhanced with ML)
- Limitations: Data/index explosion, high retrieval/indexing cost, data quality, model transparency, update latency

Geoffrey Hinton

Nobel Prize in Physics 2024

The 2010s: Deep Learning and Neural Information Retrieval

- The 2010s saw a major shift as deep learning techniques were applied to information retrieval. Neural ranking models began using shallow and deep neural networks to rank search results, moving away from traditional handcrafted features toward learned representations.
- Classical learning-to-rank models relied on manual feature engineering. Neural models learned representations that bridged the vocabulary gap between queries and documents; they required large training sets but captured semantic meaning far more effectively.

- Google introduced BERT in 2018 as a major advance in natural language understanding. BERT's ability to capture context and relationships between words allowed for more accurate interpretation of complex queries. Google integrated BERT into its search algorithm in 2019.
- Transformer architectures fundamentally changed how machines handle sequential data, enabling them to capture long-range dependencies in text. Capabilities in 2010s were still constrained lacking datasets, scalability, and performance.

Characteristics:

- Users: Web search users, e-commerce consumers, mobile/app users, data scientists & academics
- Use Cases: Web search personalization, e-commerce search & recommendation, conversational search, semantic search research
- Key Technologies: Deep learning (CNNs, RNNs, Transformers), BERT, early neural ranking models, embeddings, large-scale behavior log personalization
- Retrieval model: Neural retriever-ranker pipelines
- Limitations: High computational cost, large data requirements, latency in updates, model interpretability, scalability of training & indexing

Word2Vec

<https://opensearch.org/blog/ltr-with-opensearch-and-metarank>

The 2020s: Semantic Search and Vector Databases

- The 2020s marked the start of semantic search and vector-based retrieval. Transformer models improved information retrieval by enabling systems to understand context and meaning rather than rely solely on keyword matching.
- Vector databases are now essential infrastructure for modern search systems. They allow fast similarity search over high-dimensional embeddings and can match semantically similar content even when keywords do not overlap, for example “car” with “automobile”.
- Retrieval-Augmented Generation (RAG) combines large language models with external knowledge retrieval. It lets models access up-to-date information that is not in their training data, reducing hallucinations and improving factual accuracy.
- Agentic information retrieval systems can understand complex information needs, create multiple search strategies, and combine results from different sources. They act as intelligent intermediaries that reason about information requirements, carry out multi-step retrieval, and deliver thorough responses tailored to each context.

Characteristics:

- Users: Web search users, e-commerce consumers, mobile/app users, data scientists, knowledge workers, AI developers, enterprise teams
- Use Cases: Semantic search, retrieval-augmented generation (RAG), conversational AI, personalized recommendations, knowledge bases, intelligent virtual assistants, multi-step query resolution
- Key Technologies: Large language models, embeddings, vector search
- Retrieval model: Neural retriever-ranker, retriever-generator
- Limitations: Requires high-quality embeddings and up-to-date data, may still hallucinate if retrieval fails, computationally intensive, complexity in multi-source integration

<https://opensearch.org/platform/vector-engine>

query

response

agent

LLM

knowledge base 1

knowledge base 2

web search

code execution

tool A

tool B

reasoning

Breakthroughs Behind Modern Retrieval

The history of information retrieval is inseparable from the growth of the data it seeks to organize. Over the past three decades, the scale of digital data has expanded so dramatically that it has transformed not only retrieval methods but also the very infrastructure of computation and storage.

By the mid-1990s most digital information remained concentrated in business applications and private databases. The internet itself was still modest in scope: in 1999 it contained fewer than one billion web pages. Retrieval systems of the time, still grounded in probabilistic and vector-space models, were adequate for these collections.

But the explosive adoption of the web, broadband connections, and early mobile devices rapidly outstripped those capabilities. Social media platforms and e-commerce generated continuous streams of user content. At the same time, the emergence of cloud computing separated storage from local physical devices, allowing organizations to keep vastly larger datasets online.

- In 2010 the world's data volume was estimated at 2 zettabytes. By 2023 its estimate had grown to 120 zettabytes, is expected to exceed 181 zettabytes in 2025, and to top 400-500 zettabytes by 2028.
- This means that 90% of the world's total digital data has been created in the past 10 years.
- 1 zettabyte equals 1 billion terabytes or 12 billion 4k videos (1000x all titles listed on IMDb)

This relentless expansion means that every new search engine query competes with a background of billions of potential matches: a simple search for "ford", for example, can return more than two billion results. Determining which documents actually satisfy a user's intent is becoming increasingly difficult even for AI based ranking systems.

The velocity of data creation compounds the challenge. Real-time sources such as news or social media posts generate information that may be relevant for only a short time. News can trend globally and then fade in relevance within an hour. Retrieval systems must therefore index and rank new content close to its creation, or risk becoming obsolete. Traditional pipelines designed for nightly or weekly index updates cannot keep up with this pace. Meeting these demands has required dramatic advances in both storage and computation. Physical limits on read and write speeds illustrate the scale of the problem.

- Storage has become much cheaper and more efficient, turning a once scarce resource into an abundant one. Magnetic hard drives steadily increased capacity while lowering cost per gigabyte, followed by optical discs that provided durable, low-cost backups. Solid-state drives improved performance with fast, reliable flash memory and no moving parts. At the same time, cloud storage removed the need to own hardware by pooling vast arrays of magnetic and flash devices across global data centers.
- Compute:
 - A pivotal step came with Nvidia's introduction of CUDA, a parallel computing platform that made GPUs practical for general-purpose computing. Combined with distributed frameworks such as MapReduce, CUDA enabled researchers to process enormous datasets by distributing work across thousands or even millions of machines.
 - As of March 2023, the AMD EPYC 9654 has 96 cores and 192 threads, can perform nearly 1 trillion integer operations per second, and delivers about 550 gigaflops of floating-point performance at 400 watts.

Even with these capabilities, training a state-of-the-art large language model remains difficult. A single epoch can take several days on a single CPU. Nvidia's RTX 4090 (October 2022), built on a 5-nanometer process, delivers up to 100 teraflops with CUDA at about the same power use, enabling training speeds roughly 200 times faster than a single CPU.

1 Petabyte of storage (2024)

Illustration of price and space compression of storage

In the past decades, we have seen price drops of 50% every 14 months. Every 4 years, the costs decreased by an order of magnitude. On the other side, firms still spend the same amount of \$ to increase and replace their storage real estate. As a consequence, the amount of managed storage also grew exponentially and makes it ever more difficult to find relevant information.

1 Petabyte of storage (2015)

72x 16TB

42U Rack

2U Appliance

21x smaller in 3 years

source: <https://ourworldindata.org/>

So, how long does it take to read 1 Petabyte? All data points as of 2025:

- Fastest HDD have about 550MB/s read rate
- Fastest SATA SSD have about 550MB/s read rate
- Fastest PCI-E 5.0 NVMe SSD have about 14,000MB/s read rate
- USB 3.2 can handle up to 2,000MB/s transfer rate
- USB 4.0 can handle up to 40,000MB/s transfer rate
- PCI-E 7.0 can handle up to 256GB/s in one direction (512GB/s bi-directional)
- Ethernet (400GbE) can handle up to 50GB/s transfer rate
- Fibre Channel 256GFC can handle up to 50GB/s (per direction)
- The largest Internet exchange point (DE-CIX) operates at up to 3TB/s

Meeting these demands has required dramatic advances in both storage and computation. Physical limits on read and write speeds illustrate the scale of the problem. Engineers have relied on massive parallelism and distributed architectures to keep pace.

Distributed Computing and the Rise of Modern Foundational Models:

Distributed computing is essential for foundational models, especially large language models (LLMs). These models can exceed 500 billion parameters and require far more computing power than a single machine can provide. Training OpenAI's GPT-3 is estimated to have required about 3.14×10^{23} floating-point operations (FLOPS). Using the theoretical throughput of a single Nvidia V100 GPU, performing that many operations would have taken many years.

- Hardware: OpenAI used 10,000 V100 GPUs in a high-bandwidth cluster to train GPT-3.
- Time: Based on the V100's theoretical FLOPS, training would take about 13 days on that specific 10,000-GPU cluster using half-precision, or hundreds of years on a single GPU.
- Cost: Estimates suggest the compute cost alone for a single training run was in the millions of dollars.

Distributed computing also enables parallelism in multiple dimensions. Model parallelism splits the parameters of a single network across devices, while data parallelism processes different portions of the training dataset simultaneously. Pipeline parallelism and sharding techniques further optimize memory usage and throughput. By leveraging these approaches, modern LLMs not only become trainable but can also be fine-tuned, evaluated, and iteratively improved on enormous datasets.

Epoch AI - <https://epochai.org/blog>

The Future of Information Retrieval: Context, Agency, and Multi-Modal Intelligence

The next phase of information retrieval will focus on deeper contextual understanding, integration across data types, and agentic capabilities. Multi-modal retrieval, which combines text, images, audio, and video into unified representations, is becoming a major research area. Its use will enable systems to handle complex queries that involve multiple kinds of data and produce more relevant, insightful results than traditional keyword or vector searches.

Contextual search and user-centered retrieval will become increasingly important. Agentic systems that adapt dynamically to user goals, integrate multi-step reasoning, and combine information from diverse sources will act as intelligent collaborators rather than passive tools. These systems will mediate between users and vast information landscapes, prioritizing relevance, coherence, and user intent.

At the same time, computing and energy needs are becoming major constraints. Training and running large language models, especially multi-modal ones, requires large clusters of GPUs or specialized chips

and uses a lot of power. This creates financial and environmental limits and is driving the development of more efficient architectures, low-precision training, and distributed or federated learning.

The increasing autonomy of retrieval systems also raises concerns about the role of humans in evaluating information. Language models can generate, locate, and summarize content with minimal human intervention, but the risk is that critical assessment may diminish and the value of information may decline. Systems may present content that is accurate yet contextually irrelevant, creating a perception of understanding without genuine insight.

How Retrieval Systems Find Answers

A retrieval system addresses the following fundamental problem:

First, what do we mean with “relevant for query Q in the context of the query originator”

- Relevancy is the degree to which retrieved information matches a query and meets the query originator’s needs or intent. For example, if someone asks “Where can I get a pizza tonight?”, the user’s location is crucial to providing a relevant result. A restaurant that makes excellent pizza will not be relevant if it is too far away. Social media often presents information without an explicit query; users still expect content that fits their interests rather than random posts. Relevance is also shaped by the user’s objective: when searching for a product, it matters whether the user wants to evaluate the item or find the best place to buy it.
- Objective relevance means factual correctness. If someone searches for the capital of France, Paris is objectively relevant because it is the capital. Subjective relevance depends on individual preferences. When a user asks for “movies to watch,” what counts as relevant will vary with taste and with personal definitions of a good movie. Search engines use feedback from users to improve relevance. Clicks and longer time spent on a result signal that it was useful, and ranking algorithms learn from those signals to show results that better satisfy users.
- Query-less search is an approach where users discover content without entering an explicit query. Platforms such as Instagram Reels and TikTok analyze user preferences, behavior, and trends to recommend videos that match inferred interests, enabling discovery without typed or spoken queries. Relevance is defined implicitly by how well the recommended content matches the user’s inferred interests and intent, even though the user never specifies a query. Because there is no explicit question to measure against, the system estimates relevance from behavioral signals and contextual data. Content is considered relevant when it aligns with these inferred preferences and keeps the user engaged, for example, when the user consistently watches similar videos to completion or interacts positively with them. In other words, relevance is not judged by textual similarity to a query but by predicted satisfaction and engagement based on the user’s past behavior and the patterns the system detects.

Given a set of N documents D_0 to D_{N-1} and a query Q , find a set of documents D_{i_j} with $0 \leq j < k$ that are relevant for the query Q in the context of query originator. Rank the documents such that D_{i_0} is the most relevant document and $D_{i_{k-1}}$ is the least relevant document for query Q in the context of the query originator.

- Offline processing: concerned with analyzing documents in advance, extracting features, and organizing these features into indexes that allow for fast retrieval. Some systems also adjust their (objective) relevance ranking during this phase (see IDF or Google’s PageRank later in the course)

Because scanning billions of documents at query time is infeasible given the user’s expectation for near-instant responses, retrieval is split into offline processing and online query answering:

data sources

document analysis

metadata

feature extraction

metadata

features

ranking & scoring

scores

retrieval index

optimization & scaling

query: best song by “The Beatles”

query parsing & analysis

retrieval & ranking

result presentation

query refinement & iteration

feature (meta-)data

scores & doc ids

meta data & docs

Searching images, audio, and video is harder than searching text because of the so-called semantic gap. Users typically enter queries as keywords. Text search can match those keywords directly because queries and documents use the same representation. By contrast, images cannot be matched to keywords at the pixel level. For example, there is no clear, fixed relation between the keyword “cat” and the many ways a cat can appear in an image.

When queries and media use different forms, the retrieval system must translate between them. This mismatch is called the semantic gap:

- The semantic gap refers to the disparity between low-level features extracted from multimedia data and the high-level semantics that humans associate with that data.

100100110010101010101101011010101010101010101101

picture of a “cat”

machine interpretation

human interpretation Traditional, mostly hand-crafted pipelines for feature extraction (illustrations on the following pages)

- Since the start of information retrieval, feature extraction has been a carefully designed, hand-crafted process for describing media content precisely and concisely so data can be retrieved quickly and with high relevance. In this course, we study many of these approaches, which remain useful despite the rise of improved methods based on large language models.
- For text, common features include term vectors, bag-of-words, and n-grams. Natural language processing steps include stemming, handling synonyms, and resolving homonyms. Systems also extract embeddings and measure term discriminative power with methods such as inverse document frequency. Web and social retrieval add weights from metadata and link analytics like PageRank. Image retrieval once used simple features such as color, texture, and shape and now depends largely on neural network features from classification and representation models. Audio retrieval uses frequency and amplitude domain features, musical features such as pitch and tempo, and speech recognition. Video retrieval adds shot detection and motion analysis.

General-purpose feature extractors via prompting and large language models (illustrations on the following pages)

- The rise of generative AI has changed how people create content and how they extract useful information from it. Traditional analysis extracted features specific to each media type. Methods were often one-off and required new engineering for each domain and often for each use case scenario. For example, analyzing medical images required tailored techniques different from those used to detect faces in images.
- Generative AI and multi-modal transformers shift the emphasis from manual feature engineering to prompt engineering. A single multi-modal model can be prompted with requests such as “Describe the key visual elements in this image”, and it will generate a summary of the most important parts of the image. The same approach can be adapted to different tasks and domains by changing prompts,

and light fine-tuning methods such as LoRa (Low Rank Adaptation) can quickly adapt the model to domain-specific needs without retraining the entire model.

- This universal approach has cut the cost of content analysis by about tenfold. Because the same core infrastructure can support many media types, teams can experiment faster: they can start with generic prompts and models to validate a use case and invest in fine-tuning only after the application proves valuable.
- Prompts become the primary engineering artifact, which introduces new risks. Prompt execution is not strictly deterministic. The internal workings of large language models and multimodal transformers are not transparent, so one cannot fully explain how they generate outputs. Updates to underlying models can change behavior in subtle ways, affecting the applicability and stability of deployed solutions even when prompts do not change.

Traditional, mostly hand-crafted pipelines for feature extraction

keyword extraction

NLP

bag-of-words

[0.1,0.2,0.3,...]

text

hand-crafted

hand-crafted

hand-crafted

[0.1,0.2,0.3,...]

[0.1,0.2,0.3,...]

[0.1,0.2,0.3,...]

low level features

HOG / SIFT SURF

SVM

car

image

hand-crafted

hand-crafted

supervised

shot detection

optical flow

combine streams

car moving fast

video

supervised

hand-crafted

supervised

spectral features

tempo features

classifier

hip-hop

music

hand-crafted

hand-crafted

supervised

MFCC

Mixture of Gaussians

classifier

/ˈkɑːr/

speech

hand-crafted

unsupervised

supervised

General-purpose feature extractors via prompting and large language models

The main subject of this image is a majestic, snow-covered mountain peak, which appears to be the iconic Matterhorn in the Swiss Alps. The mountain dominates the frame, its distinctive pyramid shape rising dramatically against a clear blue sky. The setting is a high-altitude alpine environment, with the peak surrounded by other snow-capped mountains and glaciers visible in the lower portions of the image. The background is primarily composed of a vivid blue sky with a few wispy clouds. The colors in the image are striking, with the brilliant white of the snow contrasting sharply against the deep blue of the sky. The lighting appears to be natural sunlight, creating a play of light and shadow across the mountain's face that accentuates its rugged features and crevices. There are no visible people, animals, or man-made objects in the image. The focus is entirely on the natural grandeur of the mountain. The overall mood of the image is one of awe-inspiring beauty and serene majesty. There's a sense of isolation and pristine wilderness that the mountain embodies. Notable details include the jagged ridgelines of the mountain, the smooth snow fields on its flanks, and the wisps of cloud that cling to its lower slopes, suggesting high winds at the peak. The composition of the image is well-balanced, with the mountain placed slightly off-center, allowing the eye to follow its slopes from base to peak. The surrounding mountains and glaciers ...

traditional feature engineering

universal prompt engineering

Online query answering evolved rapidly in the past years

- Early systems provided simple retrieval without relevance ranking, which was enough for file searches or basic catalog queries. As users asked for finer control and higher precision, systems added filtering and ranking, trading speed for accuracy. With the web's growth, large-scale retriever-ranker architectures became standard, offering high accuracy and low latency but higher infrastructure costs.
- Demand for question answering drove the next advance. Systems began combining retrieval with reading or generation to give direct answers instead of lists of documents. These hybrid models improved the user experience but required more computation and raised operating costs. For example, a web search for a factual question, such as "Who won the Formula 1 race this weekend", now returns a direct answer instead of a list of links that the user must click and read. Often, finding the answer on the linked pages took more time than the search itself.
- The field is now entering a period of rapid change after a long steady phase. Retrieval-augmented generation and agentic extensions combine planning, tool use, and iterative reasoning, improving

accuracy and flexibility while challenging traditional cost and latency trade-offs. This faster research cycle reflects a wider range of use cases, from everyday search to complex multi-step problem solving, and points to rapid innovation in retrieval systems.

- Throughout this course, we will study models ranging from classical text retrieval approaches to modern methods that use agentic AI. In what follows, we describe the different retrieval types and explain what distinguishes them.

Retriever-only systems use a retriever component to identify documents that match a query and present them to the user without an explicit relevance ranking. This basic search functionality is widely available in file search tools and simple web applications. Without ranking, filtering, and sorting, this approach works only for small data sets where queries usually narrow results to a few items which users can quickly assess for relevance.

- ☒ Example: https://www.goodreads.com/search?q=agatha+christie&search_type=books

Retriever

query

doc 1

doc 2

doc 3

...

index

Retriever-Filter systems are similar to retriever-only approaches but add a filtering and sorting stage to refine results before presentation. Filters let users narrow results by parameters such as year or rating, and sorting can be driven by attributes like popularity or price. Relevance may affect ordering, but other criteria often dominate. This search feature is common in e-commerce applications and is often enhanced with faceted search.

- ☒ Example: <https://www.galaxus.ch/en/search?q=clothes+iron>

Retriever-Ranker systems first select a pool of candidate documents using the retriever and then apply a ranker to assign a relevance score to each candidate, returning documents by score. This is a common architecture in both classical and modern retrieval systems and is frequently enhanced with semantic search and context-sensitive ranking such as user location, objective importance, and subjective importance. Web search engines typically use this model, combining text retrieval with web-specific ranking signals.

- ☒ Example: <https://www.google.com/search?q=multimedia+retrieval+lecture> (change your location with a VPN client and submit again)

Retriever

query

doc 1

doc 2

doc 3

...

index

Filter & Sort

meta-data

criteria

Retriever

query

doc 1

doc 2

doc 3

...

index

(Filter &) Ranker

rank model

Retriever-Reader systems are designed for question-style queries that ask for a specific answer. The retriever fetches relevant documents and the reader identifies one or more passages within those documents that answer the question, returning the passages rather than a list of documents. Readers often rely on language models to locate concise answers in the result documents.

- ☒ Example: Google's 'featured snippet from the web' (this feature is now often replaced with a retriever-generator answer) <https://www.google.com/search?q=what+is+the+main+ingredient+in+tylenol> <https://www.google.com/search?q=What+did+Albert+Einstein+win+the+Nobel+Prize+for%3F> (this later query may provide an answer from the knowledge database; try "People also ask" for retriever-reader answers)

Retriever-Generator systems, also known as Retrieval-Augmented Generation or RAG, combine the retriever with a generative language model. The retriever selects relevant documents or passages and those snippets are combined with the user query into a prompt template for a large language model. The model then generates a comprehensive answer rather than extracting a single passage.

- ☒ Example: Bing copilot search ("What did Albert Einstein win the Nobel Prize for?") <https://www.bing.com/copilotsearch?q=What+did+Albert+Einstein+win+the+Nobel+Prize+for%3F>

Retriever

query

doc 1

doc 2

doc 3

...

index

(Filter &) Ranker

rank model

Retriever

query

index

Generator

LLM

documents Retriever-Synthesizer systems fetch a set of relevant documents and then instruct a language model to synthesize a condensed summary rather than extracting a direct answer. This is especially useful for exploratory or conceptual queries, where users benefit from an overview of multiple sources rather than a single factoid. Unlike Retriever-Reader systems that target pinpoint answers, the summarizer model must integrate information, reconcile conflicting statements, and produce a coherent narrative.

- ☒ Example: [perplexity.ai](https://www.perplexity.ai/search/how-does-vector-space-retrieva-DuutsrURRgeeYmrJw_0bZg) with “How does vector space retrieval work?” (normal search) https://www.perplexity.ai/search/how-does-vector-space-retrieva-DuutsrURRgeeYmrJw_0bZg

Agentic RAG systems extend the concept of Retriever-Generator and Retriever-Synthesizer with agentic capabilities. In this setup, the agent receives the query and actively decides how to fetch information, which sources to query, and whether to iterate or reformulate queries. It can plan a sequence of retrieval actions based on intermediate results, not just a single retrieval pass. Once the agent has gathered and processed the necessary information, a generative language model synthesizes a comprehensive final answer.

☒ Example: [perplexity.ai](https://www.perplexity.ai/search/how-does-vector-space-retrieva-01jcYjFZQWmwzpaCVGdfRQ) with “How does vector space retrieval work?” (pro version with multi-step reasoning) <https://www.perplexity.ai/search/how-does-vector-space-retrieva-01jcYjFZQWmwzpaCVGdfRQ>

Retriever

query

index

Synthesizer

language model

documents

Agent

query

Synthesizer

language model

scratchpad

scratch pad

Tools

knowledge base

web search

tool / API

user context

code execution

reasoning

retriever tools

- Generator-only systems use only generative models and have no explicit retriever. They produce answers from knowledge stored in their training data. General-purpose models can handle many tasks with careful prompting, but fine-tuned models are needed for specialized or business queries. A major risk is hallucination, when the model gives plausible but incorrect information. This occurs because the model is trained to always answer and to satisfy the user, even when it lacks facts. Ambiguous questions, gaps in the training data, and the lack of retrieval grounding increase this risk. In short, generator-only systems are flexible but trade reliability for broad usefulness, making them vulnerable to mistakes in specialized or high-stakes situations.
- ☒ Example: ChatGPT with “What did Albert Einstein win the Nobel Prize for?” <https://chatgpt.com/share/68c57b5f-d174-8011-a6c1-14a0eb5078fa>

query

Generator

LLM

domainknowledge

fine-tuning

prompt template

References & Links

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Semantic gap: the definition goes much beyond the scope of this introductory example, see Wikipedia

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1 - Classical Text Retrieval

Every day, billions of queries hit search engines, email clients, and document archives. The core challenge is always the same: given a short query, find the most relevant items in a large collection. This chapter introduces the classical methods that solve this problem for text.

Text retrieval was one of the earliest problems in computer science to receive serious attention. Beginning in the 1950s, researchers like Gerard Salton and Karen Sparck Jones developed methods that worked remarkably well. The reason is fundamental: queries and documents share the same vocabulary. When a user types “climate change policy,” the system can match those exact words in documents without needing to bridge different representations. This directness gave text retrieval a head start over image or audio search, where the gap between raw data and human meaning is far wider.

By the early 2000s, many considered text retrieval a solved problem. The core algorithms were mature, search engines worked well enough, and research attention shifted elsewhere. Recent developments in generative AI have changed the picture. Modern language models depend on precise, up-to-date text passages retrieved from external sources, and suddenly the quality of the retrieval stage matters more than ever. Classical text retrieval is no longer just the backbone of search engines. It is a critical component in modern AI systems.

We begin with Boolean filtering, the simplest approach, and trace the evolution toward ranked retrieval models that estimate how relevant each document is. Along the way, we develop the feature extraction pipeline that transforms raw text into searchable representations.

Retrieval Systems and Flows

The need to search large text collections has its roots in library science. As described in the opening chapter, librarians and researchers relied on card catalogs, subject indexes, and classification systems for centuries. When collections outgrew manual methods, the first computerized retrieval systems automated what librarians had always done: match a user's query against a catalog of documents. Today, the same principles apply far beyond libraries, from legal databases and patent archives to corporate knowledge bases and web search engines.

Across these applications, a retrieval system must answer three connected questions: what unit should be returned, what work can be completed before a query arrives, and whether matching documents should be filtered or ranked. These decisions determine the system architecture and the flow from source documents to results.

From Filtering to Ranking

Consider a university library with millions of articles and books. A researcher enters terms like “neural plasticity AND memory consolidation”. The simplest architecture retrieves all documents whose catalog entry satisfies this Boolean expression and returns the matching set. This retriever-only approach can evaluate each document independently and return results as soon as it finds them. No scoring or sorting is required. Figure 1.1 illustrates this basic architecture.

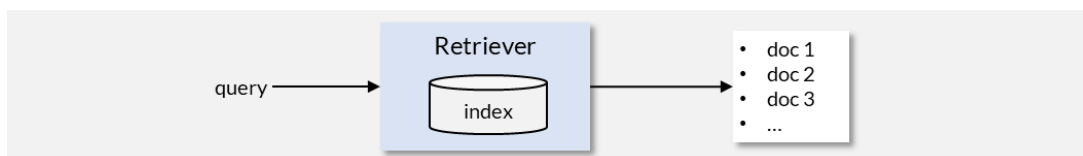


Figure 1.1: Retriever-only architecture: query enters, matching documents come out.

As the library's collection grows, a Boolean query like “neural plasticity” might return hundreds of articles. The researcher needs a way to narrow results without reformulating the query. A post-processing step adds **faceted search**: the ability to filter and sort results by metadata such as publication year, journal, or language. The researcher can toggle these filters while browsing without resubmitting the original search. Faceted search does not change which documents are considered relevant. It simply helps navigate large result sets. Figure 1.2 shows this extended pipeline.

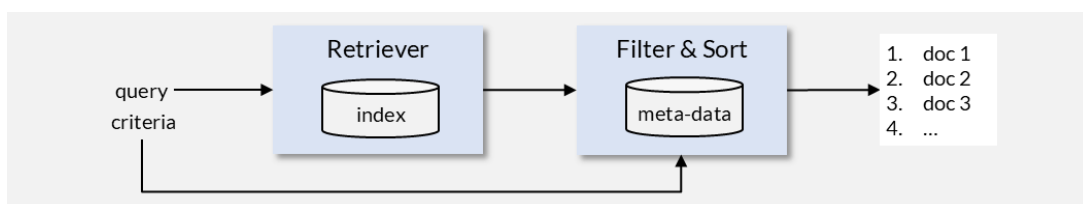


Figure 1.2: Retriever with faceted filtering: candidates are filtered and sorted by metadata.

Even with filters, an unranked list of 200 articles forces the researcher to scan them manually. In the 1970s, the Vector Space and Probabilistic retrieval models introduced a fundamentally different idea: instead of a binary relevant/not-relevant decision, the system estimates a degree of relevance and produces a ranked list. The most promising articles appear first. The architecture becomes a two-stage pipeline: a fast retriever selects candidates, and a ranker scores and orders them (Figure 1.3).

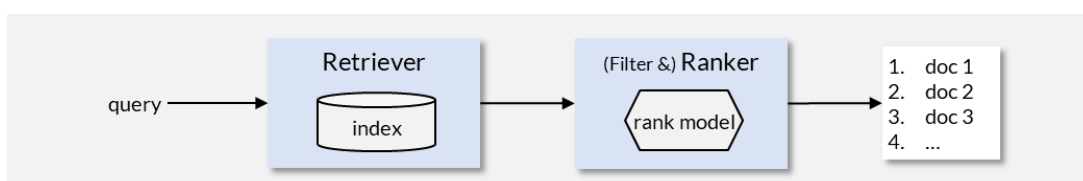


Figure 1.3: Retriever-ranker pipeline: a scoring model orders candidates by estimated relevance.

These three architectural patterns, retriever-only, retriever with faceted filtering, and retriever-ranker, represent the historical progression of text retrieval. The same patterns appear today in systems of all scales, from a researcher querying PubMed to a customer searching an e-commerce catalog.

Documents and Retrieval Granularity

Before we can search a collection, we need to define what a “document” is. In retrieval, a document is the unit that gets indexed, matched against queries, and returned to the user. It consists of three elements:

- **Document ID** - a unique identifier (ideally a UUID) for efficient storage and lookup.
- **Metadata attributes** - descriptive fields (author, publication date, language) used for filtering and faceted search, but not necessarily included in full-text matching.
- **Content attribute(s)** - the text fields indexed for search and ranking.

The critical design decision is: what constitutes a single retrieval unit?

Consider a pharmacology reference book with 800 pages. If we treat the entire book as one document, a search for “ibuprofen dosage” returns “the pharmacology textbook” and the user must search within the book manually. If instead we treat each section or page as a separate document, the system can return “page 47: Ibuprofen - Dosage and Administration” directly.

This decision about retrieval granularity defines the collection structure. Finer granularity gives more specific results but increases the number of entries the system must manage. Coarser granularity reduces index size but forces users to locate relevant passages themselves.

For classical text retrieval, the choice is typically straightforward: emails are individual documents, web pages are individual documents, book chapters or sections become individual documents. The retrieval models in this chapter assume that this splitting has already occurred and each document in the collection is a coherent, self-contained unit. We describe the concrete splitting strategies in the [feature extraction pipeline](#).

Note

Choosing the right retrieval granularity becomes significantly more complex when building systems that feed passages to language models. We revisit chunking strategies, overlapping windows, and hierarchical approaches in the chapter on Retrieval-Augmented Generation.

Offline Phase: Indexing Pipeline

Many search systems scan through data at query time. File search on a local computer, for example, reads through every file on disk whenever the user types a query. This works for a personal laptop with a few thousand files but not for a library with millions of articles. Instead, retrieval is split into two phases: an offline indexing phase that processes documents before any query arrives, and an online querying phase that handles user requests in real time.

The offline phase processes documents through a sequence of stages before any query arrives ([Figure 1.4](#)): (a) ingest raw documents from their source formats, (b) split them into retrieval units at the chosen granularity, (c) identify metadata attributes for filtering, (d) extract and normalize text into feature vectors, and (e) store these vectors in a searchable index.

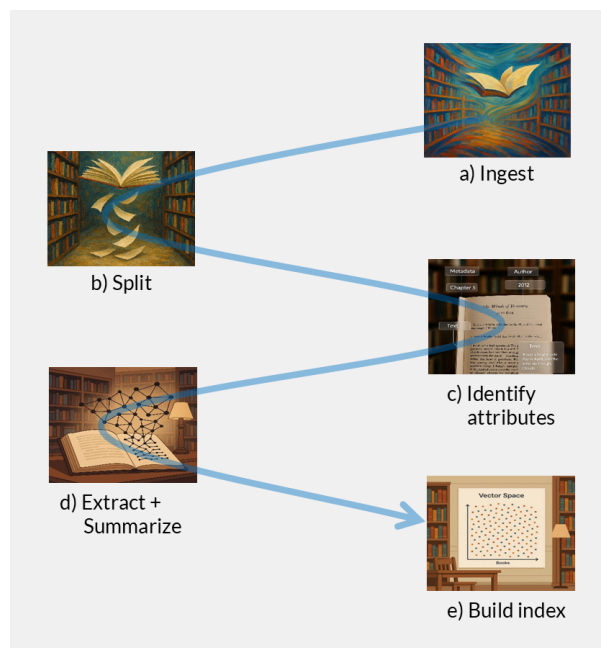


Figure 1.4: Document preprocessing pipeline: ingest (a), split (b), identify attributes (c), extract features (d), build index (e).

Online Phase: Query Processing

When a researcher types “neural plasticity AND memory consolidation” into the library search, the online phase begins. The system processes the query through a symmetric sequence of stages (Figure 1.5): (a) accept the user’s query, (b) tokenize it using the same pipeline that processed documents during indexing, (c) optionally broaden the query through spelling correction or synonym expansion, (d) compare the query’s feature vector against the vectors stored in the index to find similar documents, and (e) rank candidates by estimated relevance and return an ordered result list.

This symmetry between offline and online processing is essential: queries and documents must live in the same feature space for comparison to work.

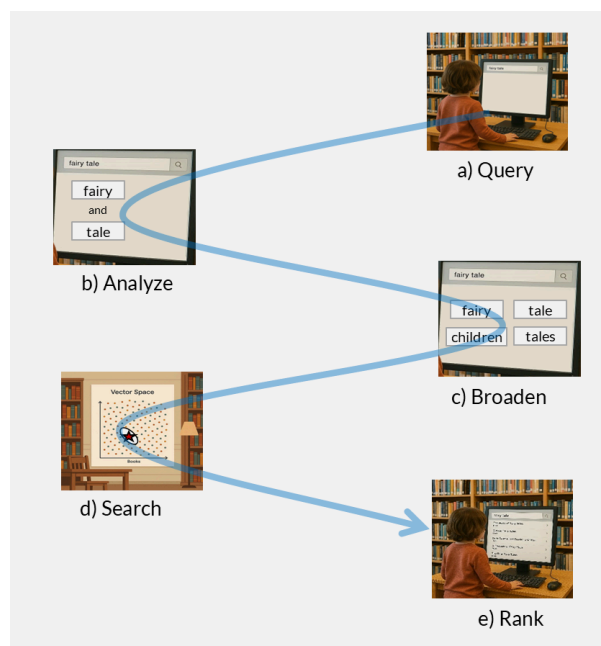


Figure 1.5: Online query pipeline: query (a), tokenize (b), expand (c), search (d), rank (e).

The central challenge in the online phase is relevance ranking: accurately estimating how important a document is given the query. The next sections develop the feature extraction pipeline that creates document representations, followed by the retrieval models that use those representations for scoring.

From Text to Searchable Vectors

Retrieval models cannot operate directly on PDF files, HTML markup, or character streams. They compare structured representations of documents and queries. The feature extraction pipeline shown in [Figure 1.6](#) creates these representations through five conceptual steps: extract text from the source format, split it into retrieval units, tokenize and normalize the text, construct a vocabulary, and summarize each unit as a feature vector. This produces two key artifacts:

- **Index** - high-dimensional feature vectors for every retrieval unit, stored together with source metadata.
- **Vocabulary** - the complete set of normalized terms used for both document and query processing.

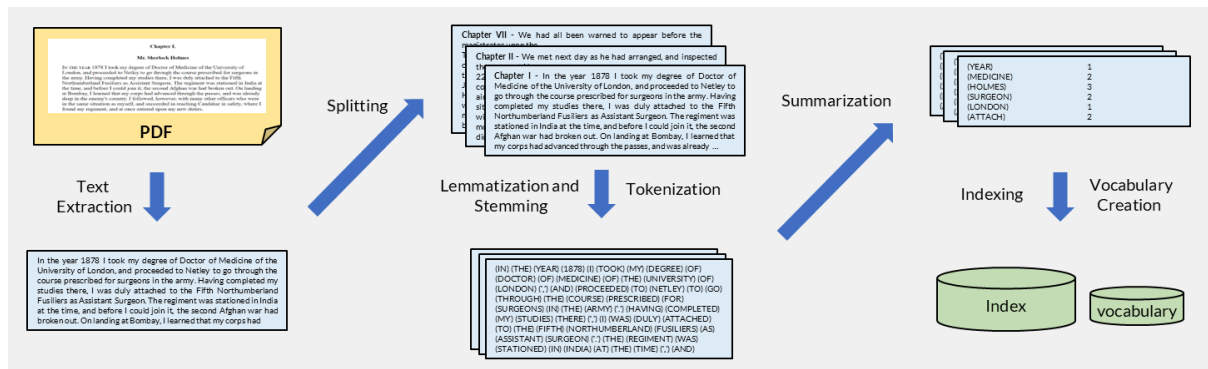


Figure 1.6: Text indexing pipeline: extract, split, tokenize, stem, summarize, then build the index and vocabulary.

Text Extraction

The first step extracts plain text and metadata from the source format. Documents arrive in many formats: PDF, EPUB, plain text, HTML, or office documents (DOCX, PPTX). Each format encodes content differently, mixing actual text with layout instructions, styling, and structural markup. The extraction step strips formatting and control sequences, isolates the character stream, and records metadata attributes (author, title, date) for later use in filtering.

Consider the HTML example in [Figure 1.7](#). The header holds structured metadata (title, keywords) that can enrich the index entry, while the body contains the main text interleaved with markup tags. Extracting useful text from HTML requires parsing the DOM tree and distinguishing content nodes from navigation, scripts, and boilerplate. Although HTML follows a well-defined standard, real-world pages use diverse layouts, making robust extraction (often called scraping) a non-trivial engineering problem.

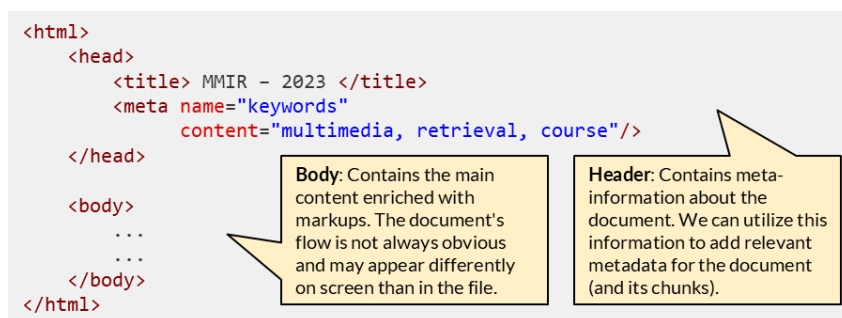


Figure 1.7: HTML document structure: metadata in the header, content in the body.

Source formats differ, so extraction normally relies on format-specific parsers and established libraries. The retrieval pipeline itself only requires that this stage return clean text and useful metadata; the implementation details can be selected for the collection at hand.

Splitting

As discussed in the previous section, retrieval granularity determines what constitutes a single retrieval unit. Once we have extracted the raw text from a document, we must split it into the units that will

be indexed and returned to the user. In classical text retrieval, the splitting strategies are pragmatic and structure-driven:

- **By document boundary:** Each file, email, or web page becomes one retrieval unit. This is the most common approach when documents are naturally self-contained.
- **By structural element:** Books are split at chapter or section headings. Legal texts are split by article or paragraph number. The document's own structure defines the boundaries.
- **By line or log entry:** In observability systems like Elasticsearch, each log line becomes an independent retrieval unit. This enables searching millions of events by timestamp, severity, or content.
- **By fixed page or paragraph count:** When no structural markup is available (e.g., scanned documents), we split at regular intervals such as every N paragraphs or every page.

These approaches share a common property: the splitting decision is made once during indexing and remains fixed. Each resulting unit is treated as an independent document by the retrieval models that follow. Figure 1.8 illustrates this for a novel split into chapter-sized retrieval units.

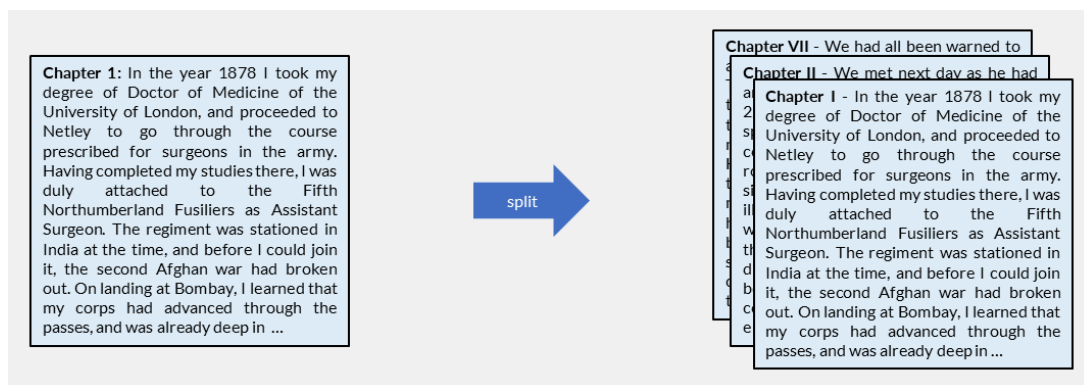


Figure 1.8: A single document divided into multiple smaller retrieval units.

Tokenization

Tokenization converts the extracted character stream into a sequence of discrete units called tokens. A **token** is an individual element in this sequence: it can be a word, a number, or a punctuation mark. A **term** is the normalized form of a token after processing (stemming, case folding). In classical text retrieval, tokens are typically whole words, and the two concepts are often used interchangeably. In later chapters, we will see that this equivalence breaks down when tokens become subword fragments or multi-word phrases.

For classical retrieval, we split text at whitespace and punctuation boundaries to produce word-level tokens. This requires decisions about edge cases: how to handle hyphenated words (“state-of-the-art”), numbers (“3.14”), abbreviations (“U.S.A.”), and possessives (“Watson’s”). In languages without explicit word boundaries (e.g., Japanese, Chinese), segmentation requires dictionary-based or statistical methods.

The most significant challenge is variation in word forms. Since retrieval models match documents to queries by comparing tokens, two tokens are either identical or unrelated. There is no notion of “almost the same”. A query for “cats” will not match a document containing only “cat” because these are distinct tokens. Similarly, “go”, “goes”, “went”, and “going” are four independent dimensions in the vocabulary despite sharing a meaning. The next subsection addresses this through stemming and lemmatization, which reduce inflected forms to a common token before indexing. [Figure 1.9](#) shows the tokenization step applied to a document collection.

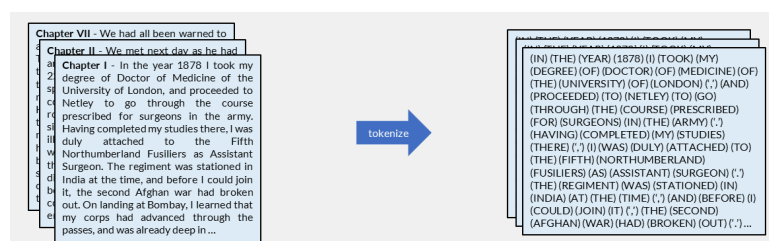


Figure 1.9: Tokenization: raw text split into individual word-level tokens.

Lemmatization and Stemming

The previous subsection established that retrieval models treat each token as an independent symbol: “cat” and “cats” are unrelated unless we normalize them to the same form. The goal of this pipeline stage is to reduce surface variation so that semantically equivalent word forms map to a single canonical token in the vocabulary.

Two approaches exist:

- **Stemming** applies rule-based suffix stripping to produce a common pseudo-stem. The stem is not necessarily a real word (e.g., “computing”, “computation”, “computer” all reduce to “comput”). Stemming is fast, requires no dictionary, and works well enough for most retrieval tasks. Its errors go in both directions: it can merge unrelated words (“universal” and “university” both stem to “univers”) or fail to merge related ones (“alumnus” and “alumni” remain distinct).
- **Lemmatization** uses a dictionary or morphological analysis to map each word to its actual root form (the lemma). “Better” maps to “good”, “went” maps to “go”. This produces more accurate results but requires language-specific resources and is computationally more expensive. Modern NLP libraries like spaCy provide lemmatization out of the box.

For classical text retrieval, stemming is the standard choice due to its speed and simplicity. The most widely used stemmer for English is the Porter Algorithm, which we examine next.

The Porter Algorithm

The most well-known rule-based stemmer for English is the Porter Algorithm (1980); see [An Algorithm for Suffix Stripping](#) in the Further Reading section of the chapter summary. It applies an ordered sequence of suffix-stripping rules to produce a shared pseudo-stem. For the retrieval pipeline, the important result is that related forms map to the same vocabulary term; the exact rule sequence is secondary.

Consider these examples:

Input	Porter stem	True root
“computing”, “computation”, “computer”	comput	compute
“retrieval”, “retrieving”, “retrieved”	retriev	retrieve
“generalization”, “generalizing”	general	generalize
“relational”	relat	relate
“agreed”, “agreeable”	agre	agree

The stems “comput”, “retriev”, and “agre” are not dictionary words. They exist only to ensure that related forms map to the same token during indexing. When a user queries “computation”, the system stems it to “comput” and matches documents containing “computing” or “computer” because those were also stemmed to “comput”.

This approach is fast (roughly 60 rules in about 400 lines of code), requires no dictionary, and works well enough for most English retrieval tasks. However, it is purely mechanical and makes errors in both directions: over-stemming merges words that should remain distinct (“operate” and “operating” correctly merge, but “operation” and “operational” may conflate with “opera”), while under-stemming fails to merge irregular forms (“be”, “was”, “been” remain three separate tokens).

Note

The internal mechanics of Porter are covered in detail in [3 - Advanced Text Processing](#), along with modern alternatives like Snowball and dictionary-based lemmatization.

Vocabulary

After tokenization and stemming, we have a set of normalized tokens across the entire collection. The **vocabulary** is the complete set of distinct terms that appear at least once. Each term becomes a dimension in the feature space used for retrieval. But not all terms are equally useful.

Stop Words

Many terms are grammatically necessary but carry no content. The article “the” appears in virtually every English document but tells us nothing about what a document is about. A search for “the” would return the entire collection, unable to differentiate between relevant and non-relevant results. Figure 1.10 shows that the 50 most frequent terms in a news corpus account for roughly one-third of all occurrences and appear in over 60% of documents.

t_j	$df(t_j)$	$tf(D, t_j)$						
the	100%	6.03%	from	90%	0.44%	when	74%	0.24%
a	99%	2.57%	be	88%	0.46%	their	71%	0.27%
of	99%	2.55%	an	88%	0.38%	his	70%	0.49%
and	99%	2.40%	have	88%	0.45%	had	70%	0.29%
to	99%	2.58%	was	85%	0.65%	been	70%	0.21%
in	99%	2.00%	not	84%	0.38%	all	69%	0.20%
for	98%	0.97%	this	83%	0.35%	which	69%	0.20%
that	96%	1.15%	are	83%	0.45%	will	68%	0.27%
on	96%	0.75%	has	83%	0.40%	out	68%	0.20%
is	95%	0.93%	who	81%	0.37%	up	68%	0.20%
with	95%	0.70%	they	78%	0.37%	if	67%	0.21%
at	93%	0.55%	he	78%	0.70%	than	66%	0.18%
by	92%	0.49%	one	77%	0.26%	were	66%	0.22%
it	92%	0.69%	said	77%	0.70%	would	65%	0.23%
as	91%	0.57%	more	75%	0.26%	can	65%	0.20%
but	91%	0.49%	about	75%	0.27%	new	64%	0.23%
			or	75%	0.31%	there	64%	0.18%

Figure 1.10: Document frequency and collection term frequency for the 50 most frequent terms in a news corpus (~20,000 documents).

Stop word lists for most languages are readily available, for example on [Kaggle \(stop words in 28 languages\)](#). However, eliminating stop words entirely creates problems. Consider the search for “it”: if we remove this term, we lose the ability to find IT books or the novel “It” by Stephen King. The modern approach retains all terms but assigns them different weights based on their discriminating power.

Zipf’s Law

The distribution of term frequencies follows a remarkably regular pattern. Let N be the total number of token occurrences in the collection and M the number of distinct terms. If we rank terms by decreasing frequency, Zipf’s law states that the probability of encountering the term at rank r is:

$$p_r = \frac{c}{r} = \frac{tf(t)}{N}, \quad \text{term } t \text{ with rank}(t) = r \quad (1.1)$$

The constant c depends only on the vocabulary size:

$$c = \frac{1}{\sum_{r=1}^M \frac{1}{r}} \approx \frac{1}{0.5772 + \ln M} \quad (1.2)$$

In other words, a few terms dominate the collection while the vast majority are rare.

For a collection with $M = 5,000$ terms, $c \approx 0.11$; with $M = 100,000$, $c \approx 0.08$. The practical implication is shown in Figure 1.11: terms fall into three zones. The most frequent terms (above the upper cut-off) appear everywhere and cannot discriminate. The rarest terms (below the lower cut-off) are highly specific but unlikely to occur in queries, making them less useful despite their discriminating potential. The terms most useful for distinguishing relevant from non-relevant documents lie in between.

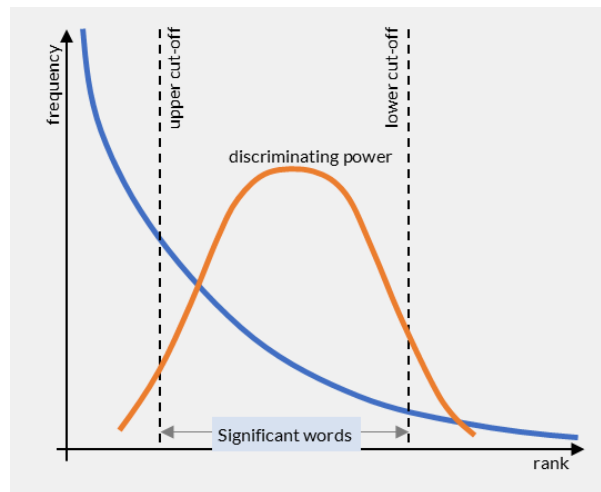


Figure 1.11: Zipf's law and term discrimination: the most useful terms for retrieval fall between the frequency extremes.

Term Discrimination and IDF

The intuition behind term weighting is straightforward: imagine removing a single term from every document in the collection. If documents suddenly become harder to tell apart, that term was helping to discriminate between them. Conversely, removing a term that appears everywhere (like “the”) changes nothing. Terms that appear in a moderate number of documents have the highest discriminating power, as confirmed empirically in Figure 1.12.

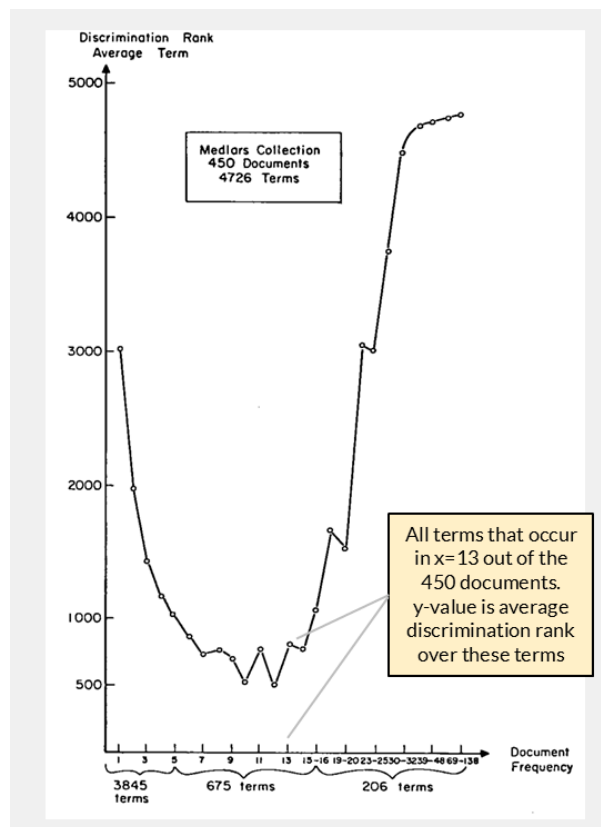


Figure 1.12: Discrimination rank vs. document frequency in the Medlars collection (450 documents). Terms occurring in 9-12 documents have the highest discrimination rank. Source: Salton, Wong & Yang (1975).

Karen Spärck Jones (1972) captured this intuition in a single formula called the **inverse document frequency** (IDF); see [A Statistical Interpretation of Term Specificity and Its Application in Retrieval](#) in the Further Reading section of the chapter summary. It builds on one statistic: the **document frequency** $df(t)$, which counts how many documents contain term t at least once. IDF then weights terms inversely to their commonness:

IDF provides the weighting mechanism we need: common terms get low weights, rare terms get high weights. Combined with term frequency, it produces the **tf-idf** weighting that we will use for some of the retrieval models in the next sections. Figure 1.13 compares IDF weights with empirical discrimination power, confirming that IDF closely approximates the true discriminating value of terms.

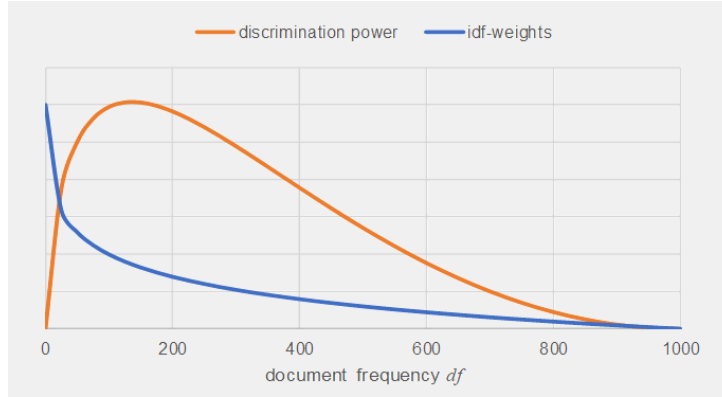


Figure 1.13: IDF weights closely track empirical discrimination power as a function of document frequency ($N = 1,000$).

Document Representation

With the vocabulary established and each term assigned an IDF weight, we can now represent documents as vectors. Each document becomes a point in an M -dimensional space where M is the vocabulary size and each dimension corresponds to one term. Figure 1.14 shows this process: tokenized text is stemmed, stop words are removed, and the remaining terms are counted to produce per-document frequency vectors.

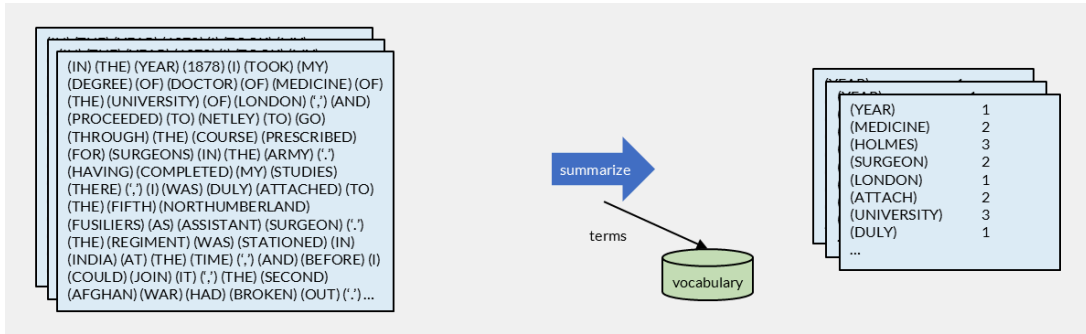


Figure 1.14: From tokenized text to term-frequency vectors: stem, remove stop words, count occurrences per document.

Set-of-Words and Bag-of-Words

Let D_i be the i -th document in a collection of N documents. The feature representation $\mathbf{d}_i \in \mathbb{R}^M$ is a vector where component $d_{i,j}$ describes how term t_j occurs in document D_i . We use $\text{tf}(D_i, t_j)$ to denote the number of occurrences of term t_j in document D_i .

The simplest option is the **set-of-words** model: for each term, record only whether it is present (1) or absent (0). A document mentioning “cat” three times looks identical to one mentioning it once:

$$d_{i,j} = \begin{cases} 1 & \text{if } t_j \text{ is present in } D_i \\ 0 & \text{otherwise} \end{cases} \quad (1.3)$$

A richer option is the **bag-of-words** model: record how many times each term appears. A document mentioning “cat” five times is represented differently from one mentioning it once:

$$d_{i,j} = \text{tf}(D_i, t_j) \quad (1.4)$$

Both models ignore term proximity and order. A document containing “New York” is represented the same way as one containing “York New”. Despite this limitation, these representations work remarkably well because the presence and frequency of specific terms are strong signals of relevance.

TF-IDF Weighting

The bag-of-words model treats all terms equally: a document with 10 occurrences of “the” scores the same as one with 10 occurrences of “algorithm”. By combining term frequency with IDF, we produce weighted vectors where informative terms contribute more:

Sparsity

In practice, a vocabulary can include millions of terms. However, most documents contain only a few hundred or thousand unique terms. The feature vectors are therefore sparsely populated: the vast majority of components are zero. Efficient storage methods like the inverted file exploit this sparsity by recording only the non-zero entries. During retrieval, the system only needs to consider documents that share at least one term with the query.

Boolean Retrieval

The previous section transformed raw text into tokens and term-frequency vectors. A retrieval model takes these representations and decides which documents match a query and, for ranked models, how strongly each document matches. We use the following small fictional library catalogue as a running example. Its wording is deliberately controlled so that the effects of preprocessing, term weighting, document length, and lexical ambiguity remain visible.

ID	Catalogue entry
D_1	The Cat and Dog in the Forest. A cat and a dog begin a woodland adventure.
D_2	Cats of the Woodland. Wild cats begin an adventure beneath ancient trees.
D_3	The Forest Hound. A loyal dog follows a woodland trail through ancient trees.
D_4	Feline Detective. A clever cat solves mysteries with a canine companion.
D_5	Random Forest for Pet Detection. A random forest model classifies cats and dogs in photographs.
D_6	Forests of Search Trees. Algorithms explore a forest of binary trees and graph paths.
D_7	Dog Dog Dog! A dog chases a ball, finds a bone, and wakes the neighbors.
D_8	Forest Forest Forest! A forest guide names trees, flowers, rivers, birds, and hidden ruins.
D_9	Cat Dog Forest.
D_{10}	Cat Dog Forest. A cat meets a dog in a forest beside rivers, mountains, castles, villages, bridges, and caves.
D_{11}	Woodland Companions. A kitten and a puppy share a moonlit adventure among old trees.
D_{12}	The Pet Bakery. A cat, a dog, and a baker make cakes, bread, biscuits, pies, and coffee.

Following [From Text to Searchable Vectors](#), we must define the preprocessing pipeline before applying a retrieval model. For this example, we lowercase the text, split it at punctuation and whitespace, and remove common stop words. We deliberately apply neither stemming nor lemmatization, and we do not expand synonyms. For example:

$D_1 \mapsto [\text{cat}, \text{dog}, \text{forest}, \text{cat}, \text{dog}, \text{begin}, \text{woodland}, \text{adventure}]$

$D_2 \mapsto [\text{cats}, \text{woodland}, \text{wild}, \text{cats}, \text{begin}, \text{adventure}, \text{beneath}, \text{ancient}, \text{trees}]$

These choices expose an important property of lexical matching: terms match by identity, not by visual or semantic similarity. During vocabulary construction, each distinct token receives its own identifier. For example, the vocabulary might map “cat” to token ID 70 and “cats” to token ID 83. A retrieval model therefore treats them as two unrelated dimensions, even though a reader immediately recognizes their relationship. For the same reason, “cat” does not match “feline” or “kitten”, and “dog” does not match “hound”, “canine”, or “puppy”. The reverse problem occurs with “forest”: the same token ID represents both a natural environment and the technical concept in “random forest”, although the meanings differ.

Standard Boolean Model

Consider the query `cat AND dog`. For each document, we test two conditions: the token “cat” is present, and the token “dog” is present. The complete predicate evaluates to true only if both conditions are met. The result set is therefore $\{D_1, D_9, D_{10}, D_{12}\}$. For `cat OR dog`, the predicate evaluates to true if at least one condition is met, producing the larger result set $\{D_1, D_3, D_4, D_7, D_9, D_{10}, D_{12}\}$.

This is the central idea of the Standard Boolean Model: a query is a predicate that maps each document to either true or false. Documents for which the predicate is true enter the result set; all others are excluded. The model filters rather than ranks, so every returned document has the same status regardless of how often or where the terms occur. This makes Boolean retrieval simple to understand, easy to calculate, and straightforward to explain.

Query Language

The query grammar defines two atomic predicates: a term is present, or a term is absent. It then defines two composition rules: AND and OR combine existing predicates into more complex expressions. Formally:

$$Q = t \quad (\text{term } t \text{ must be present}) \quad (1.5)$$

$$Q = \neg t \quad (\text{term } t \text{ must be absent}) \quad (1.6)$$

$$Q = Q_1 \vee Q_2 \quad (\text{at least one sub-query must be true}) \quad (1.7)$$

$$Q = Q_1 \wedge Q_2 \quad (\text{both sub-queries must be true}) \quad (1.8)$$

For example, $(\text{cat} \wedge \text{dog}) \vee \text{hound}$ returns a document if it contains both “cat” and “dog”, or if it contains “hound”. Boolean algebra allows any such expression to be rewritten in disjunctive normal form (DNF):

Evaluating a Query

The direct evaluation method follows the initial intuition. For each document D_i , the system checks whether every atomic predicate $\tau_{l,k}$ is true or false and then evaluates the complete expression. For $\text{cat} \text{ AND } \text{dog}$, D_1 produces $\text{true} \wedge \text{true} = \text{true}$, whereas D_4 produces $\text{true} \wedge \text{false} = \text{false}$.

A second method starts from all documents that satisfy each atomic predicate. Let $\mathbb{S}_t$ denote the set of documents containing term t . For the running collection:

$$\mathbb{S}_{\text{cat}} = \{D_1, D_4, D_9, D_{10}, D_{12}\}$$

$$\mathbb{S}_{\text{dog}} = \{D_1, D_3, D_7, D_9, D_{10}, D_{12}\}$$

The documents satisfying $\text{cat} \text{ AND } \text{dog}$ must belong to both sets. The documents satisfying $\text{cat} \text{ OR } \text{dog}$ may belong to either set:

$$\mathbb{S}_{\text{cat}} \cap \mathbb{S}_{\text{dog}} = \{D_1, D_9, D_{10}, D_{12}\}$$

$$\mathbb{S}_{\text{cat}} \cup \mathbb{S}_{\text{dog}} = \{D_1, D_3, D_4, D_7, D_9, D_{10}, D_{12}\}$$

More generally, each atomic predicate defines a set of matching documents:

$$\mathbb{S}_{l,k} =$$

The DNF structure translates directly into intersections for AND and unions for OR:

$$Q = \bigcup_{l=1}^L \bigcap_{k=1}^{K_l} \mathbb{S}_{l,k}$$

An inverted file stores these term-document sets so that the system can combine them without scanning every document. Chapter 4 develops this index structure and its efficient query-processing algorithms.

Limitations of Boolean Retrieval

Suppose the information need is a story involving a cat, a dog, and trees. The strict query $\text{cat} \text{ AND } \text{dog} \text{ AND } \text{trees}$ returns no documents. Several entries are plausible partial matches, but none contains all three exact tokens. For example, D_3 contains “dog” and “trees”, while D_1 contains “cat”, “dog”, “forest”, and “woodland”. Boolean evaluation excludes both because a missing term makes the complete AND predicate false.

Relaxing the query to $\text{cat} \text{ OR } \text{dog} \text{ OR } \text{trees}$ creates the opposite problem: it returns 11 of the 12 documents. The result includes technical entries such as D_6 because it contains “trees”, even though these are binary search trees rather than trees in an animal story. Small changes to a Boolean expression can therefore produce abrupt changes in result-set size, from no results to almost the entire collection.

The model also provides no basis for ordering the matches. For $\text{cat} \text{ AND } \text{dog}$, D_1 , D_9 , D_{10} , and D_{12} are equivalent Boolean matches. The model ignores that D_1 and D_{10} repeat both query terms, that D_9 consists only of the three topical terms “cat”, “dog”, and “forest”, and that much of D_{12} concerns baking. Term frequency, term importance, document length, and partial evidence do not affect the result.

These limitations establish the questions for the models that follow. The Extended Boolean Model introduces graded matching and ranking while retaining Boolean query structure. Probabilistic and vector-

space models provide alternative foundations for term weighting and ranking, while BM25 later combines probabilistic term importance with term-frequency saturation and document-length normalization. We will return to the same collection to see which limitation each model addresses.

Advantages: The model has simple and precise query semantics. Its decisions are easy to calculate and explain, and set operations support fast evaluation over large collections. Boolean predicates also combine naturally with structured metadata filters such as `language = English`.

Disadvantages: Users must express their information need as a Boolean expression and may obtain either too few or too many results. The model neither ranks matches nor represents partial matches. All predicates have equal influence, regardless of term frequency or discriminating power. These properties make the model better suited to exact filtering than to ranking documents by relevance.

Boolean retrieval nevertheless remains an important component of modern text and web search systems. A fast first stage often combines query terms with OR to retrieve a broad candidate set. A second stage then assigns relevance scores and ranks these candidates, allowing the user to inspect the strongest matches first. The next model takes an initial step in this direction by replacing Boolean decisions with graded scores.

Extended Boolean Model

The query `cat AND dog` gave D_1 , D_9 , D_{10} , and D_{12} the same Boolean status. Yet their evidence differs: D_1 and D_{10} contain both terms twice, while D_9 and D_{12} contain them once. Documents such as D_3 and D_4 contain only one query term and are excluded completely. The Extended Boolean Model replaces this hard boundary with a graded score. It can rank exact matches and retain partial matches with lower scores.

What Remains Boolean, and What Changes?

The query language remains the same. Users still express requirements with term predicates, AND, OR, and NOT, and the query retains its tree structure. What changes is the evaluation of that tree. An atomic predicate no longer produces only true or false. It produces a score between 0 and 1 that reflects the strength of the term evidence. Soft versions of AND and OR then combine these scores.

This extension directly addresses two limitations of the Standard Boolean Model. Term frequency and inverse document frequency can distinguish stronger from weaker evidence, while graded operators allow a document to receive a positive score even if it does not satisfy every predicate. The output is therefore a ranked list rather than an unordered set.

Formalizing Graded Evidence

Each document is represented by normalized TF-IDF weights. For term t_j in document D_i , define

$$d_{i,j} = \min\left(1, \frac{\text{tf}(D_i, t_j) \text{idf}(t_j)}{\alpha}\right), \quad 0 \leq d_{i,j} \leq 1,$$

where α is a fixed normalization value shared by the collection. A positive term predicate uses this weight directly; a negative predicate reverses it:

$$\text{sim}(\tau, D_i) =$$

The remaining question is how to combine the atomic scores. The p-norm model, introduced by Salton, Fox, and Wu; see [Extended Boolean Information Retrieval](#) in the Further Reading section of the chapter summary, interprets soft AND as closeness to the ideal point $(1, \dots, 1)$ and soft OR as distance from $(0, \dots, 0)$.

Other soft Boolean operators are possible. They differ in how strongly one high or low atomic score influences the combined result.

Evaluating the Running Example

Consider `cat AND dog` with $p = 2$. In the 12-document collection, $\text{df}(\text{cat}) = 5$ and $\text{df}(\text{dog}) = 6$, so

$$\text{idf}(\text{cat}) = \ln(12/5) \approx 0.875, \quad \text{idf}(\text{dog}) = \ln(12/6) \approx 0.693.$$

For this example, choose $\alpha = 4 \ln 2 \approx 2.773$, the largest TF-IDF weight among these query terms. We compute the two atomic scores for each candidate and combine them with the p-norm AND formula.

Document	$d_{i, \text{cat}}$	$d_{i, \text{dog}}$	p-norm AND score
D_1	0.632	0.500	0.561
D_{10}	0.632	0.500	0.561
D_7	0.000	1.000	0.293
D_9	0.316	0.250	0.282
D_{12}	0.316	0.250	0.282
D_4	0.316	0.000	0.143
D_3	0.000	0.250	0.116

The model now distinguishes repeated evidence: D_1 and D_{10} rank above D_9 and D_{12} . It also assigns positive scores to the partial matches D_3 , D_4 , and D_7 . Evaluation can still use an inverted file: the union of the query-term postings supplies candidates, after which the system computes and sorts their scores.

Limitations of the Extended Boolean Model

The ranking also exposes the heuristic nature of the method. D_7 contains “dog” four times but no “cat”, yet it narrowly outranks D_9 , which contains both requested terms. A different value of p , another normalization constant, or another soft operator can change this order. The query grammar states what AND and OR mean for true and false values, but it does not uniquely determine how graded evidence should be combined.

The model also inherits the lexical limitations of its representation. It still does not connect “cat” with “cats” or “feline”, and it cannot distinguish the meanings of “forest”. Finally, users must still translate an information need into an explicit Boolean expression. Graded evaluation softens the consequences of that expression but does not remove the burden of formulating it.

Advantages: The model retains the familiar Boolean query structure while adding partial matches and ranked output. TF-IDF weights distinguish terms by occurrence and collection frequency, and inverted files still support efficient candidate retrieval.

Disadvantages: Scores depend on heuristic choices for normalization, operators, and parameters, which can produce counter-intuitive rankings. The model retains lexical matching and explicit Boolean query formulation.

The classical p-norm model is rarely used as the primary ranker in modern search engines. Its central idea remains relevant, however: contemporary systems often combine strict Boolean filters with optional scoring clauses. This preserves explicit query constraints while allowing partial matches to receive different relevance scores. The Vector Space Model in the next section takes the next conceptual step: it replaces explicit Boolean expressions with free-text queries and ranks documents directly from their weighted term vectors.

Vector Space Retrieval

The Vector Space Model emerged from the SMART information retrieval project led by Gerard Salton and colleagues in the 1960s, formalized in Salton, Wong, and Yang's **A Vector Space Model for Automatic Indexing**, listed in the Further Reading section of the chapter summary. SMART introduced many ideas that became central to classical text retrieval: weighted terms, document and query vectors, partial matching, and ranked output. The model had a major influence on both research and practical search systems because it replaced exact logical decisions with simple numerical operations.

Consider the free-text query `cat dog forest`. Boolean OR retrieves every document containing at least one of these terms but cannot order the ten matches. The Vector Space Model uses the same tokenized collection and the same term statistics, but represents the query and documents as vectors. Their similarity becomes a score, allowing the system to place stronger matches first.

What Remains the Same, and What Changes?

As before, preprocessing determines the vocabulary, and each distinct token defines one dimension. Documents remain bag-of-words representations, so word order is ignored and lexical variants such as “cat” and “cats” remain separate. IDF still gives less weight to terms that occur in many documents.

The query changes more fundamentally. Instead of a Boolean expression, it is treated as a short document and processed through the same pipeline. No AND or OR operators are required. A document may receive a positive score when it shares only one query term, and the score determines its position in the ranked result list.

Document and Query Vectors

For a vocabulary of M terms, document D_i becomes the sparse vector $\mathbf{d}_i = (d_{i,1}, \dots, d_{i,M})$ with

$$d_{i,j} = \text{tf}(D_i, t_j) \text{idf}(t_j), \quad 1 \leq j \leq M.$$

The query becomes a vector in the same space:

$$q_j = \text{tf}(Q, t_j) \text{idf}(t_j), \quad 1 \leq j \leq M.$$

Placing the document vectors as columns produces the term-document matrix $\mathbf{A}$. This matrix is conceptually useful, but practical systems store only non-zero entries because each document contains a small fraction of the complete vocabulary.

Inner Product and Cosine Similarity

The inner product, also called the dot product, sums the weighted evidence from terms shared by query and document:

Only query dimensions contribute because $q_j = 0$ for every non-query term. Repeating a query term increases the score linearly. The score has no fixed upper bound and is often called a retrieval status value rather than a similarity probability. For the full collection,

$$\text{sim}_{\text{dot}}(Q, \mathbb{D}) = \mathbf{A}^\top \mathbf{q}.$$

Cosine similarity divides the same inner product by both vector lengths:

Cosine similarity compares vector direction rather than magnitude. For non-negative TF-IDF vectors, its value lies between 0 for no shared terms and 1 for identical directions. Document vectors can be normalized once during indexing. The system then computes cosine similarity as an inner product between unit-length vectors, avoiding repeated norm calculations at query time.

Evaluating a Query

Evaluation begins like Boolean OR. The system looks up the posting list for every query term and takes their union; documents outside this union have score zero. For each candidate, it accumulates $q_j d_{i,j}$ from the matching postings. This sum is already the inner-product score. Cosine evaluation adds one step: divide by the query norm and the document norm stored with the index. Finally, sort candidates by decreasing score.

This procedure avoids constructing the dense matrix A or comparing the query with every document. Chapter 4 develops inverted files and posting-list traversal in detail.

Running Example

For $Q = \text{"cat dog forest"}$ and $N = 12$, the query terms have

$$\text{idf}(\text{cat}) = \ln(12/5) \approx 0.875,$$

$$\text{idf}(\text{dog}) = \ln(12/6) \approx 0.693,$$

$$\text{idf}(\text{forest}) = \ln(12/7) \approx 0.539.$$

Each query term occurs once, so these values are also the non-zero components of q . Applying the same TF-IDF representation to every document gives the complete ranking evidence below.

Document	TF (cat, dog, forest)	Inner product	Cosine
D_1	(2, 2, 1)	2.784	0.659
D_2	(0, 0, 0)	0.000	0.000
D_3	(0, 1, 1)	0.771	0.102
D_4	(1, 0, 0)	0.766	0.093
D_5	(0, 0, 2)	0.581	0.059
D_6	(0, 0, 1)	0.291	0.034
D_7	(0, 4, 0)	1.922	0.232
D_8	(0, 0, 4)	1.162	0.139
D_9	(1, 1, 1)	1.537	1.000
D_{10}	(2, 2, 2)	3.075	0.342
D_{11}	(0, 0, 0)	0.000	0.000
D_{12}	(1, 1, 0)	1.247	0.137

The inner product ranks $D_{10} > D_1 > D_7 > D_9$. Repeating all query terms makes D_{10} the strongest match, while repeating only “dog” allows D_7 to outrank the compact exact match D_9 . Cosine produces a different order: $D_9 > D_1 > D_{10} > D_7$. The vector for D_9 has exactly the query direction and therefore receives the maximum score of 1. The additional non-query terms in D_{10} increase its norm and reduce its cosine score even though it contains every query term twice.

Limitations of Vector Space Retrieval

TF-IDF weights and both similarity measures are heuristic. They rank effectively in many collections, but neither formula directly estimates relevance. The inner product grows linearly with term frequency, so long documents tend to receive more query-term occurrences and authors can manipulate rankings by repeating selected terms. In the example, four occurrences of “dog” push D_7 above D_9 despite the absence of “cat” and “forest”.

Cosine removes this magnitude bias but introduces different assumptions. It prefers documents whose term-frequency ratios resemble the query and uses every document term in the normalization. Consequently, non-query terms can lower a score, although they provide no negative evidence in the inner-product numerator. This is counter-intuitive when a relevant passage is embedded in a longer document.

The representation also retains the earlier lexical limitations. Terms are independent dimensions, so the model neither connects “cat” with “feline” nor distinguishes the meanings of “forest”. Word order and dependencies such as “random forest” are absent from the bag-of-words vector.

Advantages: The model accepts natural free-text queries, supports partial matches, produces ranked output, and gives discriminating terms more influence through IDF. Sparse vectors and inverted files make evaluation simple and efficient.

Disadvantages: Weighting and similarity are heuristic. Inner products favour high term frequencies and longer documents, while cosine can penalize useful additional content and favours query-like term ratios. Neither measure includes term-frequency saturation, robust length normalization, or semantic relationships.

Modern Applications

Classical TF-IDF vectors with cosine similarity remain useful as transparent baselines and in small or specialized collections. They also support document similarity, clustering, classification, recommendation, and near-duplicate detection where a sparse lexical representation is sufficient. For primary lexical ranking, production search systems now commonly prefer BM25 because it controls term-frequency growth and document-length effects more carefully.

The geometry has become even more important than the original representation. Modern semantic search encodes text as dense embedding vectors and retrieves neighbours with inner product or cosine similarity, often through approximate nearest-neighbour indexes. Hybrid systems combine this dense vector retrieval with sparse lexical methods such as BM25. Thus, modern systems frequently retain the Vector Space Model's scoring operations while replacing TF-IDF dimensions with learned semantic features.

The Vector Space Model establishes the geometry of ranked retrieval but leaves term-frequency growth and document-length effects to heuristic choices. The next section introduces probabilistic term evidence and develops BM25 as a practical response to these limitations.

Probabilistic Ranking and BM25

The Vector Space Model ranks effectively, but its weighting and similarity measures are heuristic. Probabilistic retrieval asks a more direct question: given query Q and document D_i , how likely is the document to be relevant? The Binary Independence Model (BIR) develops this idea from binary term evidence. BM25 then retains its probabilistic term weighting while adding the term-frequency saturation and document-length normalization missing from BIR and vector-space scoring.

Binary Independence Model

The BIR model uses the same tokenized vocabulary as the previous models, but represents each query and document as a set of terms. It was formalized by Robertson and Spärck Jones; see **Relevance Weighting of Search Terms** in the Further Reading section of the chapter summary. A component is 1 when a term is present and 0 when it is absent. It assumes that terms contribute independently and that non-query terms occur equally often in relevant and non-relevant documents. Under these assumptions, only query terms present in a document affect its rank.

Let R denote relevance and NR non-relevance for query Q . Ranking by the posterior odds

$$\frac{P(R \text{ mid } D_i, Q)}{P(NR \text{ mid } D_i, Q)} \quad (1.9)$$

is equivalent to ranking by the document evidence after query-dependent constants are removed. Define

$$r_j = P(d_{i,j} = 1 \text{ mid } R, Q), \quad n_j = P(d_{i,j} = 1 \text{ mid } NR, Q). \quad (1.10)$$

The first probability measures how often term t_j occurs in relevant documents; the second measures how often it occurs in non-relevant documents. The resulting score is additive, but several probability terms must cancel before we reach that form. The derivation below explains why only query terms present in the document remain.

Relevance Feedback

Initial estimates. Before any relevance information is available, the system must produce a first ranking. BIR commonly assumes that every query term has an equal chance of occurring in a relevant document, $r_j = 0.5$. It estimates occurrence in non-relevant documents from the term's document frequency in the complete collection:

$$r_j = 0.5, \quad n_j = \frac{\text{df}(t_j) + 0.5}{N + 1}. \quad (1.11)$$

These initial values produce the first c_j weights and therefore the first result list.

Collecting feedback. The user can now mark retrieved documents as relevant or non-relevant, for example through like and dislike controls. These judgements provide a sample of both classes. Because r_j is the probability that term t_j occurs in a relevant document, we can estimate it by counting how many judged relevant documents contain the term. We estimate n_j in the same way from the judged non-relevant documents.

Updating the estimates. Suppose the user has judged K documents and marked L of them as relevant. Let k_j be the number of judged documents containing t_j , and let l_j be the number that both contain t_j and are relevant. The relevant sample therefore contains l_j occurrences among L documents. The non-relevant sample contains $k_j - l_j$ occurrences among $K - L$ documents. Without smoothing, the corresponding estimates would be

$$r_j \approx \frac{l_j}{L}, \quad n_j \approx \frac{k_j - l_j}{K - L}. \quad (1.12)$$

In practice, BIR adds pseudo-counts so that small samples do not produce probabilities of exactly 0 or 1:

$$r_j = \frac{l_j + 0.5}{L + 1}, \quad n_j = \frac{k_j - l_j + 0.5}{K - L + 1}. \quad (1.13)$$

The updated probabilities produce new c_j weights and a revised ranking. Further rounds of feedback can refine the estimates, although users may be unwilling to judge many results.

BIR marks a change in ambition, not only a change in formula. The Vector Space Model ranks by geometric heuristics that happen to work well; BIR instead tries to *explain* relevance from first principles, estimating how likely a document is to be relevant given its terms. It still represents documents as term vectors and still combines evidence with the same OR-like accumulation as VSM: any query term found in the document contributes its weight to the score. What changes is the meaning of that weight. Instead of a heuristic tf-idf product, c_j estimates the independent contribution of each query term toward relevance, grounded in how often the term separates relevant from non-relevant documents.

The restrictive assumptions behind this first version, binary term presence, term independence, and no document-length effect, are not inherent to the probabilistic idea itself. They make BIR tractable, and later probabilistic models relax them one at a time. The 2-Poisson model, for instance, models within-document term frequency directly instead of treating term presence as binary, at the cost of a more complex parameter estimation problem; see Robertson and Walker’s **Some Simple Effective Approximations to the 2-Poisson Model for Probabilistic Weighted Retrieval** in the Further Reading section of the chapter summary. Probabilistic language models take yet another route, estimating the probability that a document’s language model would generate the query. The section below develops BM25, which keeps BIR’s additive, per-term structure but approximates the 2-Poisson model’s term-frequency behaviour with a simpler saturating function.

The feedback mechanism also has a practical limitation worth mentioning. A user who rates a first result list of 10 documents gives us a sample of size $K = 10$ from which to estimate r_j and n_j , far too small to pin down these probabilities reliably, especially for terms that appear in only one or two of the judged documents. Larger judged samples would improve the estimates, but no user will realistically rate hundreds of documents to make search work. This tension between statistical need and user effort is a recurring theme in relevance feedback, not a flaw specific to BIR.

Okapi BM25

BM25 was developed at London’s City University by Stephen Robertson, Karen Spärck Jones, and colleagues as a practical approximation to the 2-Poisson model. It became the robust default for free-text retrieval that BIR itself never was. Robertson and Zaragoza’s **The Probabilistic Relevance Framework: BM25 and Beyond**, listed in the Further Reading section of the chapter summary, surveys the complete derivation and its later extensions.

What is the meaning of “Okapi BM25”?

The name comes from Okapi, the experimental retrieval system built at City University London in the 1980s and 1990s where the formula was first implemented, not from any property of the ranking function itself. “BM” stands for “Best Matching”, and “25” simply identifies it as the 25th weighting variant the researchers tried.

Three Unresolved Issues

Before introducing the BM25 formula, it helps to see exactly what the two preceding models still get wrong. Each issue below reappears as a concrete design decision in BM25.

Vector-space scoring rewards repetition without limit. Recall the running query `cat dog forest` from the Vector Space Retrieval section. Its inner product already ranks D_7 (four occurrences of “dog”, score 2.773 for that term alone) above D_9 (one occurrence of each query term, contributing 0.693 for “dog” toward its combined score of 1.537), even though D_9 is the only document that mentions “cat”, “dog”, and “forest” together. If a document repeated “dog” twenty times instead of four, its score would climb to 13.863, growing linearly and without bound. A ranking function that rewards raw repetition this directly is vulnerable to keyword stuffing: an author can inflate a document’s score simply by repeating a term, regardless of whether the document is actually about that topic.

Raw term frequency ignores what “coverage” means for document length. A term occurring once in a 3-token document (such as D_9) is stronger evidence of relevance than the same term occurring once in a 20-token document, because in the short document, that one occurrence forms a much larger share of the content. TF-IDF and the inner product treat both occurrences identically. Relevance should depend on how much of the document is about the query term, not merely how many times it appears. To reward the same relevance signal, a longer document should need proportionally more occurrences of a term than a shorter one to reach the same score.

Binary presence throws away frequency information, but the idea behind r_j and n_j is worth keeping. BIR’s document evidence is either 0 or 1, so it cannot express that D_7 mentions “dog” four times while D_3 mentions it once. At the same time, BIR contributes something valuable: the idea that a term’s ranking weight should reflect how well that term distinguishes relevant from non-relevant documents, not just how rare it is in the collection overall. BM25 keeps this probabilistic idea but combines it with a real-valued term-frequency component.

Term-Frequency Saturation

BM25 replaces the raw count in the TF-IDF product with a bounded function of term frequency:

$$\hat{tf}_{k_1} = \frac{tf(k_1 + 1)}{tf + k_1}, \quad k_1 > 0. \quad (1.14)$$

The first occurrence contributes strongly; later occurrences add progressively less; the value never exceeds $k_1 + 1$ no matter how often the term repeats. This directly answers the spamming issue: repeating “dog” twenty times can no longer produce an unbounded score. Parameter k_1 controls how quickly the function saturates. Values between 1 and 2 are common, with $k_1 = 1.2$ used in the running example. Figure 1.15 contrasts linear, square-root, and saturating term-frequency weights.

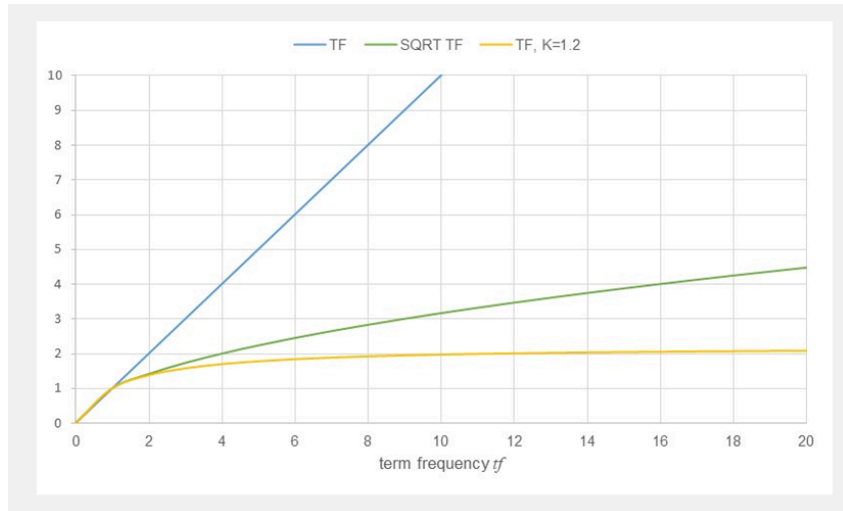


Figure 1.15: Linear, square-root, and BM25-saturated term-frequency weighting.

Document-Length Normalization

Saturation alone does not address the length issue: a document with $tf=1$ in 3 tokens and a document with $tf=1$ in 20 tokens still saturate identically. BM25 makes the saturation point depend on document length by scaling the denominator:

$$\hat{tf}_{k_1, b}(D_i, t) = \frac{tf(D_i, t)(k_1 + 1)}{tf(D_i, t) + k_1 \left(1 - b + b \frac{|D_i|}{\text{avgdl}}\right)}. \quad (1.15)$$

Here $|D_i|$ is the number of processed tokens (=document length) and avgdl is the collection’s average document length. For the running collection, $\text{avgdl} = 9.67$ tokens. A single occurrence of “dog” in a 3-token document (shorter than average) contributes 1.393, while the same single occurrence in a 20-token document (longer than average) contributes only 0.696: the short document needs less repetition to reach the same weight because that occurrence covers a larger share of its content. Parameter $b \in [0, 1]$ controls

the strength of this effect: $b = 0$ disables length normalization entirely, while $b = 1$ applies the full length ratio. Figure 1.16 shows how the same term frequency receives more weight in a short document than in a long one.

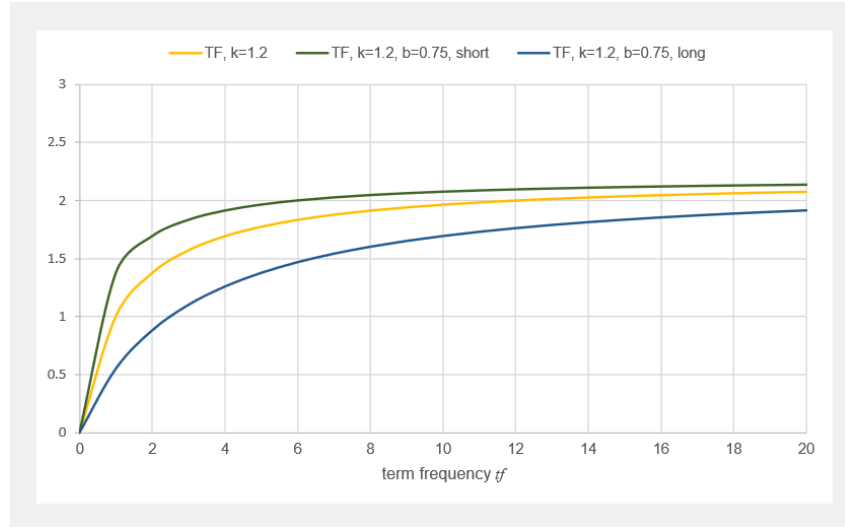


Figure 1.16: BM25 term-frequency saturation for average, short, and long documents.

Probabilistic IDF

The third issue was that BIR's idea of estimating a term's discriminating power from r_j and n_j is worth keeping even though its binary document model is not. Substituting the no-feedback BIR estimates ($r_j = 0.5$ and n_j from document frequency) into the c_j formula from the previous section gives exactly the classical BM25 weight:

$$\text{idf}_{\text{BM25}}(t) = \log \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5}. \quad (1.16)$$

This value becomes negative when a term appears in more than half of the collection. Many implementations instead use the always-positive Lucene variant

$$\text{idf}_+(t) = \log \left(1 + \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5} \right) = \log \frac{N + 1}{\text{df}(t) + 0.5}. \quad (1.17)$$

Figure 1.17 compares these variants with the smoothed classical IDF introduced earlier. All encode the same core intuition that common terms provide less evidence, but their numerical values are not interchangeable.

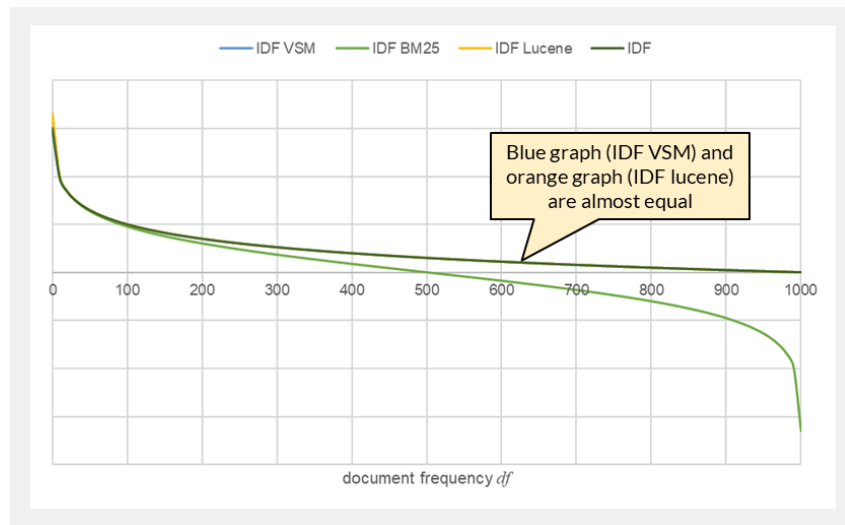


Figure 1.17: Classical, original BM25, and positive Lucene IDF variants.

A negative weight is undesirable for an additive scoring function: it would let a matching query term actively lower a document's score, the opposite of what a match should do, and it complicates summing evidence across query terms of very different commonness. The Lucene weight $\log \frac{N+1}{\text{df}(t)+0.5}$ and the smoothed classical VSM weight $\log \frac{N}{\text{df}(t)}$ are nearly identical: comparing them shows that the Lucene variant is equal to the classical one exactly when $\text{df}(t) = N/2$, slightly below it for rare terms, and slightly above it for common terms, never departing far in either direction. This is exactly the region where the original BM25 IDF turns negative, so Lucene's variant tracks the familiar classical shape almost everywhere while staying non-negative where it matters. Lucene therefore adopts it as a numerically safer stand-in for the same underlying quantity rather than as a conceptually different weighting scheme.

Note

The standard BM25 formula uses the no-feedback estimate for r_j and does not itself perform a feedback step. Nothing prevents combining BM25 with relevance feedback: a system can substitute the feedback-refined r_j and n_j from BIR into the same weight instead of the fixed $r_j = 0.5$. In practice, systems built on Lucene use the fixed no-feedback form as the default.

Complete Formula and Running Example

Using the positive IDF variant gives the practical scoring function used in this example.

For $Q = \text{cat dog forest}$, use $k_1 = 1.2$, $b = 0.75$, and the same preprocessing as before. The 12 documents have $\text{avgdl} = 9.67$ tokens. Their positive IDF values are 0.860 for “cat”, 0.693 for “dog”, and 0.550 for “forest”.

Evaluation follows the same efficient pattern as vector-space retrieval. The system takes the union of the query-term posting lists, accumulates one BM25 contribution per matching term, and sorts candidates by decreasing score. Chapter 4 develops the posting-list algorithms used for this process.

Limitations and Modern Applications

BM25 remains lexical and bag-of-words based. It neither connects “cat” with “feline” nor distinguishes the natural and technical meanings of “forest”. It also assumes that document length should influence relevance in a consistent way, so k_1 and b may require tuning for collections with unusual document types. Despite its probabilistic derivation, a BM25 score is a ranking value, not a calibrated probability of relevance.

Advantages: BM25 combines partial matching, discriminative term weighting, term-frequency saturation, and document-length normalization in an efficient additive score. It is transparent, fast, and difficult to improve upon as a lexical baseline.

Disadvantages: It retains lexical matching and term-independence assumptions, requires parameter choices, and does not directly represent phrase meaning or semantic similarity.

BM25 is the default or standard lexical ranker in systems built on Lucene, including Elasticsearch, Solr, and OpenSearch. It is also widely used as a first-stage retriever in retrieval-augmented generation and in hybrid search, where its exact lexical matches complement dense embedding retrieval.

Summary

Model Comparison

Property	Boolean	Extended Boolean	Vector Space	BIR	BM25
Query form	Boolean expression	Boolean expression	Free text	Term set	Free text
Ranked output	No	Yes	Yes	Yes	Yes
Partial matches	No	Yes	Yes	Yes	Yes
Term frequency	Ignored	Normalized TF-IDF	Linear TF-IDF	Ignored	Saturating
Document length	Ignored	Indirect normalization	Cosine or none	Ignored	Explicit parameter b
Foundation	Set theory	Soft-logic heuristic	Geometric heuristic	Relevance probability	BIR plus TF and length models
Typical role	Exact filters	Specialized soft constraints	Baseline and vector similarity	Feedback model and foundation	Production lexical ranking
Main weakness	No ranking	Operator choices	Length and repetition effects	Binary evidence and feedback burden	Lexical matching and tuning

The progression is cumulative rather than a sequence of complete replacements. Boolean predicates remain useful for mandatory filters, vector operations remain central to dense retrieval, BIR explains relevance feedback and probabilistic term weights, and BM25 provides the strongest general-purpose lexical ranking baseline.

Key Takeaways

1. Text retrieval represents documents and queries using the same vocabulary, then measures similarity to estimate relevance — this direct word-level matching makes text the least affected modality by the semantic gap.
2. The feature extraction pipeline (Extract $\rightarrow$ Split $\rightarrow$ Tokenize $\rightarrow$ Summarize) transforms raw text into searchable representations. The bag-of-words model records term frequencies; the set-of-words model records only term presence.
3. IDF weighting captures term discrimination power: rare terms that appear in few documents are more informative for distinguishing relevant from non-relevant results.
4. Retrieval models evolved from Boolean (no ranking, exact match) through the Extended Boolean Model (partial matches, simple scoring) to the Vector Space Model (tf-idf, cosine similarity) and Probabilistic Retrieval (BIR model, relevance feedback).
5. BM25 combines the best ideas: saturating term frequency, document-length normalization, and probabilistically grounded IDF — it remains the default baseline in modern search engines (Lucene, Elasticsearch, OpenSearch).

Key Formulas

Self-Check Questions

1. (Understand) Explain in your own words why the Boolean model cannot rank documents, and what the Extended Boolean Model changes to enable ranking.

2. (Analyze) Given a collection of 10,000 documents where the term “retrieval” appears in 50 documents and the term “the” appears in 9,800 documents, calculate the IDF for both terms. Why does this difference matter for relevance scoring?
3. (Analyze) A document is $3\times$ longer than the average document in the collection. How does BM25 (with $b = 0.75$) adjust the effective term frequency compared to an average-length document? What happens when $b = 0$?
4. (Evaluate) An e-commerce search system currently uses Boolean retrieval with faceted filtering. A developer proposes switching to BM25. What are the trade-offs? In which scenarios might Boolean retrieval still be preferable?

Further Reading

- Spärck Jones, K. (1972). **A Statistical Interpretation of Term Specificity and Its Application in Retrieval**. *Journal of Documentation*, 28(1), 11–21. [PDF](#) — Introduced inverse document frequency; foundational for every term-weighting scheme discussed in this chapter, including TF-IDF and BM25.
- Salton, G., Wong, A., & Yang, C. S. (1975). **A Vector Space Model for Automatic Indexing**. *Communications of the ACM*, 18(11), 613–620. [PDF](#) — The original paper introducing the Vector Space Model; still readable and surprisingly concise.
- Robertson, S. E., & Spärck Jones, K. (1976). **Relevance Weighting of Search Terms**. *Journal of the American Society for Information Science*, 27(3), 129–146. [Publisher page](#) — Formalizes the Binary Independence Model and its r_j/n_j relevance-feedback estimates covered in this chapter.
- Salton, G., Fox, E. A., & Wu, H. (1983). **Extended Boolean Information Retrieval**. *Communications of the ACM*, 26(11), 1022–1036. [Free access](#) — Introduces the p-norm model, positioning graded Boolean retrieval between the strict Boolean and vector-space models.
- Porter, M. F. (1980). **An Algorithm for Suffix Stripping**. *Program*, 14(3), 130–137. [Author’s page](#) — The classic stemming algorithm still used as a baseline in English text processing.
- Robertson, S. E., & Walker, S. (1994). **Some Simple Effective Approximations to the 2-Poisson Model for Probabilistic Weighted Retrieval**. In *Proceedings of the 17th ACM SIGIR Conference*, 232–241. [PDF](#) — Derives BM25’s term-frequency saturation and length normalization from the 2-Poisson model discussed in the Probabilistic Ranking and BM25 section.
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). **Introduction to Information Retrieval**. Cambridge University Press. [Free online edition](#) — The standard graduate textbook; chapters 1–6 cover Boolean, VSM, and probabilistic models with rigorous treatment.
- Robertson, S. E., & Zaragoza, H. (2009). **The Probabilistic Relevance Framework: BM25 and Beyond**. *Foundations and Trends in Information Retrieval*, 3(4), 333–389. [PDF](#) — Definitive survey of BM25’s theoretical foundations and practical extensions.

2 - Performance Evaluation

A student searches a university library for books that provide the computer science foundations needed for a master's degree. One search system returns many useful books mixed with unrelated titles. Another returns fewer useful books, but places them near the top. Which system is better? The answer depends on what the student needs: broad coverage, a clean result set, or the best books in the first few positions.

Performance evaluation turns claims such as “better results” into measurements that can be checked and reproduced. A fair comparison requires more than a metric. It needs a representative collection, realistic information needs, consistent relevance judgments, and a clear definition of success. Without this shared benchmark, a higher score may reflect an easier test rather than a better retrieval method.

Evaluation also applies when a text system assigns labels rather than retrieving documents. A spam filter decides whether an email is unwanted, while an AI assistant may route a query to a knowledge base, web search, calculator, or clarification step. Retrieval evaluates decisions about query-document pairs; classification evaluates labels assigned to text items. Both expose the same fundamental trade-off: finding more desired items often means accepting more incorrect decisions.

Designing a Retrieval Benchmark

A university library holds 50 books. A student submits one information need: which books give a beginning master's student solid computer science foundations? Fifteen of the 50 books are useful for that need. Two retrieval systems answer the same request, and their first five results look like this.

Rank	System A result	Useful?	System B result	Useful?
1	Computer Organization and Design	yes	Database Systems: The Complete Book	yes
2	Operating System Concepts	yes	Pattern Recognition and Machine Learning	yes
3	The Handmaid's Tale	no	Introduction to Algorithms	yes
4	Narrative of the Life of Frederick Douglass	no	On the Origin of Species	no
5	Towards a New Architecture	no	Introduction to the Theory of Computation	yes

System A does not stop at rank 5. It returns 25 books in total, half the library, and finds 12 of the 15 useful ones. System B returns only 8 books, of which 6 are useful. So System A surfaces more of what exists, and System B wastes less of the reader's attention.

Which system is better? Two students with the same assignment can reasonably disagree. One wants a short list to read end to end and prefers System B. Another wants to be certain that nothing important was missed and prefers System A. Neither is wrong, so before we can measure anything, we have to say what counts as a good result.

What Counts as a Better Result

Relevance is not a property of a book. It is a judgment about whether a book serves one person's need at one moment. "Introduction to Algorithms" is essential to our student and useless to someone looking for a novel for a train journey. Every measurement in this chapter therefore rests on a human decision taken before any system ran.

That subjectivity is unavoidable, but it is containable. Once the need is written down and the judgments are recorded, the counting becomes mechanical: anyone who applies the same procedure to the same lists obtains the same numbers. Evaluation does not remove the subjective step, it isolates it.

What counts as success then varies enormously with the need. At one extreme only the first few results matter. Someone checking when BM25 was published reads two or three entries and stops, so anything below rank 5 might as well not exist. A short and precise list wins here, in the style of System B, and whatever relevant material it leaves behind costs the reader nothing. At the other extreme, missing anything is the real failure. A patent lawyer searching for prior art will work through a long and noisy list rather than risk overlooking one document, so broad coverage wins, in the style of System A, and a very short list would be useless however precise it was.

Many information needs are in between these extremes, including our student's. A foundations reading list has to be broad enough to cover the main areas of the field, yet short enough to work through in a semester. Neither system serves that well: System A buries the useful books among thirteen novels, and System B leaves out nine of the fifteen that would have helped. Web search leans towards the first extreme, biomedical literature search towards the middle, and the lesson is the same in every domain: "better" has no meaning until the intended use is named.

Why Anecdotes Are Not Evidence

Product advertising is full of comparative claims with no shared basis. A vendor states that a search engine is forty percent more accurate, without naming the collection it was tested on, the queries that were used,

who judged the results, or what “accurate” counts. Such a claim cannot be checked, and it is usually easy to construct: pick the queries where your system happens to win, and the number follows.

Retrieval systems invite exactly this failure, because single-example demonstrations are trivially available. For almost any pair of systems, someone can find one query where the first beats the second and another query where the reverse holds. The two result lists at the start of this section are a demonstration, not evidence. They describe one need in a 50-book collection, so they support no general statement about either system.

A scientific comparison replaces the anecdote with a fixed experimental setup. The collection is frozen, the information needs are written down in advance, the relevance judgments are made independently of the systems being compared, and the measurement procedure is stated before any results are seen. Others can then rerun the same experiment and obtain the same numbers. This is what a benchmark provides: not a guarantee that the winner is better for every purpose, but a claim that is reproducible and open to challenge.

Comparable setups also make progress cumulative. When many research groups report results on the same benchmark, a new method can be placed against a decade of prior work instead of against whatever the authors chose to reimplement.

The Four Components of a Benchmark

A benchmark is built from four parts: a document collection, a set of information needs, relevance judgments, and a measurement procedure. Each part answers one question, and a weakness in any of them undermines everything measured on top.

The **document collection** defines the universe the system may search. It can hold news articles, scientific papers, legal filings, product descriptions, or, as in our running example, library records. Two properties matter most. The collection must be stable, because results are only comparable over time if the searched material does not change underneath them. It must also be representative, reflecting the document lengths, content types, and topic mix that the target users really encounter. Well-known examples include the TREC collections, which range from newspaper archives to biomedical abstracts, MS MARCO for web-style passage retrieval, and domain corpora such as PubMed Central and arXiv for scientific literature.

The **information needs** state what the users actually want. It is worth separating the need from the query string: the need is “which books give a beginning master’s student solid computer science foundations”, while the query is whatever short text the user types into the search box. Assessors judge against the need, not the keywords, which is why the need must be written out clearly enough that two people would interpret it the same way. Needs are best derived from real user behaviour such as query logs. When no log exists, large language models can draft candidate needs from samples of the collection, which expands coverage cheaply, though the drafts still require review to stay realistic.

The **relevance judgments** record, for each need, which documents satisfy it. They are the ground truth, so their quality bounds the quality of every conclusion. Human assessors work from written guidelines, receive training on example cases, and are checked for agreement, because vague criteria produce inconsistent labels that no metric can repair. Large language models can pre-screen documents and propose judgments for clear-cut cases, leaving human effort for the borderline ones. Judgments must then be frozen: revising them later silently invalidates every result measured earlier.

The **measurement procedure** turns a result list into a number. It fixes which measure is computed, at which cutoff, and how values are combined across needs. The next three sections develop these measures, starting with results treated as an unordered set, then as a ranking, then as a ranking with graded relevance. What matters here is that the procedure is chosen before the experiment, not after the results are known.

The following table collects the practical rules for each component.

Component	What it provides	Design rule	Common failure
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Component	What it provides	Design rule	Common failure
Document collection	The universe that may be searched	Freeze it for the benchmark's lifetime, make it representative of the target domain, preprocess consistently, and give every document a stable identifier	Built around one feature, so it flatters the methods that exploit that feature
Information needs	What the target users want	Derive needs from real user behaviour, state each one unambiguously, and provide 25 to 50 for a first experiment or 100 and more for a serious comparison	Wording so vague that assessors disagree and the ground truth becomes noisy
Relevance judgments	The ground truth for scoring results	Write assessor guidelines, train assessors, check their agreement, and freeze the labels once published	Judgments revised mid-project, which invalidates all earlier measurements
Measurement procedure	The rule that turns results into numbers	State the measure, the cutoff, and the averaging method before running any experiment	Measure chosen after seeing the results, which turns evaluation into advocacy

Building all four parts is expensive, which is why established benchmarks are reused heavily and why a small, carefully judged collection is often more valuable than a large, carelessly labeled one.

Judging at Scale

One practical obstacle remains. Judging every document for every need is impossible at realistic scale: fifty books can be assessed exhaustively, fifty million cannot. Pooling is the standard response. The organizers collect the documents that participating systems actually retrieved, judge that pool, and treat everything unjudged as not relevant. [Figure 2.1](#) shows the situation this assumes. Relevant documents that no participant retrieved never get judged, so the measured values differ somewhat from what exhaustive judging would produce. For a comparison that matters less than it first appears: every participant is scored against the same judged subset, so the ordering between them survives even though the individual numbers are approximations.

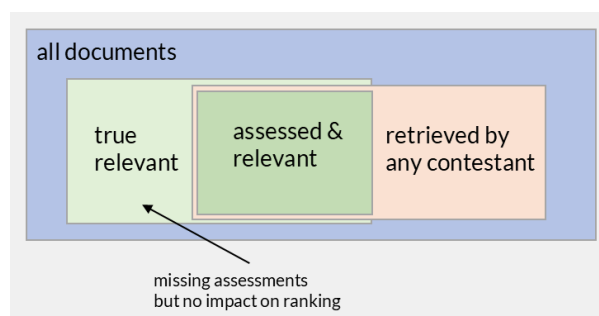


Figure 2.1: Pooled assessment when participating systems retrieve most relevant documents.

This assumption is sound while the pool covers most of the relevant material, and it weakens as the judged fraction shrinks. Pooling is one of several places where a benchmark can mislead rather than inform, and [When Evaluation Misleads](#) returns to all of them once we have the measures needed to describe what gets distorted.

Note

Sections 5 and 6 evaluate text classifiers rather than retrieval results, and the ground truth there is a test set rather than a benchmark of the kind described above. Three differences matter. The unit of evaluation is a single text item and its label, not a need-document pair. The labels are normally complete, so pooling does not apply, but class balance and label quality become the central concerns. And a classification dataset is split into training, validation, and test parts, with any decision threshold tuned on the validation part and never on the test part. Section 5 develops these points where they are needed.

With a benchmark in place, the remaining question is mechanical: how do we turn a result list into a number that reflects how well the need was served? The next section answers it for results treated as an unordered set, using the same library collection and the same two systems.

Precision and Recall

Boolean retrieval returns a set of documents with no order attached: a document either matches the query or it does not. This section evaluates that kind of result. Ranking comes in [Evaluating Ranked Results](#); until then, a result list is treated purely as a set.

Recall System A and System B from [Designing a Retrieval Benchmark](#), both answering the same need: which books give a beginning master's student solid computer science foundations? System A returned 25 books, System B returned 8. To turn "returned" and "useful" into numbers, every book in the 50-book collection falls into exactly one of four groups, depending on whether the system retrieved it and whether it is actually relevant.

Counting Outcomes

	Relevant	Not relevant
Retrieved	True Positive (TP)	False Positive (FP)
Not retrieved	False Negative (FN)	True Negative (TN)

A true positive is a relevant book the system retrieved: a hit. A false positive is a book the system retrieved that turns out not to be relevant: noise in the result. A false negative is a relevant book the system failed to retrieve: a miss. A true negative is a book the system correctly left out. [Figure 2.2](#) shows this partition as overlapping sets for both systems: the 50-book collection, the 15 relevant books, the retrieved books, and their intersection.

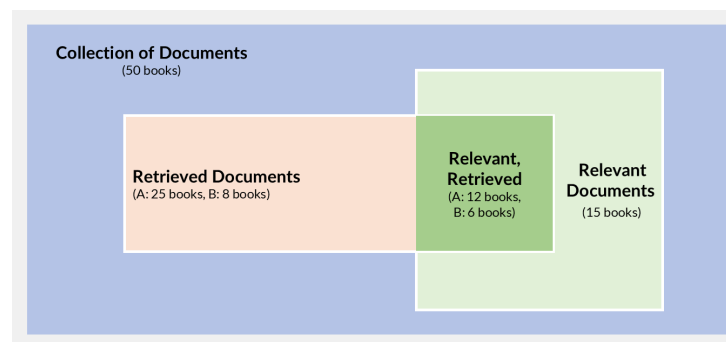


Figure 2.2: Set-theoretic view of retrieval performance for Systems A and B over the 50-book collection.

Counting each book in the collection against the two systems gives:

	TP	FP	FN	TN
System A	12	13	3	22
System B	6	2	9	33

Every row sums to 50, the size of the collection, and every count came directly from checking the 15 known relevant books against each system's result list.

Precision and Recall

The raw counts alone are hard to compare. System A has 12 true positives and System B has 6, but A also retrieved three times as many books and operates in a collection where 15 are relevant. A system retrieving the entire library would produce 15 true positives and look best by that count alone, which shows that an absolute count cannot separate a good system from one that simply returns everything. Ratios solve this: dividing by the appropriate total produces a value between 0 and 1 that is independent of how many books the system returned or how many relevant books the collection contains.

This is the trade-off from [Designing a Retrieval Benchmark](#) stated in numbers rather than in words. System B's list is three-quarters useful material; System A's list finds four-fifths of everything useful but buries it

among thirteen novels. Whether 48% precision at 80% recall is better than 75% precision at 40% recall still depends on what the user prefers (quick fact check or exhaustive library scan).

A third quantity, fallout, measures the opposite kind of mistake: how much of the irrelevant material got pulled in.

$$f = \frac{FP}{FP + TN} \quad (2.1)$$

Fallout is the fraction of non-relevant documents that the system retrieved. For our two systems, $f_A = 13/35 = 37.1\%$ and $f_B = 2/35 = 5.7\%$. Fallout measures the reading cost imposed on someone who inspects every result: the patent lawyer from Section 1 needs high recall, because missing a document is expensive, but also wants low fallout, because every irrelevant document in the list costs time and attention. System A delivers recall of 80% but pulls in more than a third of the irrelevant collection; the ideal for an exhaustive search would push recall toward 1 while keeping fallout near 0.

Balancing the Two

Comparing two numbers side by side, as above, does not say which system wins. The F_β -measure collapses precision and recall into a single value, letting a scenario's priority set the trade-off explicitly.

At $\beta = 0$ the formula reduces to precision alone; as $\beta \rightarrow \infty$ it reduces to recall alone. F_1 , the case $\beta = 1$, is the most common default and appears throughout machine learning wherever a single number is needed to compare classifiers.

The winner changes between $\beta = 0.5$ and $\beta = 1$. Neither system is better in general: a fact-checker who wants a short, clean list would set β below 1 and prefer System B, while a patent lawyer would set β above 1 and prefer System A. F_1 is a convenient default, not a neutral one, since it silently commits to weighing a false positive and a false negative equally.

Averaging Across Needs

One need produces one precision and one recall value. A real benchmark evaluates many needs, and summarising all of them into a single score requires an averaging method. The natural first idea is to compute precision and recall for each need separately and then take the arithmetic mean.

Example: Consider two additional needs alongside the CS-foundations query, answered by the same two systems:

Need	Relevant	System A: retrieved / found	System B: retrieved / found
CS foundations	15	25 / 12	8 / 6
Stage plays	5	6 / 4	4 / 3
Russian novels	1	3 / 0	1 / 1

The straightforward summary is the arithmetic mean of the per-need values:

The problem is visible: macro-averaging gives each need the same weight $1/N$ in the sum, so a need with one relevant document contributes as much to the average as a need with fifteen. A single zero from the Russian-novels miss lowers the three-need average by a full $1/3$, which is a large negative impact relative to the system's overall performance across 21 relevant documents. This is intentional when every need matters equally, but it can misrepresent a system's aggregate capability.

Micro-averaging addresses this by pooling all outcome counts before computing the ratio. Instead of averaging per-need ratios, it sums the numerators and denominators across needs:

$$p_{\text{micro}} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FP_i)} \quad r_{\text{micro}} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FN_i)} \quad (2.2)$$

Every document counts equally. Needs with more relevant documents contribute more to the result.

Same systems, same needs, opposite conclusions on recall: macro says B is better, micro says A is better. The difference reflects a genuine choice about what matters.

Architecture and the Metrics

Retrieval Systems and Flows introduced retriever-only, retriever-with-filter, and retriever-ranker architectures. Precision and recall explain why each stage exists and how each is evaluated. The retriever is optimized for recall: it casts a wide net over the collection, accepting low precision because a false positive can still be removed later, while a false negative is lost for good. The ranker or filter is optimized for precision: it reorders or removes candidates so that the top of the list is dense with relevant material. Even a user who cares only about a precise top-ten result depends on the retriever's recall, because the ranker can only surface documents that the retriever found in the first place.

Precision, recall, and their averages describe a result set with no notion of order. The next section brings rank back into the picture: whether a system found the right documents matters less if they never appear near the top of the list.

Evaluating Ranked Results

Precision and recall treat a result set as a bag: a document is either in the set or not, and the metrics do not change if the order is shuffled. A ranked retrieval system does more than decide which documents to return; it decides which ones to show first. A user who reads from the top and stops after five items gets a very different experience depending on whether the relevant books are at ranks 1 and 2 or at ranks 16 and 18. Evaluation must account for position, not only membership.

Recall the first five positions from [Designing a Retrieval Benchmark](#):

Rank	System A result	Useful?	System B result	Useful?
1	Computer Organization and Design	yes	Database Systems: The Complete Book	yes
2	Operating System Concepts	yes	Pattern Recognition and Machine Learning	yes
3	The Handmaid's Tale	no	Introduction to Algorithms	yes
4	Narrative of the Life of Frederick Douglass	no	On the Origin of Species	no
5	Towards a New Architecture	no	Introduction to the Theory of Computation	yes

System A retrieves 25 books total (12 of 15 relevant); System B retrieves 8 (6 of 15 relevant). The set-based precision numbers from [Precision and Recall](#) already showed that B is more precise overall, but ranking adds a second dimension: how quickly does the user reach useful material? System A places two relevant books at the top, then three misses; System B delivers four of five.

Precision at Rank k

The simplest rank-aware metric evaluates only the first k positions. Precision at rank k counts how many of the top- k documents are relevant, ignoring everything below.

$P@k$ answers one practical question: if the user reads only the first k items, what proportion is useful? It does not require knowing how many relevant documents exist in the collection, which makes it easy to compute and interpret. It also does not penalize a system for missing relevant documents below rank k : a system that places one relevant book at rank 1 and nothing else achieves $P@1 = 1.0$ regardless of how many relevant books it missed.

System B dominates at every cutoff. A student who reads the first five results gets four useful books from B but only two from A. $P@k$ captures this user experience directly.

Reciprocal Rank and MRR

Some information needs have a single correct answer, or the user is satisfied as soon as one relevant result appears. A library patron asking “Who wrote Dracula?” needs exactly one fact; seeing it at rank 1 is ideal, at rank 20 is frustrating. The reciprocal rank captures this notion.

$RR = 1$ when the first result is relevant; $RR = 0.5$ when the first relevant result is at rank 2; $RR \rightarrow 0$ as the relevant document is pushed further down. If no relevant document appears in the result list at all, $RR = 0$.

MRR is widely used in question answering and known-item search, where one good answer suffices. It ignores what happens after the first relevant document: a system that places five relevant books at ranks 1 through 5 scores the same $RR = 1$ as one that places a single relevant book at rank 1 and nothing else. For needs where multiple relevant documents matter, as in our CS-foundations example with 15 relevant books, MRR is the wrong tool.

The Precision-Recall Curve

$P@k$ evaluates one fixed cutoff. A more complete picture plots precision against recall at every rank where a relevant document appears. As the system returns more documents, recall increases (more relevant documents are found) and precision typically decreases (more non-relevant documents accumulate). Plotting precision against recall produces the precision-recall curve.

At each rank i in the result list, define:

$$P_i = \frac{|\{d_1, \dots, d_i\} \cap \text{Rel}|}{i} \quad R_i = \frac{|\{d_1, \dots, d_i\} \cap \text{Rel}|}{|\text{Rel}|} \quad (2.3)$$

A point (R_i, P_i) is recorded only at ranks where a relevant document is retrieved: those are the only ranks where recall changes. Between two relevant documents, recall stays constant while precision drops (the denominator i grows with no new relevant hit), so these in-between ranks do not add new curve points.

Reading the Curve

Figure 2.3 annotates the key regions of a precision-recall curve. The upper-left corner (high precision, low recall) is where a fact-checker operates: a short, clean result list with few mistakes. The lower-right corner (high recall, low precision) is where a patent lawyer operates: an exhaustive search that tolerates noise to avoid missing anything. The diagonal arrow labelled “system efficiency” points toward the ideal corner $(1, 1)$: the closer a curve pushes toward that point, the better the system balances both goals simultaneously.

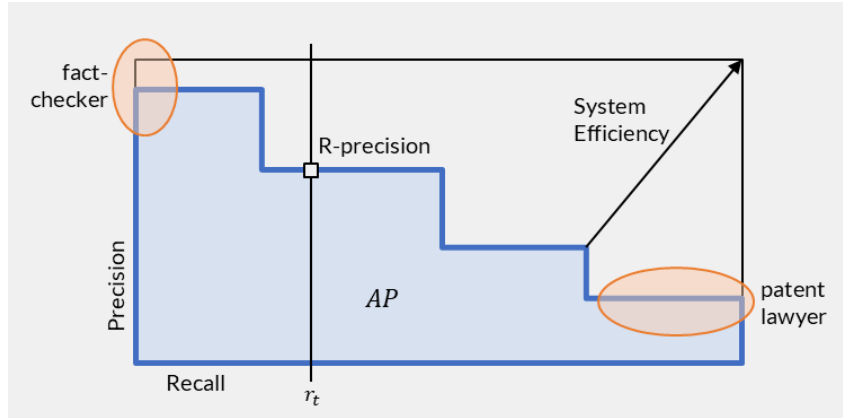


Figure 2.3: Precision-recall curve with key evaluation landmarks: the shaded area is Average Precision (AP), R-precision marks the point where the number of results equals the number of relevant documents, and the two user profiles illustrate the trade-off between precision-oriented and recall-oriented search.

The shaded area under the curve represents Average Precision (AP), which the next subsection defines formally. R-precision, marked on the curve at the recall level $r_t = R / |\text{Rel}|$ where R is the number of relevant documents, provides a single-point summary.

Interpolation for Cross-System Comparison

The 12 points in the table above form a smooth, largely decreasing scatter for one system and one need. The need for interpolation arises when we want to compare two systems or average curves across multiple needs: each system produces points at different recall levels, so we cannot directly subtract or average their precisions. Interpolation defines precision at any recall level r as the maximum precision achieved at any recall level $r' \geq r$:

$$P_{\text{interp}}(r) = \max_{r' \geq r} P(r') \quad (2.4)$$

This produces a non-increasing step function evaluated at standard recall levels $(0, 0.1, 0.2, \dots, 1.0)$. Two systems evaluated on the same 11 recall points can be compared point by point or averaged across queries. For System A, interpolated precision at recall 0 is 1.0 (the best precision it ever achieves), while at recall 0.9 and 1.0 it is 0 (the system never reaches those recall levels).

R-Precision

R-precision links precision to recall through a single number. If the collection contains R relevant documents for a need, R-precision is the precision after exactly R documents have been retrieved.

$$R\text{-Prec} = P@R = \frac{|\{D_1, \dots, D_R\} \cap \text{Rel}|}{R} \quad (2.5)$$

A perfect system would place all R relevant documents in the first R positions, achieving R-precision of 1.0. The measure adjusts itself to the difficulty of the need: a need with 15 relevant documents (our CS-foundations example) evaluates at cutoff 15, while a need with 2 relevant documents evaluates at cutoff 2.

R-precision is simple and self-adjusting, but it reduces an entire ranking to one point. The metrics that follow evaluate the full curve.

Average Precision

Average precision (AP) collapses the precision-recall curve into a single number. Geometrically, it approximates the area under the precision-recall curve: the shaded region in [Figure 2.3](#). A system whose curve stays high across all recall levels has a large area and a high AP; one whose precision collapses early has a small area and a low AP.

The computation is straightforward: at each rank where a relevant document appears, record the precision at that rank. Then take the mean of those values, using the total number of relevant documents in the collection as the divisor.

Dividing by $|\text{Rel}|$ rather than by the number of relevant documents actually found means that missed documents contribute implicit zeros: they never appear in the ranking, so they add nothing to the numerator, but the denominator still counts them. This is what makes AP sensitive to recall, not only to precision at the top.

AP uses the raw (non-interpolated) precision values. Interpolation serves a different purpose: it defines precision at arbitrary recall levels so that curves from different systems can be compared point by point or averaged. AP does not require that step because it sums only at the fixed rank positions where relevant documents actually occur.

AP rewards both placing relevant documents high (each precision term is larger when fewer non-relevant documents precede it) and finding more relevant documents (more terms contribute to the sum). This makes it the single best summary of a ranking for a given need: unlike $P@k$ it accounts for the full list, and unlike MRR it values every relevant document, not only the first.

Mean Average Precision

Evaluating a system on one need gives one AP value. A benchmark with multiple needs produces an AP per need, and mean average precision (MAP) averages them.

$$MAP = \frac{1}{|\mathbb{Q}|} \sum_{i=1}^{|\mathbb{Q}|} AP_i \quad (2.6)$$

MAP is a macro-average: each need contributes equally regardless of how many relevant documents it contains. This is the same averaging strategy discussed in [Precision and Recall](#), and the same caveats apply: a need with one relevant document has as much influence on MAP as a need with fifteen.

MAP is the standard primary metric in TREC and most retrieval benchmarks. A single MAP value summarizes an entire system's behaviour across all evaluated needs. When two systems are compared, a paired test (such as a paired t -test or Wilcoxon signed-rank test) on the per-need AP values determines whether the difference is statistically significant, not just the result of a few easy or hard queries.

Choosing a Metric

Each metric answers a different question about the ranking:

Metric	Question it answers	Sensitive to
$P@k$	How clean are the first k results?	Precision at a fixed cutoff
MRR	How quickly does the user find one good result?	Position of the first relevant document
R -Prec	How well does the system fill the first R slots?	Balance of precision and recall at a natural cutoff
AP	How good is the overall ranking for one need?	Both precision and recall, position-weighted
MAP	How good is the system across many needs?	Per-need AP, macro-averaged

$P@k$ and MRR suit settings where users inspect only the top few results: web search, question answering, voice assistants. AP and MAP suit settings where the full ranking matters: patent search, systematic reviews, benchmark comparisons. R-precision provides a single summary that adjusts to the number of relevant documents per need.

All of these metrics treat relevance as binary: a document is relevant or it is not. The next section relaxes this assumption, introducing graded relevance where some documents are more useful than others, and metrics that reward a system for placing highly relevant documents above marginally relevant ones.

Evaluating Graded Relevance

All metrics so far treat relevance as binary: a document is relevant or it is not. Average precision, for instance, gives System A's rank-1 result ("Computer Organization and Design") the same credit as System B's rank-1 result ("Database Systems: The Complete Book"), because both are relevant. Yet for a student seeking CS foundations, the two books are not equally useful. The grading from [Designing a Retrieval Benchmark](#) distinguishes core foundations (grade 3), important specializations (grade 2), and useful peripheral reading (grade 1). "Database Systems" is a grade-3 core text; "Computer Organization and Design" is grade-1 peripheral reading. A metric that sees only "relevant" cannot tell these apart.

Of the 15 relevant books in the collection, 3 are graded as core foundations, 4 as important specializations, and 8 as useful peripheral reading. The complete grade distribution:

Grade	Meaning	Count	Examples
3	Core foundations	3	Database Systems, Pattern Recognition, Data Science from Scratch
2	Important specialization	4	Operating Systems, AI: A Modern Approach, Programming Languages, Theory of Computation
1	Useful peripheral reading	8	Algorithms, Computer Networks, SICP, Cryptography, ...
0	Not relevant	35	Fiction, poetry, drama, general non-fiction

Consider the first five results from each system, now labelled with grades:

Rank	System A	Grade	System B	Grade
1	Computer Organization and Design	1	Database Systems: The Complete Book	3
2	Operating System Concepts	2	Pattern Recognition and Machine Learning	3
3	The Handmaid's Tale	0	Introduction to Algorithms	1
4	Narrative of the Life of Frederick Douglass	0	On the Origin of Species	0
5	Towards a New Architecture	0	Introduction to the Theory of Computation	2

Binary AP cannot distinguish these two top-5 lists beyond counting "2 relevant at top for A, 4 for B". Graded evaluation does more: it rewards B for placing its two highest-value books (both grade 3) in the most visible positions, while A placed a peripheral text (grade 1) at rank 1 and pushed its grade-3 titles to ranks 6, 12, and 18.

A second limitation of AP is subtler. AP rewards a system for placing any relevant document higher, but it does not penalize a system that places relevant documents in a sub-optimal order among themselves. If two relevant documents appear at ranks 3 and 5, AP does not care which of the two is more important. A graded metric can: it assigns more credit when a highly relevant document sits above a marginally relevant one.

Cumulative Gain

The simplest graded metric just sums the relevance grades of the documents returned up to rank k :

CG_k measures how much total value the user has gathered after reading k results. It is easy to compute but ignores order entirely: swapping a grade-3 book from rank 10 to rank 1 does not change CG_{10} .

Discounted Cumulative Gain

To make position matter, we discount contributions from lower ranks. The idea is that a user reading from the top gets diminishing benefit from each additional position: finding a grade-3 book at rank 1 is more valuable than finding the same book at rank 10, because by rank 10 the user may have already stopped reading, or the earlier results have already partially satisfied the need.

The logarithmic discount encodes a model of user patience: the benefit of each additional rank position decreases, but never reaches zero. A document at rank 10 still contributes, just less than the same document at rank 1.

Discounting with Binary Relevance

DCG works even when all relevance grades are binary (0 or 1). In that case, it reduces to summing the discount factors at positions where a relevant document appears. This isolates the effect of rank position from the effect of graded relevance, which helps clarify what each component contributes.

Normalized DCG

DCG values depend on the number of relevant documents that exist and on the magnitude of the grades, so raw DCG scores cannot be compared across different queries. A query with five grade-3 documents in the collection can produce a much higher DCG than one with two grade-1 documents, regardless of system quality.

Normalization divides the actual DCG by the best possible DCG for that query: the score achieved by an ideal ranking that places documents in descending order of their grades.

$nDCG$ always falls between 0 and 1. A value of 1 means the system produced the best possible ranking for that query at cutoff k . A value of 0 means no relevant document appeared in the first k positions.

What nDCG Reveals That AP Does Not

The contrast between AP and nDCG on our running example is instructive:

Metric	System A	System B	Winner
AP (binary, full list)	0.473	0.359	A
nDCG@10 (binary)	0.577	0.702	B
nDCG@10 (graded)	0.446	0.759	B

AP favours System A because AP rewards finding many relevant documents (A finds 12 vs. B's 6), and the denominator $|\text{Rel}| = 15$ heavily penalizes B's 9 misses. The nDCG metrics tell a different story because they evaluate only the top 10 positions: within that window, B places more relevant, higher-grade material near the top.

Switching from binary to graded nDCG widens B's lead further. Under binary evaluation, A's rank-1 book and B's rank-1 book both score 1.0 (both are "relevant"). Under graded evaluation, B's grade-3 book at rank 1 scores 3.0 while A's grade-1 book scores only 1.0. Graded relevance captures what a user experiences: not all useful results are equally useful.

Because the logarithmic discount shrinks toward zero, each additional position contributes less to the sum. Beyond a certain depth, nDCG effectively stops changing: a document at rank 50 contributes only $1/\log_2(51) \approx 0.18$ times its grade, while rank 1 contributes the full grade. For this reason, very large values of k add little discriminative power between systems and are rarely used in practice. Standard cutoffs align with how far users actually read: $k = 10$ for a web search results page (the BEIR and MS MARCO benchmarks report nDCG@10), $k = 20$ for TREC Web Track, and $k = 3$ for voice assistants or mobile interfaces.

Practical Notes

nDCG is the standard metric in web search evaluation (where users rarely look past the first page) and in leaderboard-style benchmarks like MS MARCO and BEIR. The cutoff k is chosen to match the application. The grading scale affects what nDCG rewards. Our linear scale (0, 1, 2, 3) treats the jump from grade 0 to grade 1 as equivalent to the jump from grade 2 to grade 3. An exponential gain variant replaces rel_i with $2^{rel_i} - 1$, which amplifies the difference between high and low grades: a grade-3 document contributes $2^3 - 1 = 7$ while a grade-1 document contributes only $2^1 - 1 = 1$.

The exponential variant is the default in many evaluation toolkits and benchmark leaderboards. When comparing published nDCG scores, always check whether linear or exponential gain was used; the two are not comparable.

Evaluating Text Classifiers

Retrieval systems do not exist in isolation. A search engine for images relies on classifiers that detect faces, identify objects, or assign scene categories before any query arrives. An audio retrieval system uses a classifier to segment speech from music. A filtering stage in a retrieval pipeline decides “does this candidate pass the criteria?”: a binary classification decision embedded inside retrieval. These classifiers are components of the retrieval pipeline, and they need their own evaluation. The confusion matrix and its metrics appear every time a system makes a binary or multiclass decision, whether that decision is the final answer or a preprocessing step that feeds into ranking.

The underlying evaluation question is familiar: how often does the system make the right decision? Precision and recall still apply. What changes is the framing. There is no “retrieved set” to measure against a collection; instead, the system assigns exactly one label to each item in a test set, and we compare those predictions against ground-truth labels. Three properties make classification evaluation distinct from retrieval evaluation:

1. **No ranking.** The system outputs a label, not a position. There is no notion of “rank 3 is better than rank 10”.
2. **Prevalence matters.** The proportion of positive items in the test set affects how we interpret every metric. A rare disease affects 1% of patients; a common spam folder holds 40% junk.
3. **Error costs are often asymmetric.** A false positive in spam filtering deletes a legitimate email; a false negative lets spam through. These are not equally bad.

The Confusion Matrix

The confusion matrix organizes every prediction the system made into a 2x2 table for binary classification. Each item in the test set falls into exactly one cell, depending on the system’s prediction and the true label.

		Actual Condition (as observed)	
		Positive (P)	Negative (N)
Predicted Condition (as computed)	True (T)	True Positive (TP)	False Positive (FP)
	False (F)	False Negative (FN)	True Negative (TN)

Figure 2.4: The binary confusion matrix. Rows represent predictions, columns represent actual conditions. Green cells are correct predictions; orange cells are errors.

The row sums give the number of items the system labelled positive ($T = TP + FP$) and negative ($F = FN + TN$). The column sums give the actual counts: $P = TP + FN$ positive items and $N = FP + TN$ negative items in the test set. The total population is $P + N$.

From these four counts, we derive the same metrics introduced for retrieval, plus several that become important only in classification:

Precision and recall are identical to the retrieval definitions from [Precision and Recall](#), just viewed from a different angle: “retrieved” becomes “predicted positive”, and “relevant” becomes “actually positive”. Specificity is new; in retrieval we called its complement “fallout” ($FPR = 1 - \text{Specificity}$). Accuracy had no retrieval equivalent because retrieval operates on a ranked list, not a fixed yes/no decision for every item.

		Actual Condition (as observed)			
		Population	Positive (P)	Negative (N)	
Predicted Condition (as computed)	True (T)	True Positive (TP)	False Positive (FP)	Positive Predictive Value (PPV), Precision	False Discovery Rate (FDR)
	False (F)	False Negative (FN)	True Negative (TN)	False Omission Rate (FOR)	Negative Predictive Value (NPV)
		True Positive Rate (TPR), Sensitivity, Recall, Hit Rate	False Positive Rate (FPR), Fall-Out	Accuracy (ACC)	
		False Negative Rate (FNR), Miss Rate	True Negative Rate (TNR), Specificity		Error Rate (ERR), Misclassification Rate

$TPR = \frac{TP}{P}$	$FPR = \frac{FP}{N}$	$PPV = \frac{TP}{T}$	$FDR = \frac{FP}{T}$	$PPV + FDR = 1$
$FNR = \frac{FN}{P}$	$TNR = \frac{TN}{N}$	$FOR = \frac{FN}{F}$	$NPV = \frac{TN}{F}$	$FOR + NPV = 1$
$TPR + FNR = 1$	$FPR + TNR = 1$			

$ACC = \frac{TP + TN}{P + N}$	$ERR = \frac{FP + FN}{P + N}$	$ACC + ERR = 1$
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Figure 2.5: The confusion matrix and its derived metrics. Column-normalized rates (TPR, FPR, FNR, TNR), row-normalized predictive values (PPV, NPV), and overall accuracy, with all complementary pairs summing to one.

Example: Spam Filtering

A university email server processes 1000 messages in one day. Of these, 400 are spam ($P = 400$) and 600 are legitimate ($N = 600$). The spam filter makes the following decisions:

	Actual Spam (400)	Actual Legitimate (600)
Predicted Spam	TP = 360	FP = 30
Predicted Legitimate	FN = 40	TN = 570

All four numbers look healthy: the filter catches 90% of spam, and 92% of what it flags is genuinely spam. But consider the 30 false positives: those are 30 legitimate emails moved to the junk folder. If one of them is an acceptance letter for a grant proposal, the cost of that single FP dwarfs the annoyance of 40 spam messages reaching the inbox.

When Accuracy Misleads

Accuracy counts all correct decisions equally. When the two classes are roughly balanced (as in the spam example with 400 vs. 600), accuracy gives a fair summary. When one class dominates, accuracy becomes misleading.

		Actual Condition (as observed)		
		Population (2030)	Positive (P = 30)	Negative (N = 2000)
Predicted Condition (as computed)	True (200)	True Positive (TP = 20)	False Positive (FP = 180)	$PPV = \frac{20}{200} = 10\%$
	False (1830)	False Negative (FN = 10)	True Negative (TN = 1820)	$NPV = \frac{1820}{1830} = 99\%$
		$TPR = \frac{20}{30} = 67\%$	$TNR = \frac{1820}{2000} = 91\%$	$ACC = \frac{1840}{2030} = 91\%$

Figure 2.6: Confusion matrix for an imbalanced dataset ($P = 30$, $N = 2000$). High accuracy (91%) masks low precision (10%) and moderate recall (67%).

Consider a screening test for a rare disease. In a population of 2030 people, 30 have the disease ($P = 30$) and 2000 do not ($N = 2000$). Figure 2.6 shows the test outcomes: $TP = 20$, $FP = 180$, $FN = 10$, $TN = 1820$.

The problem is prevalence: only $30/2030 = 1.5\%$ of the population is positive. With so few positives, even a small false-positive rate ($FPR = 180/2000 = 9\%$) produces many more false positives than true positives in absolute terms. Accuracy is dominated by the 2000 true negatives and hides the fact that the test is nearly useless for confirming the disease.

Asymmetric Error Costs

The spam and screening examples already show that FP and FN carry different costs. A third scenario makes this even starker: biometric face unlock on a smartphone.

The system decides whether the face in front of the camera matches the phone's owner. Two errors are possible:

- **False Positive:** an impostor's face is accepted. The phone unlocks for a stranger. This is a security breach.
- **False Negative:** the owner's face is rejected. The owner must retry or use a PIN. This is a minor inconvenience.

The costs are radically different: a single FP compromises all data on the device, while a FN costs five seconds. The system must therefore achieve an extremely low false-positive rate (FPR), even if this means a noticeable false-negative rate (FNR).

The choice of which error to minimize is not a mathematical question; it is a design decision that precedes evaluation. The metrics merely reveal whether the system meets the requirement. In the next section, we explore how moving a decision threshold trades FPR against FNR continuously, and how ROC curves visualize this trade-off.

Multiclass Classification

Binary classification assigns one of two labels. Many tasks assign one of K labels: an image classifier distinguishes "indoor", "outdoor", and "portrait"; an AI assistant routes user queries to the correct handler. The confusion matrix generalizes to a $K \times K$ table where rows are predicted classes and columns are actual classes.

Per-Class Metrics via One-vs-Rest

To compute precision, recall, and specificity for one class, collapse the $K \times K$ matrix into a binary view: "belongs to class C " vs. "does not belong to class C ". Figure 2.7 shows this collapse for two classes from the intent-routing example.

		Actual Class		
		Clarify ($P=20$)	Not Clarify ($N=180$)	
Predicted Class	Queries (200)			
	Clarify (12)	True Positive ($TP=5$)	False Positive ($FP=7$)	$PPV = \frac{5}{12} = 42\%$ precision
	Not Clarify (188)	False Negative ($FN=15$)	True Negative ($TN=173$)	$NPV = \frac{173}{188} = 92\%$
		$TPR = \frac{5}{20} = 25\%$ recall	$TNR = \frac{173}{180} = 96\%$ specificity	$ACC = \frac{188}{200} = 94\%$ accuracy
		Actual Class		
		KB ($P=80$)	Not KB ($N=120$)	
Predicted Class	Queries (200)			
	KB (92)	True Positive ($TP=70$)	False Positive ($FP=22$)	$PPV = \frac{70}{92} = 76\%$ precision
	Not KB (108)	False Negative ($FN=10$)	True Negative ($TN=98$)	$NPV = \frac{98}{108} = 91\%$
		$TPR = \frac{70}{80} = 88\%$ sensitivity	$TNR = \frac{98}{120} = 82\%$ specificity	$ACC = \frac{168}{200} = 84\%$ accuracy

Figure 2.7: One-vs-rest collapse of the intent-routing confusion matrix for the `clarify` class (top) and the `knowledge_base` class (bottom), with derived metrics.

For the `clarify` class (the rare intent with 20 actual queries), the system correctly identifies only one in four queries that actually need clarification. Most of the rest (12 out of 15 missed) get misrouted to `knowledge_base`, where the system attempts to answer a question it does not understand. This is the most dangerous failure mode: the user gets a confident but wrong answer instead of a helpful follow-up question.

For the dominant `knowledge_base` class (80 of 200 queries), the system routes most KB queries correctly (high recall) but also absorbs queries that belong elsewhere (22 false positives), diluting precision. The high recall on the dominant class is what inflates the accuracy of the 4x4 confusion matrix to 76.5%.

Aggregating Across Classes

With K classes, we have K precision values and K recall values. Summarizing them into one number uses the same macro/micro distinction from [Precision and Recall](#):

Macro-averaging computes the metric per class, then takes the arithmetic mean:

$$\text{Precision}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \text{Precision}_k \quad (2.7)$$

Each class contributes equally, regardless of size. For our intent router:

Class	Precision	Recall
<code>knowledge_base</code>	0.761	0.875
<code>web_search</code>	0.800	0.800
<code>calculator</code>	0.833	0.750
<code>clarify</code>	0.417	0.250

$$\text{Precision}_{\text{macro}} = \frac{0.761 + 0.800 + 0.833 + 0.417}{4} = 0.703 \quad (2.8)$$

$$\text{Recall}_{\text{macro}} = \frac{0.875 + 0.800 + 0.750 + 0.250}{4} = 0.669 \quad (2.9)$$

Micro-averaging pools TP, FP, and FN across all classes before computing the ratio:

$$\text{Precision}_{\text{micro}} = \frac{\sum_k TP_k}{\sum_k (TP_k + FP_k)} = \frac{70 + 48 + 30 + 5}{92 + 60 + 36 + 12} = \frac{153}{200} = 0.765 \quad (2.10)$$

$$\text{Recall}_{\text{micro}} = \frac{\sum_k TP_k}{\sum_k (TP_k + FN_k)} = \frac{70 + 48 + 30 + 5}{80 + 60 + 40 + 20} = \frac{153}{200} = 0.765 \quad (2.11)$$

Both denominators equal the total number of queries (200): the precision denominator sums all row totals, the recall denominator sums all column totals, and in a single-label setting these are the same. The numerator is always the sum of the diagonal (correct predictions). So micro-precision, micro-recall, and accuracy all collapse to the same value: diagonal sum divided by total.

This explains why accuracy is the dominant single-number metric in classification: it is the micro-averaged precision and recall. In retrieval, micro-averaging played a secondary role because the macro view (one score per query, then average) better matched evaluation practice. In classification, the micro view is the natural default, and accuracy is its name.

Macro-averaging tells the opposite story. It gives each class equal weight: mean precision drops to 70.3% and mean recall to 66.9%, pulled down by `clarify`'s dismal 25% recall. If every intent matters equally (and it should, because a misrouted `clarify` query produces a wrong answer), macro-averaging is the right summary.

All metrics in this section assume a fixed decision: the system outputs a label and we count outcomes. But most classifiers internally produce a continuous score (a probability, a distance, a similarity), and the label emerges only after applying a threshold. Moving that threshold changes the balance between FP and FN.

The next section explores this continuous perspective through score distributions, threshold selection, and ROC curves.

Scores, Thresholds, and ROC Curves

The previous section evaluated classifiers after the decision was already made: the system assigned a label, and we counted outcomes. But most classifiers do not jump directly to a label. They produce a continuous score, a confidence value between 0 and 1, and the label emerges only after comparing that score to a threshold. The spam filter assigns each email a spam probability: 0.92 for obvious junk, 0.03 for a message from a colleague, 0.47 for a newsletter that could go either way. At threshold 0.5, the newsletter stays in the inbox; at threshold 0.4, it moves to junk.

Different thresholds produce different confusion matrices, different precision/recall values, and different TPR/FPR trade-offs. A single classifier is not one point in metric space; it is a family of points, one for each threshold. This section explores how to visualize that family and choose a single operating point.

Score Distributions and the Threshold

A binary classifier assigns a score $s(x) \in [0, 1]$ to each item x . Items from the positive class tend to receive higher scores; items from the negative class tend to receive lower scores. If the two score distributions were perfectly separated (all positives above some value, all negatives below), any threshold between them would classify perfectly. In practice, the distributions overlap.

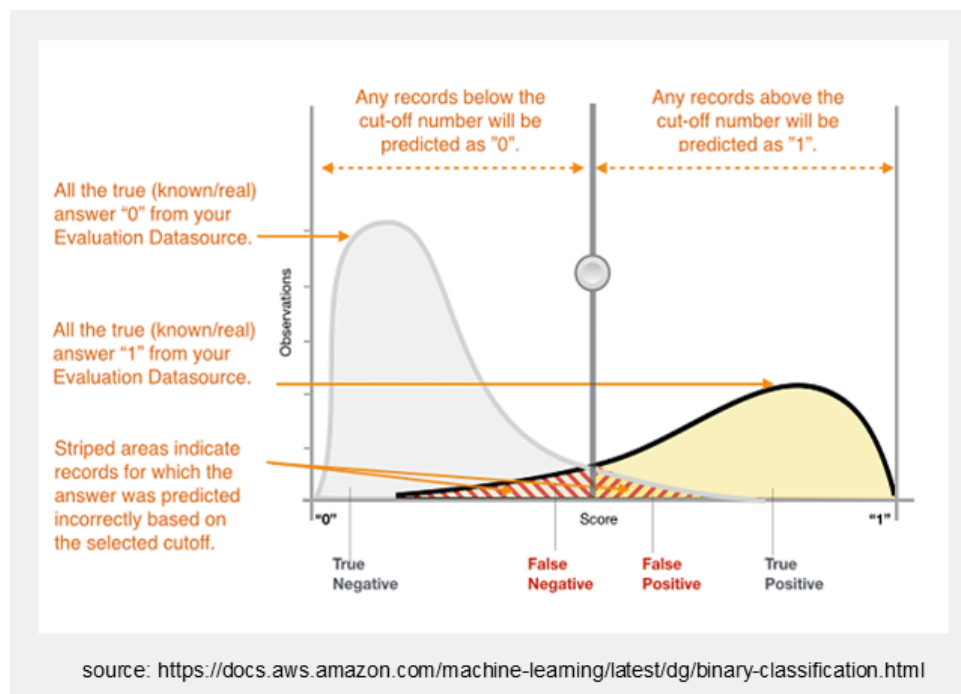


Figure 2.8: Score distributions for the negative class (grey) and positive class (yellow) with a decision threshold. The overlap region produces classification errors regardless of where the threshold is placed.

The threshold T partitions the score axis into two regions: items with $s(x) \geq T$ are predicted positive, items with $s(x) < T$ are predicted negative. Moving the threshold changes the balance between true positives, false positives, true negatives, and false negatives:

- **Shifting T to the left** (lower threshold): more items cross into the “predicted positive” region. TPR increases (fewer false negatives), but FPR also increases (more false positives). The system becomes more sensitive but less specific.
- **Shifting T to the right** (higher threshold): fewer items are predicted positive. FPR decreases (fewer false positives), but TPR also decreases (more false negatives). The system becomes more specific but less sensitive.

This is the fundamental trade-off underlying every threshold-based classifier. No single threshold is universally best; the right choice depends on the error costs of the application.

Formalizing the Threshold as Areas Under the Distributions

Let $f_n(x)$ denote the score distribution of the negative class and $f_p(x)$ the score distribution of the positive class. Each classification rate corresponds to an area under one of these curves, partitioned by the threshold T :

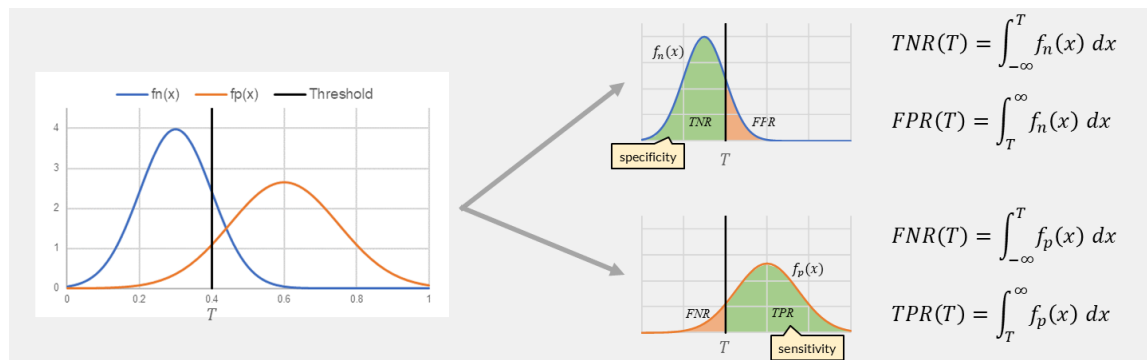


Figure 2.9: Decomposition of the score distributions into the four classification rates. TNR and FPR partition the negative-class curve $f_n(x)$; FNR and TPR partition the positive-class curve $f_p(x)$.

The integrals make the trade-off precise. Shifting T to the left expands the integration range for TPR (the area under f_p to the right of T grows), but simultaneously expands the integration range for FPR (the area under f_n to the right of T also grows). No threshold eliminates the overlap region; it can only choose how to distribute the overlap errors between false positives and false negatives.

The ROC Curve

Instead of evaluating one threshold, the ROC (Receiver Operating Characteristic) curve evaluates all of them simultaneously. It plots $TPR(T)$ on the vertical axis against $FPR(T)$ on the horizontal axis, sweeping T from high to low. The result is a curve from $(0, 0)$ to $(1, 1)$ that shows the full trade-off landscape.

Figure 2.10 shows 20 emails (10 spam, 10 legitimate) sorted by descending spam score. At each score, we set the threshold there and count cumulative TP, FP, FN, TN. The right panel plots the resulting ROC curve.

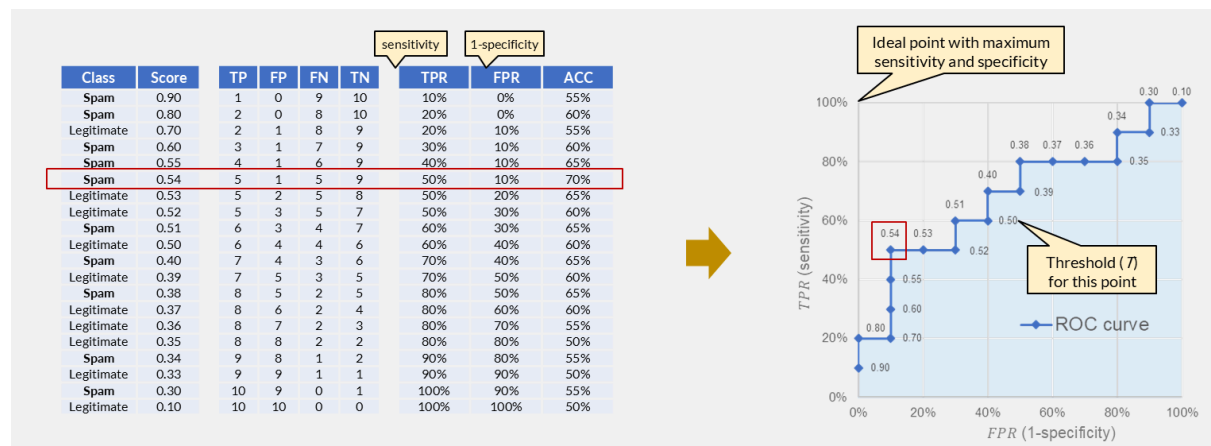


Figure 2.10: ROC curve construction from a ranked list of 20 emails. Each row sets the threshold at that score; the curve traces TPR vs. FPR as the threshold sweeps from high to low. The highlighted point ($T = 0.54$) has the highest accuracy but is not necessarily the best operating point.

Key Landmarks

Point	Meaning
(0, 0)	Threshold so high that nothing is predicted positive (always predict negative)
(1, 1)	Threshold so low that everything is predicted positive (always predict positive)
(0, 1)	Perfect classifier: all positives detected, no false positives

Point	Meaning
Diagonal	Random classifier: $TPR = FPR$ at every threshold (no discrimination)

A curve that hugs the upper-left corner represents a strong classifier: it achieves high TPR before FPR rises appreciably. A curve that follows the diagonal adds no information beyond random guessing.

Reading the ROC Space

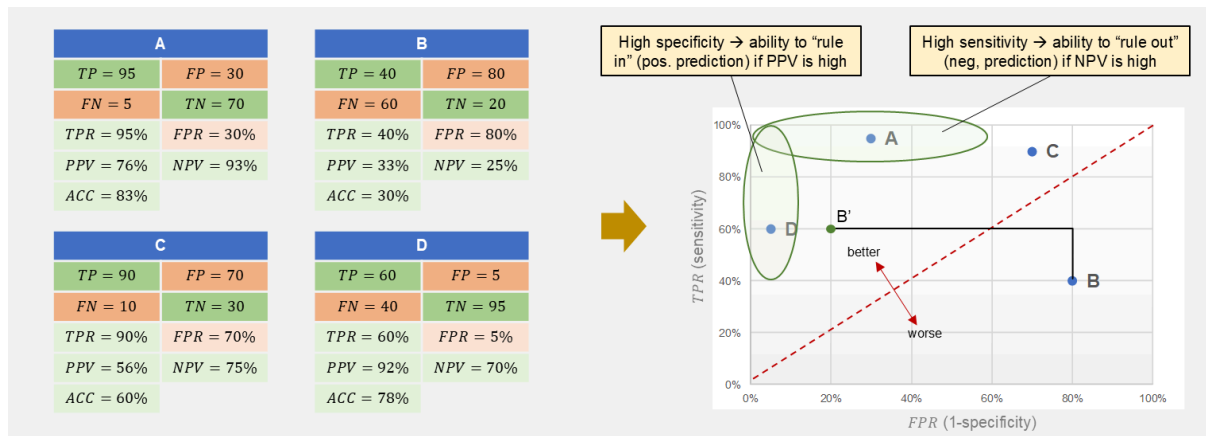


Figure 2.11: Four classifiers (A-D) mapped from their confusion matrices to ROC space. The upper-left region represents high sensitivity with low false-positive rate; the diagonal represents random performance.

Figure 2.11 places four classifiers at different operating points:

- **Classifier A** (TPR = 95%, FPR = 30%): high sensitivity, moderate false-positive rate. If a negative prediction from A is almost certainly correct (high NPV), A can “rule out” the condition: a negative result is trustworthy.
- **Classifier B** (TPR = 40%, FPR = 80%): below the diagonal. This classifier is worse than random. Flipping its predictions gives B' (TPR = 60%, FPR = 20%), which is above the diagonal and usable.
- **Classifier C** (TPR = 90%, FPR = 70%): high sensitivity but very high false-positive rate. It catches most positives but at enormous cost in false alarms.
- **Classifier D** (TPR = 60%, FPR = 5%): high specificity, moderate sensitivity. If a positive prediction from D is almost certainly correct (high PPV), D can “rule in” the condition: a positive result is trustworthy.

The biometric face-unlock from [Evaluating Text Classifiers](#) would appear as a point near the left edge of ROC space: $FPR = 0.2\%$, $TPR = 98\%$. The requirement “FPR below 0.5%” constrains the operating point to a narrow vertical strip on the left side of the plot.

AUC: Area Under the ROC Curve

The ROC curve shows performance across all thresholds. Summarizing it into a single number gives the Area Under the Curve (AUC).

AUC	Interpretation
1.0	Perfect separation: every positive scores higher than every negative
0.5	Random: scores carry no discriminative information
< 0.5	Worse than random (flip predictions to improve)

AUC is useful for comparing classifiers without committing to a threshold: a system with higher AUC has better score separation overall. However, AUC does not say which operating point the system will use in production, and two classifiers with the same AUC can have very different curves (one may be better at low FPR, the other at high TPR).

Choosing an Operating Point

The ROC curve shows what is possible. Selecting one point on it requires a decision criterion from outside the mathematics:

Maximize accuracy. Compute $(TP + TN)/(P + N)$ at each threshold and pick the maximum. This works when classes are balanced and error costs are symmetric. In the spam example from Figure 2.10, threshold 0.54 gives $TP = 5$, $TN = 9$, $\text{accuracy} = 14/20 = 70\%$, the highest value in the table. But as we discussed, the spam filter should rather optimize for a low FPR while maintaining acceptable TPR, not simply maximize accuracy.

Constrained optimization. Fix one rate and optimize the other. The face-unlock requirement “ $FPR \leq 0.2\%$ ” defines a vertical line on the ROC plot; the operating point is the highest TPR value at or to the left of that line. This is the standard approach when one error type has a hard cost limit.

Youden’s J-statistic. Maximize $J = TPR - FPR$, which is the vertical distance from the diagonal (random classifier). The point with the highest J is geometrically closest to the upper-left corner and balances sensitivity against specificity without committing to a cost model.

Equal error rate (EER). Find the threshold where $FPR = FNR$ (equivalently, $FPR = 1 - TPR$). This is the point where the ROC curve crosses the anti-diagonal from $(0, 1)$ to $(1, 0)$. EER is commonly reported in biometric and speaker-verification systems as a single-number summary that does not require choosing which error is worse.

In all cases, threshold selection should be performed on a validation set separate from the test set used to report final metrics. Optimizing the threshold on the test set inflates reported performance.

When Evaluation Misleads

Two results arrive on the same afternoon. A retrieval group reports that their new ranker raises nDCG@10 on a public benchmark from 0.41 to 0.43. A colleague reports that their spam filter reaches 99.2% accuracy on a held-out test set. Both numbers were computed correctly with the measures developed in the previous sections. Both might still be worthless.

Now that the measures are in hand, we can look at what they do not tell us. Every number in this chapter rests on a chain of decisions: which documents were collected, which needs were written down, who judged them, how the labels were split, and which measure was reported. A weakness anywhere in that chain reaches the final score, and the score itself never reveals it. Reading evaluation results critically means knowing where the chain tends to break.

Incomplete Ground Truth

Recall the pooling arrangement from [Designing a Retrieval Benchmark](#): at realistic scale, only the documents that participating systems actually retrieved get judged, and everything unjudged counts as not relevant. That convention is what makes large benchmarks affordable, and it is also the first place scores go wrong.

In the **dense case** (TREC-style pooling), the benchmark assessed all documents that participating systems submitted in a given year. The judgments are thorough for those systems. But when a new algorithm is evaluated against the same reused collection, it may retrieve relevant documents that no original participant found. Those documents were never assessed and count as non-relevant. The new system is penalized for being original: it found material the pool missed, and the benchmark cannot reward it. A careless evaluator might even conclude the new system is worse, when in fact it discovered answers nobody had seen before.

In the **sparse case** (MS MARCO-style benchmarks with millions of queries), each query has only one or a few assessed relevant documents. [Figure 2.12](#) shows this situation. Any document that equally qualifies as a correct answer but was never selected for assessment penalizes rather than rewards the system that retrieves it. The bias is systematic: retrieval systems trained on such benchmarks tend to replicate what was selected for assessment rather than what is relevant in a broader sense.

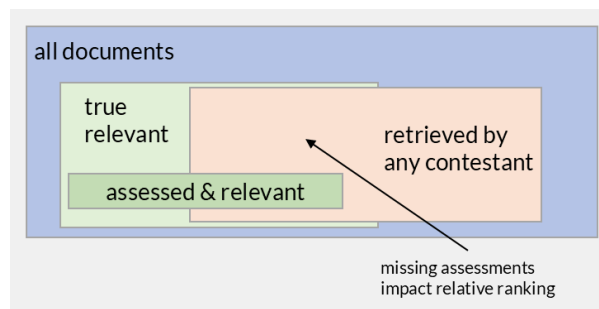


Figure 2.12: Sparse assessment: only a fraction of genuinely relevant documents were ever judged. Systems retrieving unjudged relevant material are penalized.

Recall suffers most, and in a direction that depends on who ran the experiment. Precision at rank 10 needs only the top ten documents judged, which pooling usually delivers. Recall needs the total amount of relevant material in the collection, and that is exactly what incomplete judging never establishes. For the systems that built the pool, recall comes out too high, because the divisor counts only the relevant documents somebody found. For a system evaluated later against the reused collection, the error can run the other way, since its own relevant results may be unjudged and scored as misses.

Noisy Ground Truth

Even judged labels are opinions. Assessors work from written guidelines and still disagree on borderline cases, so a benchmark encodes one panel's reading of each need. This is tolerable for comparison, since every system is scored against the same labels, but it puts a floor under how small a difference we can meaningfully interpret. A gap of one point in a measure whose ground truth two qualified assessors would not fully reproduce is not a result.

Classification test sets have the same problem in a different shape. Their labels are usually complete, so pooling bias does not apply, but completeness is not correctness. Whoever labelled the messages had to decide whether an aggressive marketing newsletter counts as spam, and different labellers draw that line differently. A classifier that learned one labeller's line will look better than one that learned another's, independently of which is more useful.

Rare classes make this worse because small counts are unstable. In an intent router with four routes, a test set may contain hundreds of knowledge-base queries and a dozen calculator queries. Per-class recall for the calculator route then moves by eight percentage points when a single example changes label, and a macro-averaged score built from such classes inherits that noise. Before trusting a per-class number, check how many test items it was computed from.

Optimizing Against the Test

A benchmark that stays fixed for years stops being a test and becomes a target. Recall how a TREC-style evaluation runs: participants receive the collection and the topics, run their systems on their own machines, and submit ranked lists that the organizers judge afterwards. Holding the topics is what makes the risk possible. A research group can read the topics, notice that several hinge on ambiguous words, add a hand-built list of expansion terms, adjust the stop word list, add exceptions to the stemmer, and keep every change that raises the score. Each step on its own looks like sound engineering. The result is a system that has absorbed the properties of this topic set and this collection: which words happen to be ambiguous here, and which documents happen to be relevant. The score rises and the method does not transfer. No dishonesty is required, because the selection pressure alone is enough.

Issuing a new topic set for each cycle limits the damage while a benchmark is live, and TREC does this. The cost is comparability, since results measured on this year's topics cannot be placed directly against last year's. The concentrated risk appears after a cycle closes, when the collection, its topics, and its judgments become a reusable test collection that papers measure against for years. Armstrong, Moffat, Webber, and Zobel examined a decade of published ad-hoc retrieval results obtained that way in **Improvements That Don't Add Up**. Dozens of papers reported gains and many claimed statistical significance, yet the improvements did not accumulate into visible progress, and the baselines were often weaker than the median system from the original TREC cycle. Reported improvement and real improvement had come apart.

Classification has a sharper and more mechanical version of the same failure. A decision threshold tuned on the test set produces a score that cannot be trusted, because the threshold was chosen with knowledge of the answers. This is why the validation split exists, and why the threshold selection in **Scores, Thresholds, and ROC Curves** must happen on validation data. The extreme case is contamination: if the test items and their labels were part of a model's training data, the model can score well by recalling an answer rather than by solving the task. Large language models make this problem acute, because they are trained on web-scale corpora that include publicly available exam questions. When GPT-4 is evaluated on the SAT or the MMLU benchmark, the questions and their correct answers were likely present somewhere in the training data. Researchers have shown that GPT-4 can guess missing answer options in MMLU questions with a 57% exact-match rate, far above chance, which strongly suggests memorization rather than reasoning (**Investigating Data Contamination in Modern Benchmarks for Large Language Models**, 2023). A high score on a contaminated benchmark tells us the model saw the answers before, not that it can generalize to unseen problems.

Withholding the labels is a partial defence. MS MARCO keeps the labels of its evaluation set private and scores submissions centrally, so participants cannot tune against them directly. The leaderboard's own designers describe the limit in **Fostering Coopetition While Plugging Leaks**: every submission returns a score, and every score leaks a little information about the hidden set. Enough attempts turn a blind test into a slowly revealed one, which is why the official metric carries a deliberate artifact meant to frustrate reverse-engineering.

A genuinely blind evaluation reverses the direction in which data travels. Instead of sending the topics to the participant, the participant sends the system to the evaluator: code or a container executed against topics it has never seen, with judgments made only after all results are collected. This is known as evaluation

as a service, or software submission. **TIRA** implements it by keeping the test data and ground truth out of public reach and running each submitted system in a sandbox that prevents data leaks. The protocol costs the organizers considerably more effort, which is why it remains the exception. One caveat survives even then: a blind topic set shows that a system never saw these needs, but for a model pretrained on web text it cannot show that the documents themselves were absent from training.

The Test Set Is Not Production

Benchmarks are frozen so that results stay reproducible, and freezing is what makes them drift away from the world they were drawn from. Vocabulary shifts, topics appear that the collection predates, document formats change. The stability that makes a benchmark useful for comparison steadily reduces how much it says about current traffic.

For classification the mismatch is measurable, and prevalence is the clearest case. Suppose a spam filter detects 90% of spam and correctly passes 98% of legitimate mail. On a balanced test set of 500 spam and 500 legitimate messages, it produces 450 true positives and 10 false positives, so precision is $450/460 \approx 97.8\%$. Deploy the same filter, unchanged, on a mailbox where only 5% of messages are spam. Out of 500 spam and 9,500 legitimate messages it still catches 450 and still misclassifies 2% of legitimate mail, but 2% of 9,500 is 190 false positives, so precision falls to $450/640 \approx 70.3\%$.

Nothing about the classifier changed. Recall and specificity are identical in both settings. Only the class balance moved, and precision, the measure the user actually experiences, dropped by 27 percentage points. A test set whose prevalence does not match deployment therefore reports a precision that deployment will not reproduce.

One Number Is Not a Result

A single score on a single collection is weak evidence, for a reason that has nothing to do with retrieval. Per-need scores vary enormously: the same system can reach average precision above 0.8 on one need and near zero on another. An average over 50 needs conceals that spread, so two systems differing by a point in the mean may be indistinguishable once the variation between needs is taken into account. Reporting the spread, and testing whether an observed difference exceeds it, matters as much as the measure itself.

The problem compounds across a community. When hundreds of groups tune against one public benchmark, the leading entry is partly the winner of a large search over method variants, and some of its margin is the luck of that search rather than a property of the method. This is the same multiple-comparisons effect that makes a single significance test misleading when many hypotheses were tried, and it is one mechanism behind the missing accumulation that Armstrong and colleagues documented.

Effectiveness Is Not the Only Axis

Finally, relevance is one dimension of quality among several. A production system must also answer quickly, sustain many concurrent requests, and stay affordable, and these goals pull against each other. Later chapters on vector search make the trade-off explicit: approximate nearest-neighbour methods deliberately accept slightly worse results in exchange for large gains in speed and cost, and that exchange is often the right decision. A benchmark reporting only effectiveness cannot express it, which is why operational targets such as latency, throughput, and cost per query belong in the performance goals next to the retrieval measures.

None of this makes evaluation futile. It makes evaluation a claim with a scope. The following questions establish that scope for any reported score:

- How complete are the judgments, and could a better system have been penalized for retrieving unjudged documents?
- Who produced the labels, and how much would a second panel agree?
- Did the authors see the test data before reporting, and was any threshold or hyperparameter chosen using it?
- How closely do the collection and the class balance match the setting where the system will run?

- How large is the difference relative to the variation across needs, and how many alternatives were tried before this one was reported?
- What does the score cost in latency and compute?

A result that survives these questions is worth acting on.

Summary

Metric Lookup Table

Metric	Formula	What it rewards	When to use
Precision	$\frac{TP}{TP+FP}$	Clean results (few false positives)	Any binary decision; especially when FP is costly
Recall	$\frac{TP}{TP+FN}$	Completeness (few misses)	Exhaustive search; patent, legal, medical
F_β	$\frac{(\beta^2+1) \cdot P \cdot R}{\beta^2 \cdot P + R}$	Balances P and R with tuneable weight	Single-number summary; $\beta = 1$ default
$P@k$	Precision in top k	Clean top-of-list	Web search, short result pages
MRR	Mean of $1/\text{rank}_{\text{first relevant}}$	First relevant result high	QA, known-item search
R-Precision	$P@R$	Balance at a natural cutoff	Self-adjusting single-point summary
AP	Mean of P_k at relevant positions, divided by $ \text{Rel} $	Both high precision and high recall, position-weighted	Full ranking evaluation per query
MAP	Mean of per-query AP	System-level ranking quality	Benchmark comparison (TREC standard)
$nDCG@k$	$DCG_k / IDCG_k$	High-grade documents placed early	Web search, graded relevance, user experience
Accuracy	$\frac{TP+TN}{P+N}$	Overall correct decisions	Balanced classification; equals micro-P/R
Specificity (TNR)	$\frac{TN}{TN+FP}$	Few false alarms	Medical tests, biometric security
AUC	Area under ROC curve	Score separation across all thresholds	Comparing classifiers without committing to a threshold

Key Takeaways

1. Evaluation requires a benchmark: a collection, information needs, relevance judgments, and a performance goal. Without any component, a number is not interpretable.
2. Precision and recall are in tension. Improving one typically costs the other, and the F_β -measure makes the trade-off explicit.
3. Ranked evaluation adds position: AP and nDCG reward systems that place relevant (or high-grade) material where users will see it.
4. Graded relevance distinguishes “somewhat useful” from “essential”. nDCG captures what binary metrics cannot: the quality of the relevant documents, not only their presence.
5. Classification evaluation reuses precision and recall but adds accuracy, specificity, and the confusion matrix. Accuracy equals micro-precision equals micro-recall in single-label settings.
6. Prevalence distorts accuracy: when one class dominates, a trivial baseline can score higher than a real classifier. Always check class balance before trusting accuracy.
7. A continuous score becomes a binary decision only after choosing a threshold. The ROC curve shows all possible trade-offs; AUC summarizes overall score separation.
8. No benchmark score is a claim about generalization. Incomplete judgments, data contamination, prevalence mismatch, and overfitting to static test sets all limit what a number can mean.

Key Formulas

$$P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN} \quad (2.12)$$

Precision and recall: the fundamental pair from which all other metrics derive.

$$AP = \frac{1}{|\text{Rel}|} \sum_{k: d_k \in \text{Rel}} P_k \quad (2.13)$$

Average precision: area under the precision-recall curve, the standard per-query measure.

$$nDCG_k = \frac{\sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}}{IDCG_k} \quad (2.14)$$

Normalized discounted cumulative gain: the standard for graded relevance evaluation.

$$AUC = P(s(x_+) > s(x_-)) \quad (2.15)$$

Area under the ROC curve: the probability that a random positive scores higher than a random negative.

Self-Check Questions

1. (Understand) A system achieves 95% accuracy on a dataset where 95% of items belong to class A. Is this system useful? What additional metric would reveal the problem?
2. (Analyze) System X has MAP = 0.60 and System Y has MAP = 0.55, both measured on 50 queries. Under what circumstances might Y actually be the better system?
3. (Evaluate) A face-unlock system reports AUC = 0.99 and a spam filter reports AUC = 0.99. Are they equally good? What additional information do you need to judge each for its intended use?
4. (Apply) Given a ranked list of 10 documents where relevant documents appear at positions 1, 4, and 7 (out of 8 relevant in the collection), compute $P@5$, AP, and explain why AP is less than the average of the three precision values.
5. (Analyze) Explain why macro-averaged recall across 4 information needs can disagree with micro-averaged recall, and give a concrete scenario where the disagreement is large.

Further Reading

- Manning, C. D., Raghavan, P., & Schütze, H. (2008). **Introduction to Information Retrieval**, Chapter 8: Evaluation in Information Retrieval. *Cambridge University Press*. [Free online version](#). The standard textbook treatment of precision, recall, MAP, and benchmark design.
- Järvelin, K., & Kekäläinen, J. (2002). **Cumulated Gain-Based Evaluation of IR Techniques**. *ACM Transactions on Information Systems*, 20(4), 422-446. [Free PDF](#). Introduced DCG and nDCG for graded relevance evaluation.
- Hanley, J. A., & McNeil, B. J. (1982). **The Meaning and Use of the Area under a Receiver Operating Characteristic (ROC) Curve**. *Radiology*, 143(1), 29-36. Established the probabilistic interpretation of AUC and its statistical properties.
- Fawcett, T. (2006). **An Introduction to ROC Analysis**. *Pattern Recognition Letters*, 27(8), 861-874. [Free PDF](#). The standard tutorial on ROC curves, AUC, and threshold selection for binary classifiers; covers construction, interpretation, and common pitfalls.
- Armstrong, T. G., Moffat, A., Webber, W., & Zobel, J. (2009). **Improvements That Don't Add Up: Ad-Hoc Retrieval Results Since 1998**. *Proceedings of CIKM 2009*, 601-610. [Author copy](#). Documented how benchmark overfitting prevents genuine progress from accumulating.
- Oren, Y., Bandel, E., Bhatt, G., Shmidman, A., & Sharvit, Y. (2023). **Investigating Data Contamination in Modern Benchmarks for Large Language Models**. [arXiv:2311.09783](#). Demonstrated that LLMs can recall benchmark answers from training data, with GPT-4 guessing missing MMLU options at 57% exact-match rate.

Benchmarks and Evaluation Campaigns

- **TREC (Text REtrieval Conference)**. Annual evaluation campaigns run by NIST since 1992. The source of pooled test collections, shared topics, and the MAP-based evaluation methodology used throughout this chapter.

- **MS MARCO**. Large-scale passage and document retrieval benchmark built from real Bing queries. Standard proving ground for neural retrieval models; reports MRR@10 and nDCG@10.
- **BEIR**. Zero-shot retrieval benchmark spanning 18 heterogeneous datasets. Tests whether a retrieval model generalizes beyond its training domain; reports nDCG@10.

3 - Advanced Text Processing

A user searches a bookshop's website for "car repair" and gets nothing useful. The catalog holds hundreds of matching titles, but their spines read "automobile repair", "auto repair", "Cars" (the Pixar film), or "car maintenance". The search index stores each of these as a separate token. "Car" and "automobile" never meet.

A different user asks "Who won the F1 race on the weekend?" and gets nothing useful either. The answer sits in a race-results page that reads "Verstappen took the win at Silverstone on Sunday, 6 July". That page shares almost no tokens with the query. "Who" does not appear in it; "weekend" does not appear either (the report says Sunday and gives a date); even "race" is absent because the article calls it a "Grand Prix". The user asked a question and expected a fact back. The retriever received a bag of five keywords and matched them against pages that also contain the word "weekend".

A third user types "Bücher von Goethe" and again gets unrelated results. Nothing in the pipeline noticed that "Bücher" is German for "books", that "Goethe" is a person's name, or that the sentence is a request for works by an author. All the structure the user provided was there in plain text and none of it was used.

The three failures come from the same source. Classical retrieval treats text as a bag of exact tokens: one surface form is one term and a query is a bag of query terms. That model works for keyword searches over clean text and breaks in two directions. On the document side, one concept has many surface forms and recall collapses. On the query side, natural-language input has structure, meaning, and intent that do not map to tokens in the answer at all, and the retriever ignores every layer of that information.

Advanced text processing addresses both. This chapter builds the classical toolbox that turns raw strings into features a retrieval and routing system can actually use: tokenization, normalization, sentence segmentation, stop words, stemming and lemmatization, phrase and compound detection, part-of-speech tagging, named entity recognition, and simple classifiers for language and intent. Together they close the recall gap on the document side and extract enough structure from queries that a retrieval system can route them, filter them, and rank the results.

The three failure examples do not all fully resolve in this chapter. The bookshop's "car repair" query and the German "Bücher von Goethe" query are handled end to end here. The F1 question is harder: the classical pipeline can turn it into a targeted news-search query with a date filter, but delivering "Verstappen won" as a two-word answer rather than a list of pages is the work of semantic search and retrieval-augmented generation in later chapters. The F1 case will follow us through the book.

Tokenization and Normalization

The chapter opener showed a bookshop returning nothing for “car repair” because “automobile repair” is stored under a different token. That failure begins at the very first stage of the retrieval pipeline: how raw text is split into tokens and normalized before any matching happens. This section walks through that stage, starting with the simple regex from 1 - [Classical Text Retrieval](#) and ending with a rule-based language detector that decides which downstream pipeline to invoke.

The naive tokenizer and where it breaks

Chapter 1 introduced a one-line tokenizer:

```
def word_tokenize(text: str) -> list[str]:
    text = re.sub(r'^\w\-', ' ', text)
    return [t for t in text.split(' ') if t]
```

It substitutes any non-word character with a space, then splits on whitespace. It works on well-behaved English prose. Give it a sentence that exercises real-world text and it fails in several distinct ways at once:

I buy my parents' 10% of U.K. startup for \$1.4 billion. Dr. Watson's cat called Mrs. Hersley and it was w.r.o.n.g., more to come ...

Running the naive tokenizer produces:

```
['I', 'buy', 'my', 'parents', '10', 'of', 'U', 'K', 'startup',
 'for', '1', '4', 'billion', 'Dr', 'Watson', 's', 'cat', 'called',
 'Mrs', 'Hersley', 'and', 'it', 'was', 'w', 'r', 'o', 'n', 'g',
 'more', 'to', 'come']
```

Four systematic problems appear:

- **Possessives are handled inconsistently by shape.** parents' (trailing apostrophe only) becomes just parents: the apostrophe and the following space collapse into one space and the apostrophe leaves no trace. Watson's (apostrophe followed by s) becomes two tokens, Watson and a stray s, because the apostrophe becomes a space that splits the two letters apart. For retrieval, folding “Watson's” into “Watson” is often desirable (users search for the name, not the possessive), so the stray s is dropped in a subsequent cleanup step and both possessive forms end up equivalent. For syntactic analysis this is harmful, because the link between “Watson” and “cat” is broken and the shape-dependent behaviour is easy to overlook.
- **Numbers, currencies, and percentages fall apart.** 10% becomes 10, \$1.4 billion becomes 1, 4, billion. Small integers survive; anything realistic does not.
- **Abbreviations disintegrate.** U.K., Dr., Mrs., and the artificial w.r.o.n.g. all break at their internal periods. Searching for U.K. is impossible against this index.
- **Punctuation vanishes.** Fine for retrieval, but downstream sentence analysis loses its cues.

For a retrieval-only pipeline over English, this behaviour is not fatal: users rarely search for “U.K.” with the period, and possessive stripping helps recall. But the distinctions are gone for good, and any task that needs sentence structure or numeric precision needs a better tokenizer up front.

Modern tokenization with nltk and spaCy

Both nltk and spaCy ship word tokenizers built from rules combined with abbreviation and exception lists learned from text corpora. Running each on the demonstration sentence:

```
import nltk
import spacy

sentence = ("I buy my parents' 10% of U.K. startup for $1.4 billion. "
           "Dr. Watson's cat called Mrs. Hersley and it was w.r.o.n.g., more to
```

```
come ...")

nltk.word_tokenize(sentence)
# ['I', 'buy', 'my', 'parents', "'", '10', '%', 'of', 'U.K.', 'startup', 'for',
# '$', '1.4', 'billion', '.', 'Dr.', 'Watson', "'s", 'cat', 'called', 'Mrs.',
# 'Hersley', 'and', 'it', 'was', 'w.r.o.n.g.', ',', 'more', 'to', 'come', '...']

[t.text for t in spacy.load('en_core_web_sm')(sentence)]
# ['I', 'buy', 'my', 'parents', "'", '10', '%', 'of', 'U.K.', 'startup', 'for',
# '$', '1.4', 'billion', '.', 'Dr.', 'Watson', "'s", 'cat', 'called', 'Mrs.',
# 'Hersley', 'and', 'it', 'was', 'w.r.o.n.g.', '.', ',', 'more', 'to', 'come', '...']
```

The two outputs agree on every token except one: nltk treats w.r.o.n.g. as a single abbreviation, spaCy splits the trailing period as sentence-terminating punctuation. On real text the two libraries usually agree; artificial or novel abbreviations are where they diverge.

Compared with the naive regex, the trained tokenizers fix several of the earlier failures but not all of them, and the two possessive shapes still behave differently:

- **Possessives.** parents' produces parents and a bare '. Watson's produces Watson and 's. Both tokenizers keep the apostrophe visible; only the 's form carries a marker that distinguishes it from any other apostrophe. Retrieval pipelines usually drop both apostrophe tokens after this step; syntactic pipelines keep them because they carry grammatical information.
- **Numbers.** Decimals stay together (1.4), which fixes the naive regex's worst behaviour. But the currency and percent symbols split off as their own tokens: \$1.4 becomes ['\$ ', '1.4'] and 10% becomes ['10 ', '%']. A downstream step that wants "\$1.4 billion" as one money entity needs to recombine, which is the job of the named entity recognizer covered later in the chapter.
- **Abbreviations.** U.K., Dr., and Mrs. are kept whole with their trailing periods, because both libraries carry curated abbreviation lists. w.r.o.n.g. is the divergent case: nltk keeps it whole, spaCy splits its final period. Real abbreviations rarely differ between the two libraries.
- **Punctuation.** ., ,, and ... are preserved as their own tokens. Retrieval pipelines filter them out; syntactic pipelines keep them.

For retrieval, we still want a leaner token list than the tokenizer alone provides. The standard cleanup after tokenization is:

1. Drop single-letter tokens and pure punctuation tokens.
2. Drop numbers and currency tokens unless the collection is numeric (a scientific-paper index would keep them).
3. Drop possessive tokens (" 's") once the possessive information has been used.

The chapter's demo notebook composes these three cleanup steps on top of a regex tokenizer and compares the intermediate outputs side by side.

Case, Unicode, and accents

A tokenizer separates words. Normalization decides which surface forms of the same word collapse to the same token. Consider three variants of one city name:

Zurich, Zürich, ZURICH

They refer to the same city. To match a query for Zurich against a document that spells the city Zürich, the index has to normalize both spellings to a shared form before storing them. Three transforms handle almost all cases in European text:

- **Case folding** applies `text.lower()`, or the locale-aware equivalent for Turkish İ/ı. The three variants collapse to `zurich`, `zürich`, `zurich`.

- **Unicode normalization** applies `unicodedata.normalize('NFKC', text)` to canonicalise characters that have multiple valid encodings, such as the ligature `fi` (one codepoint) unfolding to `fi` (two ASCII letters).
- **Accent folding** strips diacritics after normalization, mapping `ü` -> `u`, `é` -> `e`, `ß` -> `ss`. The library `unidecode` does this for arbitrary Unicode; `nltk`'s stemmers apply narrower per-language rules.

After all three, the three variants above become `zurich`. Users can type any spelling and match any document.

In practice, modern text-processing stacks apply Unicode normalization by default because it has no downside, case-fold when the retrieval scenario is case-insensitive (most web search), and fold accents only when the user population cannot easily type them.

Sentence segmentation

Some tasks need whole sentences as input, not tokens. Part-of-speech taggers work sentence by sentence because word ambiguity resolves in context. RAG systems (see the retrieval-augmented generation chapter) build larger chunks by grouping consecutive sentences until they hit a size limit. Both need a reliable answer to the question: where does one sentence end and the next begin?

Splitting on `.` is the naive approach and fails on the same abbreviations we saw earlier. In “Dr. Watson’s cat called Mrs. Hersley.”, the periods after `Dr` and `Mrs` are not sentence boundaries; only the trailing period is. Multi-line quoted dialogue, ellipses, and decimal numbers add more edge cases.

Two libraries handle this well:

- **Punkt** (NLTK) is an unsupervised sentence segmenter that learns abbreviations and sentence-start patterns from a training corpus. NLTK ships pre-trained models for English and several other languages. It is fast and accurate on newspaper-style prose; it struggles on narrative dialogue where punctuation appears both inside and outside quotation marks.
- **spaCy** segments sentences as a side effect of its full linguistic pipeline. It is heavier than Punkt but handles complex sentence structures more consistently and supports many languages out of the box.

```
from nltk.tokenize import sent_tokenize
sent_tokenize("Dr. Watson's cat called Mrs. Hersley. She was Egyptian.")
# ["Dr. Watson's cat called Mrs. Hersley.", "She was Egyptian."]
```

Sentence segmentation is one of the few pieces of a classical pipeline that stayed useful into the LLM era. Every RAG system built on top of embeddings still needs to decide where its chunks begin and end, and sentence boundaries are the most reliable low-cost signal for that decision.

Word boundaries are not always spaces

Not every writing system uses spaces to separate words, and not every “word” in a modern document is meant to be a single token. Both cases break the “split on whitespace” assumption at the root.

Scriptio continua languages, including Chinese, Japanese, Thai, and classical Greek, do not use spaces at all. A single sentence is a continuous string of characters:

莎拉波娃现在居住在美国东南部的佛罗里达。

莎拉波娃	现在	居住	在	美国	东南部	的	佛罗里达
Sharapova	now	lives	in	US	southeast	in	Florida

Code identifiers pack multiple words into a single token by convention: `QueryParser`, `assertEquals`, `word_tokenize`. A code-search index that stores only whole identifiers cannot answer a query for `parser`.

Spoken-language transcripts come from a continuous phoneme stream. Speakers do not pause between words; the pauses that exist mark breath, hesitation, or emphasis, and rarely align with word boundaries. Even the pause at a sentence end is unreliable in fluent speech. The segmenter has to infer word boundaries

from the phoneme sequence itself, aided by a pronunciation dictionary and a language model, and non-native pronunciation or unclear articulation shifts where those boundaries land.

There are two ways to break these continuous streams into useful tokens:

- **Dictionary lookup with a probabilistic tie-breaker.** The dictionary lists every candidate word starting at the current position, and a trained model (historically an HMM, today usually a neural network) picks the most likely segmentation. This works well when the vocabulary is closed and the domain is stable. It breaks on names, brand terms, and loanwords.
- **Sub-word tokens used directly for retrieval.** Instead of reconstructing words, we index overlapping character n-grams. For languages that *do* have whitespace, a common choice is trigrams with a # boundary marker indicating that a trigram starts at a word boundary (Cavnar & Trenkle 1994):

```
This course teaches multimedia retrieval.
#Thi his #cou our urs rse #tea eac ach che hes #mul ult lti tim ime med edi dia ...
```

For scriptio-continua text where no spaces exist, the # markers are simply omitted and the entire character stream is shingled into overlapping trigrams. Either way, queries are encoded identically:

```
Q = "teach multtimedia"
#tea eac ach #mul ult ltt tti tim ime med edi dia
```

Even with a different inflected form (“teaches” versus “teach”) and a typo (“multtimedia”), 10 of the 12 query trigrams match trigrams from the document. A retrieval model that supports partial matching with a proximity bonus finds the passage without any explicit stemming or spell-check.

This workaround for unsegmentable text turned out to be a foundational idea. Byte Pair Encoding (BPE) and WordPiece take the same principle (match parts of words when whole-word matching fails) and learn the token vocabulary from a corpus instead of fixing it to trigrams. Both power today’s large language models, and the semantic-search chapter covers them in detail.

Language detection

Both indexing documents and understanding queries depend on knowing what language the input is in. A German document benefits from Snowball’s German rules and the German stop-word list; running it through the English pipeline strips accents but leaves compounds and inflections untouched. A query like “Bücher von Goethe” from the chapter opener needs to be recognized as German before the retrieval system can restrict results to German-language books or apply German-specific normalization.

For long documents, a small number of hand-crafted rules is enough:

- **Alphabet diversity** looks at which scripts appear. Latin, Cyrillic, Greek, Arabic, Devanagari, Thai, and CJK (Chinese, Japanese, Korean) are easy to tell apart even from a single character.
- **Character diversity** narrows down within a script. ä, ö, ü are strong signals for German (and Turkish, Estonian, Finnish); ç for French, Portuguese, or Turkish; ñ for Spanish; ß almost exclusively for German.
- **Stop-word counts** track how many of the top hundred function words of each candidate language appear. English “the”, “of”, “and”; German “der”, “die”, “und”, “ist”; French “le”, “la”, “de”, “et”. Whichever language’s stop-word list matches most terms wins.
- **Vocabulary counts** extend the stop-word idea to a longer list of frequent content words.

For document-side indexing this is usually enough: a paragraph or a page provides many opportunities for the rules to fire, and the winning language emerges quickly. Once the language is known, the pipeline branches: a German document goes to the German Snowball stemmer, an English document to the English WordNet lemmatizer, and so on.

Short queries break the rules, though. “Mein computer” mixes German with an English loanword; “pain” is a French word for bread and an English word for suffering. Neither has enough evidence for the rules to fire reliably. That regime needs a statistical classifier over character n-grams, developed with Naive Bayes in [Intent Routing and Classification](#).

Where this sits in modern retrieval

Tokenization and normalization are the least glamorous part of the retrieval stack and still the largest source of quality issues in production. Modern lexical retrievers like Lucene, Solr, and Elasticsearch expose the whole pipeline as a chain of configurable analyzers: a tokenizer, a lowercase filter, a Unicode-normalization filter, a stop-word filter, a stemmer, sometimes a compound splitter. Dense retrieval sidesteps most of these choices by learning them implicitly in the embedding model, but hybrid stacks (BM25 combined with dense retrieval) still need the lexical side, and the lexical side still starts with these steps. Getting them wrong at ingestion time is expensive to fix later because the index has to be rebuilt.

The next section takes tokens as given and looks at the second half of the recall gap: collapsing “car”, “cars”, “carrying”, and “carried” so a document mentioning any of them matches a query for any other.

Stemming and Lemmatization

The bookshop catalog stores documents word by word. A user searches “car repair” and misses titles like “cars”, “repairing cars”, and “repaired vehicles” because each surface form is a distinct token in the index. Tokenization from the previous section fixed splitting; it did not fix the fact that one root word has many written shapes. This section closes that gap. First it decides which tokens are worth indexing at all, then it collapses inflected surface forms of the same word onto a common stem or lemma.

Stop words: what to keep out

Half a dozen very frequent words dominate any English text. “The”, “of”, “and”, “to”, “is” together account for roughly a quarter of tokens in typical prose. We already saw in [From Text to Searchable Vectors](#) that these high-frequency terms appear in over 60% of documents yet carry almost no content signal. A **stop-word list** is the standard shortcut: a fixed set of high-frequency function words to discard before indexing and querying. English stop-word lists usually contain 100-200 words. The filtering logic is straightforward:

```
STOPWORDS = {'a', 'an', 'the', 'and', 'or', 'in', 'of', 'to', ...}

def remove_stopwords(tokens: list[str], stopwords: set = None) -> list[str]:
    sw = stopwords if stopwords is not None else STOPWORDS
    return [t for t in tokens if t not in sw]
```

Every major language has its own list. `nltk.corpus.stopwords.words('german')` returns 232 German function words; French, Spanish, Italian, and the other main European languages are provided by both `nltk` and `spaCy`.

Dropping stop words works well most of the time and fails visibly the rest of the time. The cost is a small class of queries where the stop word is content-bearing:

- A search for **Stephen King’s “It”** finds nothing if the tokenizer drops “it” as a stop word. The novel’s title is the stop word.
- A search for **“IT security”** loses “IT” (information technology) for the same reason. The remaining “security” retrieves every document about locks, guards, and stock exchanges.
- **“To be or not to be”** consists entirely of stop words. Every word disappears.
- **“The Who”** (the band) has both tokens on typical stop lists.

BM25 already handles high-frequency words gracefully through two mechanisms that earlier bag-of-words models lacked: inverse document frequency drives the weight of corpus-wide common terms toward zero, and term-frequency saturation prevents any single term from dominating a document’s score no matter how often it repeats. Together these make aggressive stop-word removal less necessary. The modern default is to keep stop words in the index (storage is cheap) but skip them when generating phrase collocations, computing term-based similarities, or building compact features for a classifier.

Stemming: rule-based reduction

Once stop words are out, we still have “car”, “cars”, “carrying”, “carried”, “carrier”. They all point to the same concept, but a bag-of-words index sees five distinct tokens. **Stemming** applies a set of ordered rules that strip suffixes off inflected forms so that variants of one root end up as the same token.

The three widely-used English rule-based stemmers are Porter, Lancaster, and Snowball. All three take a token, apply an ordered sequence of suffix-stripping rules, and return a stem. The stem does not need to be a real English word; it only needs to be the same for all variants of the same lemma.

The Porter algorithm

The Porter algorithm is the classical English stemmer, published by Martin Porter in 1980. Chapter 1 introduced it as a black box; here we look at the internal structure.

Porter defines a **vowel** as A, E, I, O, U, and additionally Y when the preceding character is a consonant (so “Y” is a vowel in “rhythm” but a consonant in “yellow”). All other characters are consonants. Let C be a sequence of consonants and V a sequence of vowels. Every English word can be written in the form:

$$[C](VC)^m[V] \quad (3.1)$$

where the exponent m is the **measure** of the word. TREE has $m = 0$ (consonant sequence TR followed by vowel sequence EE, no VC pair in between), TROUBLE has $m = 1$, and TROUBLES has $m = 2$. Porter uses m to prevent over-stemming: rules that would strip “ATE” from “OPERATE” are gated behind $m \geq 1$ so they do not also strip the “ATE” from “MATE”.

The algorithm applies five steps in sequence, containing roughly 60 suffix-replacement rules in total. The entire implementation fits in about 400 lines of Python (or the equivalent in Java/C). Each step targets a different layer of English morphology:

Step	Goal	Example rules	Example
1a	Strip plurals	SSES arrow.r SS, IES arrow.r I, S arrow.r $\emptyset$	caresses arrow.r caress, ponies arrow.r poni, cats arrow.r cat
1b	Strip past tense / progressive	$(m > 0)$ EED arrow.r EE, (v) ED arrow.r $\emptyset$, (v) ING arrow.r $\emptyset$	feed arrow.r feed, plastered arrow.r plaster, motoring arrow.r motor
2	Simplify derivational suffixes	$(m > 0)$ ATIONAL arrow.r relational arrow.r relate, digitizer arrow.r digitize ATE, $(m > 0)$ IZER arrow.r IZE	
3	Continue simplification	$(m > 0)$ ICATE arrow.r IC, triplicate arrow.r triplic, formalize arrow.r formal $(m > 0)$ ALIZE arrow.r AL	
4	Remove residual suffixes	$(m > 1)$ ION arrow.r $\emptyset$, adoption arrow.r adopt, platonism arrow.r platon $(m > 1)$ ISM arrow.r $\emptyset$	
5	Clean up trailing letters	$(m > 1)$ E arrow.r $\emptyset$, double consonant arrow.r single rate arrow.r rate, controll arrow.r control	

Within each step, rules are tried in order and only the first match fires. The conditions in parentheses prevent rules from applying to stems that are too short: $(m > 0)$ requires at least one vowel-consonant pair in the remaining stem, and (v) requires at least one vowel anywhere. This is what keeps Step 1b from stripping “ED” off a two-letter word.

The full rule set stems on the order of 95% of English text correctly. Porter’s original paper (Porter, 1980) lists every rule; the `nltk.PorterStemmer` implementation is faithful to that specification.

```
import nltk
porter = nltk.PorterStemmer()

for w in ['car', 'cars', 'carrying', 'carried', 'carrier']:
    print(w, '->', porter.stem(w))

# car      -> car
# cars     -> car
# carrying -> carri
# carried  -> carri
# carrier  -> carrier
```

Note that “carrying” and “carried” collapse to `carri` (not to `carry`), and “carrier” is left intact. The stems are not linguistically pretty, but they map three variants of the verb “carry” onto the same token, which is what the retrieval index needs.

Lancaster and Snowball

Lancaster (Paice, 1990) is a more aggressive stemmer for English. It applies iterative rule-based suffix removal until no rule fires, producing shorter stems than Porter. Speed is its main advantage: on average it does fewer rule scans per word. Its main risk is collisions: unrelated words can reduce to the same short stem.


```

lancaster = nltk.LancasterStemmer()
for w in ['one', 'only', 'organization', 'organize']:
    print(w, '->', lancaster.stem(w))

# one          -> on
# only         -> on
# organization -> org
# organize     -> org

```

“one” and “only” both stem to “on”, so a document about “the only book on Rome” collides with one about “book number one”. For a scenario that favors recall the collisions are usually acceptable; for a name search or a precise query they are a problem.

Snowball is Martin Porter’s own multi-language framework, published later, that generalizes the Porter design. It supplies stemmers for around 20 European languages including English, German, French, Spanish, Italian, Dutch, and Russian. On English, Snowball is essentially a revised Porter with a handful of small corrections; on other languages, it is often the only rule-based option available.

```

snowball_en = nltk.SnowballStemmer("english")
snowball_de = nltk.SnowballStemmer("german")

snowball_de.stem("Wagen"), snowball_de.stem("Wägen") # ('wag', 'wag')
snowball_de.stem("Reparatur"), snowball_de.stem("reparieren") # ('reparatur', 'repar')

```

The German stemmer folds umlauts (Wägen -> Wagen -> wag) and reduces “reparieren” to repar, which does not match reparatur. Rule-based stemmers approximate the linguistic stem but do not compute it: for German, the noun “Reparatur” and the verb “reparieren” share a root but have different stems under any pure suffix-stripping rule set.

Lemmatization: dictionary-based reduction

Rule-based stemming is fast and needs no external data, but it produces linguistically incorrect stems and cannot handle strong inflections. English “go” and “went” share no letters at all, so no suffix rule can map them together. **Lemmatization** takes a different approach: look the word up in a dictionary and return its recorded base form.

The three components of a dictionary-based lemmatizer are:

1. A **rule set** for regular inflections (-s, -ed, -ing, plurals with -ies).
2. An **exception list** for irregular forms: went -> go, ate -> eat, mice -> mouse, children -> child.
3. A **dictionary** of all base forms in the language, used to check whether a candidate form is valid.

The algorithm proceeds in order:

1. Take the current token and its part-of-speech (POS) tag (verb, noun, adjective, ...). POS tagging is covered later in this chapter; what matters here is that “meeting” as a noun lemmatizes to meeting, while “meeting” as a verb lemmatizes to meet.
2. If the token is in the dictionary as-is, it is already a base form. Return it.
3. If the token is in the exception list for its POS, return the base form listed there.
4. Apply the regular-inflection rules for its POS, checking the dictionary after each rule. Return the first match.
5. If no rule produces a dictionary word, return the token unchanged. This handles names, misspellings, and loanwords.

WordNetLemmatizer in nltk and the built-in lemmatizer in spaCy are the two standard implementations. Their exception lists cover several thousand irregular forms each.

```

wn = nltk.WordNetLemmatizer()
wn.lemmatize('carried', 'v') # 'carry'
wn.lemmatize('carrier', 'n') # 'carrier'

```

```
wn.lemmatize('mice', 'n')      # 'mouse'
wn.lemmatize('went', 'v')      # 'go'
wn.lemmatize('better', 'a')    # 'good'
```

Note the last case: “better” lemmatizes to “good” because the exception list encodes the adjective’s comparative form. No rule-based stemmer can do this.

spaCy runs a full linguistic pipeline that assigns POS tags automatically, so the lemmatizer is applied without the caller having to pass a tag:

```
nlp = spacy.load('en_core_web_sm')
for token in nlp("She carried the mice yesterday and went home"):
    print(token.text, '->', token.lemma_)

# She      -> she
# carried  -> carry
# the      -> the
# mice     -> mouse
# yesterday -> yesterday
# and      -> and
# went     -> go
# home     -> home
```

For retrieval, the practical difference between rule-based stemming and dictionary lemmatization is that a lemmatizer gives you the linguistic base form and a stemmer gives you an arbitrary reduction that just happens to be consistent within the class. Both work for matching a query to a document. Lemmatization is heavier because it needs POS tagging plus a dictionary lookup; stemming is a handful of string operations.

Multi-lingual comparison

Every stemmer and lemmatizer described above assumes one language. Running an English stemmer on French text strips accents and collapses “les” to “l” without doing anything useful. The right choice depends on the language of the input, which is what the language detector from the previous section is for.

Snowball is the go-to rule-based option outside English because it covers 20 languages under a single framework. spaCy is the go-to lemmatizer for the same set. On the same French sentence, the two diverge on inflected verb forms:

```
Input:      "Nous avons aperçu les châteaux"

Snowball:   ['nous', 'avons', 'aperçu', 'le', 'château']
spaCy:      ['nous', 'avoir', 'apercevoir', 'le', 'château']
```

Snowball preserves accents in the French output (visible on *aperçu* and *château*) but it does not recognize inflected verb forms: *avons* becomes *avon* by simple suffix stripping rather than mapping to *avoir*, and *aperçu* stays unchanged rather than lemmatizing to the infinitive *apercevoir*. spaCy’s French lemmatizer handles both because it has the verb tables. Both correctly reduce the plural *châteaux* to *château* while keeping the circumflex. On entity-heavy queries the difference is small; on prose retrieval where verbs carry the topic, it moves the recall needle.

In German, nouns inflect for case, gender, and number, and German also forms diminutives with “-chen” and “-lein”. The two tools handle these differently:

```
Input:      "die Häuser und ihre Häuschen"

Snowball:   ['die', 'haus', 'und', 'ihr', 'hausch']
spaCy:      ['der', 'Haus', 'und', 'ihr', 'Häuschen']
```

Both reduce the plural “Häuser” to the singular (“haus” for Snowball, which also lowercases; “Haus” for spaCy). They part ways on the diminutive “Häuschen”: Snowball over-stems it to “hausch”, a token that

matches neither “Haus” nor any real word, while spaCy keeps “Häuschen” as its own lemma. For queries against German text, spaCy’s dictionary lemmatizer avoids these spurious stems, though the small models are weak on strong verbs and do not always map inflected verb forms to a common infinitive.

Neither approach handles compound splitting, which is the concern of the next section: “Wolkenkratzer” (skyscraper) stays as a single token under both stemmers.

Where this sits in modern retrieval

Rule-based stemming and dictionary lemmatization are still the default text-processing configuration for lexical retrievers. Lucene, Solr, and Elasticsearch ship analyzers for every major language that combine tokenization, stop-word filtering, and stemming or lemmatization in the same order this section presented them. Stemming is the option chosen when performance matters or when no dictionary is available; lemmatization is preferred when accuracy on strongly-inflected languages or on strong-verb English matters more than throughput.

Dense retrieval learns these normalizations implicitly during embedding training. A well-trained embedding model puts “carried”, “carrying”, and “carrier” near each other in vector space without an explicit stemming step. This does not make the lexical stack obsolete: hybrid retrieval combines BM25 (which needs stemming or lemmatization to keep recall high) with dense retrieval (which does not), and the two together consistently outperform either alone. The classical stemmers of this section are still on the query path of every serious production search stack in 2025.

The next section takes stemmed and lemmatized tokens as given and looks at what happens when meaning crosses word boundaries: “New York” is not “New” plus “York”, and “Wolkenkratzer” is not “Wolke” plus “kratzer”.

Phrases and Compounds

Tokenization decides what a token is; stemming and lemmatization decide which tokens are equivalent. Both operate at the level of one word. This section handles the two cases where a one-word view of meaning breaks down. Phrases pack multiple words into one concept: “New York” is not “New” plus “York”, and a query for “New” retrieves the wrong documents. Compounds do the opposite: “Bücherregal” (bookcase) is one German token built from “Bücher” (books) and “Regal” (shelf). A shopper on an e-commerce site who searches for “Regal” will not see the bookcase listings until we split the compound into its parts. Both cases need explicit handling before the retrieval model sees the tokens.

Bi-grams, tri-grams, phrases

The naive bag-of-words model treats a document as a multiset of independent tokens. “New York City”, “Salt Lake City”, and “prime minister” become bags of three or two unrelated tokens. Searching for “New York” as two separate terms still retrieves every document that mentions the city, so recall is not the problem. Precision is: the query also matches documents where “new” and “York” occur unrelated (“a new book from York University”), and a plain bag-of-words index has no way to tell those apart from documents where the two words sit side by side. Without a proximity or phrase mechanism, the genuine “New York” documents cannot be ranked above the incidental co-occurrences. Indexing “New York” as a single token `new_york` restores precision and gives the ranker a clean signal.

This generalises to n-grams: a bi-gram joins two adjacent words, a tri-gram three, and so on. Indexing the bi-grams and tri-grams of “the city of New York City” produces

```
the, city, of, new_york_city, new_york, york_city, city
```

with `_` marking a phrase token. We keep the original single-word tokens and emit every overlapping n-gram rather than only the longest one. This is deliberate: a query for “New York” must still match even though this document only ever contains the longer “New York City”, and a query for the single word “city” must still match as well. Keeping all of them costs some index space but avoids missing any of these queries. A phrase query for “New York” now hits `new_york` directly, without proximity constraints. The question that follows is which n-grams are worth indexing. Adding every bi-gram in the corpus is prohibitive (it turns a 100k-token vocabulary into a 10-billion-pair vocabulary) and most of them are noise: “it is”, “of the”, “and to”. We need a scoring function that selects the small subset of bi-grams that mean something the individual tokens do not.

Naive frequency and the stop-word problem

The obvious approach is to count how often each bi-gram appears in the corpus and keep the most frequent ones. On any English text this immediately breaks:

of the	12,847
in the	9,203
to the	7,451
and the	6,988
for the	5,102
...	

The top of the list is dominated by pairs of stop words. Filtering these out helps but does not solve the problem: even after stripping stop-word pairs, high-frequency co-occurrences of common words (“said Holmes”, “young man”, “could see”) still bubble to the top and squeeze out the phrases we actually want. Frequency alone cannot tell us whether two words appear together because they mean something together or because they are both common.

Two better scoring functions capture that distinction: pointwise mutual information (PMI) and the likelihood ratio (LHR).

Pointwise mutual information

The intuition behind **pointwise mutual information** is straightforward. Two words that mean something together should co-occur more often than their individual frequencies would suggest under independence. In *A Study in Scarlet*, a corpus of about 43,200 tokens, “said” appears 207 times and “Holmes” 98 times. If the two were independent, we would expect the bi-gram “said Holmes” to occur about $207 \cdot 98/N \approx 0.5$ times. It actually occurs 12 times, roughly 25 times more than chance. That is a real signal, but a modest one: “said” precedes dozens of different names throughout the book.

Compare “Sherlock Holmes”. “Sherlock” appears 52 times, and every one of those 52 occurrences is followed by “Holmes”. Independence would predict about $52 \cdot 98/N \approx 0.1$ co-occurrences; the observed count of 52 is more than 400 times higher. “Sherlock” is effectively bound to “Holmes”.

Two things help read the formula. First, $\log_2(N)$ is the same constant for every bi-gram, so ranking by PMI is the same as ranking by $\log_2 \text{tf}(t_1, t_2) - \log_2 \text{tf}(t_1) - \log_2 \text{tf}(t_2)$. Second, look at what maximizes the score. PMI is largest when all three counts equal 1: a word that occurs once, sitting next to another word that occurs once, giving $\text{pmi} = \log_2 N$. Even when the two words and their pair all occur n times together and never apart, the score is $\log_2(N/n) = \log_2 N - \log_2 n$, which only falls as n grows. So PMI rewards rarity, and this is the mirror image of the stop-word problem: instead of frequent function words dominating, two rare words that happen to land side by side by chance float to the top. In *A Study in Scarlet*, 203 different bi-grams reach the maximum score of $\log_2 N \approx 15.4$, every one of them a pair of words that each occur exactly once and only next to each other (“aqua tofana”, “admired treated”, “ambitious title”). All of them outrank “Sherlock Holmes”. A single chance adjacency is no evidence of a real collocation; there is simply too little frequency to be confident the two words prefer each other. The cure is a minimum-frequency filter: require a bi-gram to occur at least, say, three times before scoring it at all, which drops the low-evidence rare pairs.

Likelihood ratio

The likelihood ratio test attacks the same problem from a hypothesis-testing angle rather than an information-theoretic one. It asks: how much better is the “these words are dependent” hypothesis than the “these words are independent” hypothesis, given the observed counts?

Let $\text{tf}_1 = \text{tf}(t_1)$, $\text{tf}_2 = \text{tf}(t_2)$, and $\text{tf}_{12} = \text{tf}(t_1, t_2)$. Two hypotheses:

- H_1 (independence): the probability that t_2 follows any word is a single value p , whether the previous word is t_1 or not. The maximum-likelihood estimate is $p = \text{tf}_2/N$.
- H_2 (dependence): the probability that t_2 follows t_1 is $p_1 = \text{tf}_{12}/\text{tf}_1$, and the probability that t_2 follows any word other than t_1 is $p_2 = (\text{tf}_2 - \text{tf}_{12})/(N - \text{tf}_1)$, with $p_1 \neq p_2$.

Assuming a binomial distribution for the “next word is t_2 ” events, the likelihood of each hypothesis is:

$$L(H_1) = b(\text{tf}_{12}; \text{tf}_1, p) \cdot b(\text{tf}_2 - \text{tf}_{12}; N - \text{tf}_1, p) \quad (3.2)$$

$$L(H_2) = b(\text{tf}_{12}; \text{tf}_1, p_1) \cdot b(\text{tf}_2 - \text{tf}_{12}; N - \text{tf}_1, p_2) \quad (3.3)$$

with $b(k; n, x) = \text{binom } nkx^k(1-x)^{n-k}$.

LHR is more robust than PMI on sparse data because it directly compares two probabilistic models rather than comparing observed to expected counts. It does not have PMI’s rare-word bias, so infrequent bi-grams score less aggressively. The trade-off is that LHR ranks frequent grammatical pairs highly, because those pairs really are non-independent even though they are useless as phrases. The bi-gram “I am” is the clearest example: “am” occurs 41 times and 39 of them follow “I”, so the dependence is overwhelming, yet “I am” is worthless as an index phrase. This is why LHR still needs a stop-word filter: without one, “of the”, “to be”, “had been”, and “I am” crowd the top of the ranking alongside the genuine names.

PMI or LHR: which to use

The two measures rank the same corpus differently because they measure different things. PMI measures the *strength* of association: how much more often two words appear together than chance predicts, regardless of how often that is. LHR measures the *evidence* for association: how confidently the data rules

out independence, which grows with frequency. “Sherlock Holmes” makes the contrast concrete: it ranks 45th of 115 under PMI, held down because “Holmes” is common, but first under LHR, which rewards the sheer weight of 52 co-occurrences.

Because they optimise different quantities, the two fail in opposite directions, and a genuine phrase can score well on one and poorly on the other:

- **Rare but tight** phrases occur a handful of times, always as a unit. They score high on PMI and low on LHR. In our results these are PMI’s top rows: “Lauriston Gardens” (the address of the murder), “Audley Court”, “chemical laboratory”. LHR buries them below its frequency-driven head, so PMI is what surfaces them.
- **Frequent but loose** pairs such as “young man” or “could see” occur often but are not real phrases. They score moderately on LHR but low on PMI, which correctly discounts them because the individual words are common.
- **Frequent and tight** phrases such as “Sherlock Holmes” or “Jefferson Hope” score high on both and need no adjudication.

No single measure is uniformly best. Some practical guidance:

- **For a phrase vocabulary that maximises recall** (finding as many genuine phrases as possible), take the top- k from both measures and union them: LHR contributes the well-attested head, PMI the rare-but-specific tail of named entities and technical terms that users actually search for. A union only adds candidates, so plan a downstream prune, whether a human review pass or a secondary filter, to drop the frequent-but-loose pairs that LHR lets through.
- **For a compact, high-precision list**, use LHR alone with the stop-word filter and a frequency floor. Its preference for well-attested pairs is an asset here, and you accept missing some rare phrases.
- **The filters matter more than the choice of measure.** A minimum-frequency floor removes PMI’s chance pairs and a stop-word filter removes LHR’s grammatical pairs; without both, neither measure produces a usable list. Scale the frequency floor to the corpus: 3 works for a single novel, but a large collection needs a higher floor to keep the vocabulary manageable.
- **An alternative to the union** is to use one measure to rank and the other to filter, for example rank by LHR but drop any pair whose PMI falls below a threshold. This keeps LHR’s ordering while using PMI to veto the frequent-but-loose pairs, trading some recall for precision.

Choosing a threshold and indexing strategy

Once a scoring function is chosen, we pick a threshold and add every bi-gram above it to the vocabulary. The threshold is not a precise cut-off; it trades index size against phrase coverage. Missing a bi-gram is not fatal: the document is still retrievable through its individual terms, so recall is unaffected. What we lose is the precision and ranking boost the phrase token would have given, which lets more incidental co-occurrences through as noise. For many use cases, such as fact-checking or broad information search, that extra noise is acceptable.

Applying the vocabulary changes how we tokenize, without changing the single scan through the text. Once the accepted bi-grams are known, we tokenize each document, and every query the same way, in one left-to-right pass. At each position we emit the single token, and we do one extra vocabulary lookup to check whether the current token together with the next one forms an accepted bi-gram; if it does, we emit the phrase token as well and move on to the next position, so overlapping phrases are still caught. Keeping the single tokens alongside the phrase is deliberate, as the chapter opener showed: a query for “Holmes” must still match a document that only contains “Sherlock Holmes”.

A compact implementation keeps two vocabularies: the ordinary single tokens, and a set of accepted token pairs stored as Python tuples.

```
bigrams = {("new", "york"), ("york", "city")} # accepted pairs from PMI/LHR (see
code above)

def apply_phrases(tokens: list[str], bigrams: set[tuple[str, str]]) -> list[str]:
    result = []
```



```

    for i, tok in enumerate(tokens):
        result.append(tok)                                # always keep the single
token
        if i + 1 < len(tokens) and (tok, tokens[i + 1]) in bigrams:
            result.append(f"{tok}_{tokens[i + 1]}")        # add the phrase token
    return result

apply_phrases("the city of new york city".split(), bigrams)
# ['the', 'city', 'of', 'new', 'new_york', 'york', 'york_city', 'city']

```

The two overlapping phrases “new york” and “york city” are both emitted, next to the single tokens, exactly the mix the chapter opener described. The accepted-pair set itself comes from the collocation finder: rank with PMI or LHR, apply the frequency and stop-word filters, take the top- k , and store each surviving bi-gram as a tuple.

The construction extends to tri-grams and quad-grams: keep a set of triples, check a window of three tokens, and emit a three-word phrase when it matches. Returns diminish quickly, though. Any four-word sequence is rare enough that the scoring functions lose statistical power, and the extra vocabulary rarely earns its storage. Most production systems stop at bi-grams and handle the occasional longer phrase with proximity search at query time. nltk also provides `TrigramCollocationFinder` and `QuadgramCollocationFinder` with matching `TrigramAssocMeasures` and `QuadgramAssocMeasures`.

Compounds

A phrase glues two tokens together. A compound goes the other way: it hides several words inside a single token. German makes this productive: “Wolkenkratzer” (skyscraper) combines “Wolke” (cloud) and “Kratzer” (scratcher). “Rindfleischetikettierungsüberwachungsaufgabenübertragungsgesetz” (a Mecklenburg-Vorpommern law from 1999-2013 delegating the supervision of beef labelling) chains together seven concepts as one token. Finnish and Dutch have the same construction. Programming languages have their own version: `word_tokenize`, `assertEquals`, and `QueryParser` each pack a whole phrase into one identifier.

Compounds break retrieval in an asymmetric way. A user who types “Etikettierung Gesetz” cannot match a document indexed under the full compound. The reverse also fails: a query for “Rindfleischetikettierungsüberwachungsaufgabenübertragungsgesetz” only finds documents that spelled it exactly the same way, which is optimistic for a 63-letter word. The fix is to index both the compound and its parts, giving the retriever a chance to match either.

Splitting compounds

Compound splitting proceeds in two stages: generate candidate splits, then score them.

For **generating candidates**, we use language-specific rules. In English, hyphens and hyphenation syllables give the splits: “must-have” splits into “must” and “have”, “skyscraper” into “sky”, “scrap”, “er”. In German, syllable boundaries following German hyphenation rules produce candidates: “Wolkenkratzer” becomes (“wol”, “ken”, “krat”, “zer”), and every way of grouping consecutive syllables into words yields a candidate split: (wolken, kratzer), (wolken, krat, zer), (wol, kenkratzer), and so on. German also uses binding letters like “s” between compound parts (“Schifffahrtskapitän” needs to be tested both as “Schifffahrt-Kapitän” and “Schifffahrts-Kapitän”), so the candidate generator must try both variants.

For **scoring candidates**, we discard any split containing a piece that is not in the language’s dictionary, then score the survivors by how frequent their parts are. Crucially, the score is not a winner-take-all decision. Just as we indexed both “new york” and its parts, and both the long beef-labelling law and its constituents, we do not have to commit to a single decomposition of a compound: we can keep several plausible splits and index the union of their parts alongside the full compound. The score’s job is to rank plausibility and prune weak candidates, not necessarily to crown one winner.

The intuition: the dictionary filter first removes any split containing a piece that is not a real word, so (“Wol”, “Kenkratzer”) is discarded outright because neither piece is a German word. Among the splits that

survive, the score ranks plausibility by the frequency of the parts. We then either keep the single top split, or, favouring recall, keep several plausible splits and index the union of their parts. The average is used rather than the sum so that splits with different numbers of parts are comparable.

Where this sits in modern retrieval

Phrase detection and compound splitting are both still active production techniques. Lucene ships two compound-word filters. The `HyphenationCompoundWordTokenFilter` is the two-stage method above in production form: a hyphenation grammar, the same pattern file that breaks words across lines at the end of a text line, proposes candidate break points in place of our syllable rules, and a dictionary keeps only the fragments that are real words. The matching sub-words are added to the token stream next to the original compound, exactly the union-indexing we described. Its sibling, the `DictionaryCompoundWordTokenFilter`, skips the hyphenation step and tests substrings directly against the word list, which is slower and produces more spurious fragments, so Elasticsearch recommends the hyphenation variant for German, Dutch, and other compounding languages. Solr exposes the same filters through the Redlink extension, and actively maintained dictionaries such as `uschindler/german-decompounder` provide the word lists these filters consume.

Dense retrieval learns the compound structure implicitly through sub-word tokenizers like BPE and WordPiece, but the learning is imperfect. A 2023 study (arXiv:2305.14214) showed that SentencePiece, the tokenizer most widely used with multilingual language models, still splits German compounds into semantically incoherent fragments a significant fraction of the time. Hybrid retrieval stacks that combine BM25 with dense embeddings therefore keep the explicit decompounder on the BM25 branch, letting the neural side learn what it can while the classical side handles the compounds the classical way.

Phrase detection has an analogous story. Bi-gram indexing appears in modern lexical retrievers wherever named entities matter: product-search stacks index brand-name bi-grams, patent search indexes technical compounds, and academic search indexes conference names. PMI and LHR remain the standard offline scoring functions for building those bi-gram vocabularies.

The next section leaves the document-recall problem behind and turns to the query side of the two-way failure from the chapter opener. Given a natural-language query, how do we extract enough structure from it that the retriever can do more than bag-of-words matching?

Query Understanding

The first three sections closed the recall gap on the document side. A well-processed collection now has one token per concept, phrases identified, and compounds split. The chapter opener's other failure remains: the query "Bücher von Goethe" arrives as three tokens and the retriever has no idea that "Bücher" and "books" are the same concept, that "Goethe" names a person, or that the sentence is a request for works by a specific author. This section adds three tools that extract that kind of structure from free-text: word-relationship expansion, part-of-speech tagging, and named entity recognition. The section ends with the practical concern of spelling: a query that arrives misspelled needs to be repaired before any of these tools can help.

Word relationships: synonyms and homonyms

Stemming and lemmatization collapse different surface forms of the *same* word onto one token. They do nothing about the looser assumption we used so far: that one normalized token is independent of another normalized token. Every retrieval model in the book to this point has relied on it, treating tokens as unrelated dimensions with no notion that two of them might mean similar things or that one might mean two things. Real language breaks that assumption in two directions. A **synonym** is two different words for the same concept ("buy" and "purchase"); a **homonym** is one word carrying two concepts ("bank" the institution and "bank" the riverside). Lemmatization helps with neither, because "buy" and "purchase" reduce to unrelated stems and "bank" keeps a single stem whichever sense is meant. This section handles both with classical lexical tools; the semantic search chapter returns to the independence assumption in full and replaces it with learned representations that place related tokens near each other.

Synonyms break recall. A user searching a product catalog for "purchase history" misses documents indexed under "buy" or "buying", even though they hold exactly what the user wants, purely because the tokens differ. Two responses:

- **Synonym expansion (classical).** Attach a synonym list to each token. When the document contains "purchase", also index "buy" and "acquire". When the query contains "buy", also search for "purchase" and "acquire". The synonym lists come from curated resources (WordNet for English, thesaurus databases for other languages) or from domain-specific glossaries (medical terminologies, legal dictionaries).
- **Dense embeddings (modern).** Instead of maintaining synonym lists, learn a vector space where "buy" and "purchase" naturally end up near each other. This is the approach of the semantic search chapter and it subsumes the synonym problem, at the cost of an embedding model and a vector index.

Domain vocabularies show the same problem more sharply. The chapter opener's F1 query, "Who won the F1 race on the weekend?", needs to match documents that call the same event a "Grand Prix" and never use the tokens "F1" or "race" at all. WordNet does not capture this mapping because it is not a linguistic synonym but a domain shorthand. The fix is a domain-specific synonym list maintained by the search team: F1 maps to {Formula 1, Formula One, Grand Prix}, race maps to {Grand Prix, GP}. Product-search systems maintain similar lists for brand aliases ("iPhone" -> "Apple iPhone", "iPhones") and category synonyms ("laptop" -> "notebook").

Hand-maintained lists work well for the common, high-value terms a search team can anticipate, but they cannot keep up with the long tail: driver names the list has never seen, one-off event aliases, or code words specific to a community. This is where dense retrieval starts to pay for itself, because an embedding model learns which words appear in similar contexts without any manual list. Modern production stacks combine both: hand-crafted lists for the head, embeddings for the tail. The full argument comes back in the semantic search chapter.

Homonyms are the other direction, and the harder one. A query for "bank" cannot know whether the user means the financial institution or the riverside, and "lead" could be the metal or the verb meaning to guide. Two responses again:

- **Context from surrounding words (classical).** The words around a homonym usually fix its sense. Part-of-speech tagging is the simplest case: "lead" the noun (the metal) and "lead" the verb (to guide)

carry different tags, so tagging at index and query time separates them. Broader lexical context helps further: a text that also mentions “money”, “account”, or “loan” points to the financial “bank”, while “river”, “water”, or “shore” points to the riverside. This works whenever there is enough surrounding text to read.

- **Contextual embeddings (modern).** Sentence encoders and large language models represent a word by the company it keeps, so “bank” in “the river bank” and “bank” in “the savings bank” receive different vectors. They resolve the sense from context without any hand-written rules, which is why they handle homonyms far more reliably than lexical rules do. This is the approach developed in the semantic search chapter.

Both approaches lean on context, so both fail when the query is too short to supply any: a bare query “bank” gives the system nothing to disambiguate on. There the only options are to return results for both senses or to ask, which is what web search does with “did you mean” suggestions. Genuine ambiguity with no context is inherent to language, and no technique fully removes it.

Hypernyms and hyponyms

WordNet provides two more relationships that expand queries in a different direction. A **hypernym** is a broader category (“animal” is a hypernym of “cat”), and a **hyponym** is a narrower one (“cat” is a hyponym of “animal”, and “mammal” is both a hyponym of “animal” and a hypernym of “cat”). These relationships form a hierarchy, and every English noun in WordNet sits somewhere in that hierarchy.

Three retrieval uses:

- **Faceted search.** A user searching for “animals” is offered “cats”, “dogs”, “birds” as sub-facets that drill down into the hypernym tree. If the initial query returns too few results, the user is offered the parent facet (“mammals”) that broadens it. Amazon’s category tree and Wikipedia’s category system are both hypernym hierarchies dressed up as navigation UIs.
- **Query expansion with weighted hyponyms.** A search for “cat” is silently expanded to include hyponyms like “kitten”, “tabby”, “siamese”, each with a lower weight than the original term. Documents about specific cat breeds surface for a general “cat” query.
- **Relevance ranking with hypernym paths.** Instead of expanding the query, the scoring function is adjusted so a document mentioning “labrador” contributes something to a query for “dog” even without an explicit expansion.

```
from nltk.corpus import wordnet

for synset in wordnet.synsets('cat', pos='n')[:3]:
    print(synset.name(), '->', synset.definition())
    print('  hypernyms:', [h.name() for h in synset.hypernyms()])
    print('  hyponyms: ', [h.name() for h in synset.hyponyms()][:3])

# cat.n.01 -> feline mammal usually having thick soft fur and no ability to roar
#  hypernyms: ['feline.n.01']
#  hyponyms:  ['domestic_cat.n.01', 'wildcat.n.03']
```

WordNet covers English; multilingual equivalents such as Open Multilingual WordNet exist for around 30 languages but with less coverage than the English version.

Part-of-speech tagging

Sentences are made of words that fall into grammatical classes: noun, verb, adjective, adverb, pronoun, article, and so on. A **part-of-speech tagger** assigns one of these labels to each token in a sentence, using surrounding words to disambiguate cases where the same surface form belongs to different classes (“run” as a noun in “a good run” versus “run” as a verb in “I run daily”). Figure 3.1 shows a POS-tagged parse tree for a simple English sentence, with the POS tags at the leaves and phrase-level structure above them.

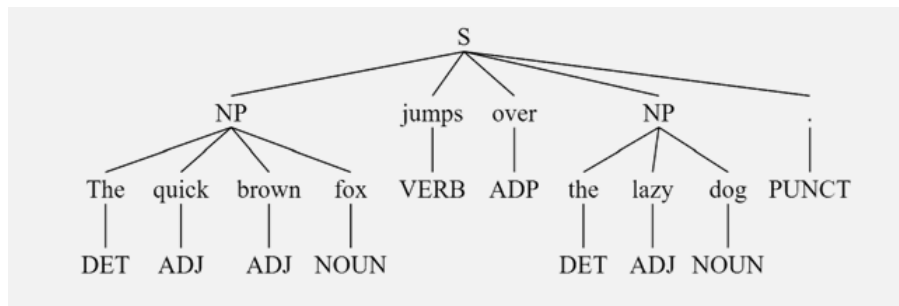


Figure 3.1: Constituency parse tree for “The quick brown fox jumps over the lazy dog.” POS tags (DET, ADJ, NOUN, VERB, ADP, PUNCT) appear at the leaves; phrase-level constituents such as NP (noun phrase) sit at intermediate nodes.

POS tagging supports three retrieval-side uses:

- **Selective stop-word filtering.** Not every occurrence of “it” is a pronoun. In the phrase “the IT department”, “IT” is a noun and should be kept; in “it is easy”, it should be dropped. POS-aware stop-word filtering keeps content-bearing occurrences and removes function-word ones. This is how the “IT security” case from the stop-word discussion in the previous section is handled correctly.
- **Lemmatizer disambiguation.** WordNet’s lemmatizer needs the POS tag: `wordnet.lemmatize("meeting", "n")` returns “meeting” while `wordnet.lemmatize("meeting", "v")` returns “meet”. Feeding the lemmatizer the wrong tag produces the wrong lemma.
- **Question-form analysis.** The query “Who is Albert Einstein?” has a WH-word (“who”), a copula verb (“is”), and a proper noun (“Albert Einstein”). Combining that structure with the recognition that “Albert Einstein” is a person’s name (see the next subsection) tells the retriever this is a person-lookup query, and the right response is not a keyword search but a lookup in a people database or a redirect to Wikipedia. The intent-routing section returns to this example.

Three families of taggers have been used historically:

- **Rule-based taggers** apply hand-written rules based on suffixes and surrounding words. Fast and simple, but each language needs its own rule set and rules cannot easily be extended.
- **Statistical (HMM) taggers** model the sentence as a sequence of hidden POS states emitting observed tokens. Transition and emission probabilities are learned from a POS-tagged corpus; decoding uses the Viterbi algorithm. Dominant approach from the 1990s to the mid-2010s.
- **Neural taggers** run a small transformer or LSTM over the token sequence and output a POS tag per token. This is the current default in spaCy and in most Hugging Face pipelines.

All three families reach around 97-99% accuracy on English news text. On informal genres (social media, product reviews) the neural taggers pull ahead because they handle out-of-vocabulary words better.

See Also

The Hidden Markov Model formulation used by statistical taggers is developed in the ML foundations appendix, alongside the Viterbi decoding algorithm. This section does not repeat the HMM math.

```

import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp("Who is Albert Einstein?")
for token in doc:
    print(f"{token.text:12s} {token.pos_:8s} {token.tag_:6s} {token.dep_}")

# Who      PRON      WP      nsubj
# is       AUX       VBZ     ROOT
# Albert   PROPN     NNP     compound
# Einstein PROPN     NNP     attr
# ?        PUNCT     .       punct
  
```

Two tag sets appear in practice. The `pos_` field uses the coarse Universal tag set (around 17 categories: NOUN, VERB, ADJ, ADV, PROPN, PRON, DET, ADP, ...). The `tag_` field uses the fine-grained Penn

Treebank tag set (around 45 categories: VBZ for third-person singular verb, NNP for proper noun, WP for WH-pronoun). The Universal tags are enough for retrieval-side filtering; the Penn tags are needed for building parse trees.

Named entity recognition

The tokens “Albert Einstein” in the previous example are not just two proper nouns; they name a specific person, and the retrieval system that recognizes them as such can act on that recognition. **Named entity recognition (NER)** classifies spans of tokens into entity types: person, location, organization, product, date, money, and so on. NER is a natural extension of POS tagging: it runs after tokenization and POS tagging (or jointly with them in modern neural pipelines) and outputs entity spans.

The approaches behind NER mirror the POS taggers above. The earliest systems were rule-based, combining gazetteers (lists of known people, places, and organizations) with hand-written patterns such as a capitalized word following a title like “Dr.” or preceding a suffix like “Inc.”. Statistical sequence models replaced them: a conditional random field (CRF) labels each token from features of the token and its neighbours, trained on annotated corpora such as CoNLL-2003. Modern systems fine-tune a transformer for token classification, tagging each sub-word with its entity type. This is the current state of the art and what spaCy’s larger models and Hugging Face NER pipelines use.

```
doc = nlp("Jack Higgins deposits £50,000 with BestBank in London.")
for ent in doc.ents:
    print(f"{ent.text:15s} {ent.label_}")

# Jack Higgins      PERSON
# £50,000           MONEY
# BestBank          ORG
# London            GPE
```

For query understanding, four categories dominate the retrieval-side use cases:

- **Person names.** “Who is Albert Einstein?” should trigger a person-lookup: prioritize Wikipedia, IMDb, MusicBrainz, and other person-oriented sources rather than running a keyword search across the whole web.
- **Time and date.** “Who won the F1 race last weekend?” contains a temporal expression. Retrieval should restrict to news articles from the past few days and rank recent items above older ones.
- **Locations.** “What to do in Basel?” should prioritize regional content and augment the results with a map. The location is a hard filter, not just another keyword.
- **Product brands.** “Where can I buy the latest iPhone?” should boost shopping and product-review pages, and can trigger a price-comparison widget.

Every NER category exposes a routing decision. The classifier and router are the subject of the next section; this section only extracts the entities themselves.

NER also generates candidate phrases automatically: consecutive tokens tagged as PERSON produce a bi-gram or tri-gram like “Albert Einstein” that should be indexed as a unit, exactly as in the previous section’s phrase-detection discussion. This is often faster and more targeted than PMI or LHR scoring over the whole corpus, because named entities are exactly the phrases that matter for entity-oriented search.

For tooling, spaCy and Stanza provide NER for many languages out of the box, nltk offers a lighter `ne_chunk`, and the Hugging Face token-classification pipeline exposes fine-tuned transformer models such as `dslim/bert-base-NER`. Beyond libraries, the major cloud providers offer NER as a managed API: Amazon Comprehend, Google Cloud Natural Language, and Azure AI Language all return typed entities (often alongside key phrases and relations) without training a model. This convenience is why they anchor many production document-processing pipelines, extracting parties and amounts from contracts, fields from invoices, or names and dates from scanned forms.

Chunking with rule-based grammars

For queries that do not fit neatly into a POS-plus-NER analysis, a lightweight **chunker** groups consecutive tokens matching a small grammar. The classic pattern is:

```
NP -> DET? ADJ* NOUN
```

which matches a noun phrase like “a red car” or “the lazy dog”. Chunking gives the retriever access to noun-phrase-level tokens without running a full parser. `nltk.RegexpParser` implements this cheaply:

```
grammar = r"NP: {<DT>?<JJ>*<NN>}"
chunker = nltk.RegexpParser(grammar)
tree = chunker.parse(nltk.pos_tag(nltk.word_tokenize("a red car")))
```

More elaborate grammars produce dependency parses that connect the noun phrase “red car” to the adjective “red” that modifies it. Stanford CoreNLP and spaCy’s dependency parser produce these for free once POS tags are available. For most retrieval scenarios noun-phrase chunking is enough.

Spell correction and “did you mean?”

Every query understanding pipeline has to handle typos. A user typing “Bücher von Goehte” (transposed letters) gets no results if the retriever demands an exact string match on “Goethe”, regardless of how good the rest of the pipeline is. The classical response is a two-part strategy: correct at both indexing time and query time.

- **At indexing time**, keep the original token but also add the auto-corrected form. A document containing the misspelling “Goehte” is indexed under both “Goehte” and “Goethe”, so users find it whether they type the correct or the incorrect spelling.
- **At query time**, run the query through the spell-checker. If a token is not in the dictionary, offer corrections and either search all of them silently or ask “Did you mean ‘Goethe’?”. Modern search engines do both: they auto-run the corrected query and let the user click back to the original spelling if the correction was wrong.

Spell correction is especially tricky for names. “Britney” is often misspelled as “Britny”, “Brittney”, or “Britnee”, which argues for folding them all back to “Britney”. The catch is that several of these forms are themselves legitimate names that other people genuinely carry, so a single variant can be at once a typo of one name and the correct spelling of another. A spell-checker that rewrites every variant to one canonical “Britney” then merges distinct people and loses anyone who is really spelled the other way. This is the “did you mean Britney, or Britnie?” problem. The pragmatic compromise is to keep the original spelling in the index and, at query time, expand to the common variants rather than silently rewriting to a single form.

Two standard algorithms underpin classical spell correction:

- **Edit distance** (Damerau-Levenshtein). Score candidate corrections by the minimum number of insertions, deletions, substitutions, and transpositions to reach the query. Widely used but slow to compute exhaustively; production systems restrict the candidate set with a phonetic prefilter or a small edit-radius trie.
- **Phonetic codes** (Soundex, Metaphone). Reduce each word to a compact code that captures its rough pronunciation. Different spellings of the same-sounding word share a code and are candidates for one another. Essential for name variants that no edit-distance approach can handle.

Query expansion and query rewriting

Several of the transformations described so far share a family resemblance. Synonym mapping, hypernym expansion, spell correction, alias substitution, and temporal normalization all take the tokens the user typed and change them so the resulting query matches more of the collection. In the IR literature these transformations fall under two umbrella names:

- **Query expansion** adds tokens to the original query without removing the ones the user typed. Synonym expansion, weighted-hyponym expansion, and adding all common spelling variants of a

name to a query all fall here. The original query terms usually stay at full weight and the added terms come in at a lower weight, so a document that matches the user's exact words still ranks above one that matches only via expansion.

- **Query rewriting** transforms the query text itself, replacing or restructuring tokens. Spell correction ("Goethe" -> "Goethe"), alias substitution ("F1" -> "Formula 1"), temporal normalization ("last weekend" -> a date range), and compound splitting on the query side ("Wolkenkratzer" -> "Wolken", "Kratzer") are query rewrites. The rewritten query goes to the retriever; the original may or may not be preserved for logging.

The line between the two blurs in practice. A rewrite that adds tokens alongside the originals (F1 OR "Formula 1" OR "Grand Prix") is an expansion in disguise. An expansion that later drops the original terms in favour of the added ones is a rewrite. What matters is that the retriever sees a different query from the one the user typed, and both directions are standard tools.

This chapter has been building both throughout:

- **Section 1** — language detection informs both expansion (which stop-word and synonym list to use) and rewriting (which stemmer to apply). Case, Unicode, and accent normalization are rewrites at the character level.
- **Section 2** — stemming and lemmatization are rewrites at the token level ("carried" -> "carri" for Porter, "carried" -> "carry" for WordNet).
- **Section 3** — compound splitting rewrites one token into several on both the document side and the query side.
- **Section 4** — synonym and hypernym mappings are expansions; spell correction is a rewrite.
- **Section 5** — temporal normalization and domain-alias substitution are rewrites applied during intent routing.

Two techniques not covered here belong in the same family and are worth naming so students recognize them elsewhere:

- **Pseudo-relevance feedback**, also called blind feedback. This is the relevance-feedback and query-expansion mechanism of the Binary Independence Model from [Probabilistic Ranking and BM25](#), run without any user judgements. Instead of asking the user which results are relevant, it assumes the top-*k* retrieved documents are relevant and applies the same term-discrimination weighting to choose expansion terms, the high-scoring terms like "woodland" in that chapter's worked example. It is a retrieval-model technique rather than a text-processing one, which is why it belongs with vector-space and BM25 ranking.
- **LLM-based query rewriting**. A language model reformulates the query. Common patterns include HyDE (write a plausible answer and search for that instead of the question), multi-query (generate several rewrites and union their results), and step-back prompting (generalize the query one level before searching). These belong to the retrieval-augmented generation chapter.

The chapter opener's F1 query is a plain example of query rewriting. "Who won the F1 race on the weekend?" is rewritten by dropping the WH-pronoun "who" and the verb "won" as non-content, expanding "F1" and "race" to "Formula 1" and "Grand Prix", and turning "the weekend" into a date range. What reaches the retriever is a scoped keyword query rather than a question, and the next section walks that rewrite through the full routing pipeline. Classical rewriting stops there: it can retrieve the right race-report page but cannot produce the short answer "Verstappen". That last step needs a reader or generator to extract the answer from the retrieved text which is covered in a later chapter (retrieval-augmented generation).

Where this sits in modern retrieval

Every classical technique in this section still runs somewhere in a modern search stack. Web search engines apply POS tagging, NER, and spell correction to every query before any retrieval or ranking. Product-search systems for e-commerce use NER heavily to extract brand names, category constraints, and numerical facets ("under \$50"). Enterprise search over document management systems relies on synonym expansion using domain-specific terminologies.

The one classical technique that dense embeddings largely subsume is synonym expansion: a well-trained embedding model puts “purchase” and “buy” near each other in vector space, so the query “purchase history” retrieves documents about “buying” without an explicit synonym list. Hypernym reasoning is partially learned but less reliably: an embedding model may or may not know that a labrador is a dog, depending on the training data. POS tagging, NER, and spell correction remain explicit steps even in fully neural pipelines because they produce structured outputs (tags, spans, corrected strings) that downstream systems can act on discretely. Large language models can substitute for all of them in a single prompt, but for high-throughput retrieval the classical stack is still cheaper by orders of magnitude.

The final section takes everything this section produces (language ID, POS tags, entities, corrections) and turns it into a routing decision: which backend should the query go to?

Intent Routing and Classification

Recall the second query from the chapter opener: “Bücher von Goethe”. By this point in the chapter, the pipeline has enough machinery to see it clearly. The tokenizer splits it into three tokens. The rule-based language detector recognizes German characters and function words. POS tagging identifies “Bücher” as a plural noun, “von” as a preposition, and “Goethe” as a proper noun. Named entity recognition promotes “Goethe” to a person entity. What has not been decided yet is what to do with all this structure. Should the query go to a book catalogue, a web index, an author database, or a general-purpose LLM? This section turns the extracted features into that routing decision.

The end-to-end query pipeline

A query enters the retrieval system as a raw string. Before any matching happens, it flows through the classical pipeline this chapter has been building:

```
raw query
-> tokenize (section 1)
-> normalize (case, Unicode) (section 1)
-> segment sentences (section 1, if multi-sentence)
-> detect language (section 1 rules + this section's classifier)
-> stop-word filter (section 2, language-specific)
-> stem or lemmatize (section 2, language-specific)
-> split compounds (section 3, if applicable)
-> POS-tag (section 4)
-> extract named entities (section 4)
-> spell-correct (section 4)
-> classify intent (this section)
-> route to backend (this section)
```

The order matters. Language detection has to come before any language-specific step; POS tagging has to come before selective stop-word filtering; NER has to come before intent classification because the entity types are input features to the classifier. What comes out at the end is a structured representation of the query that a downstream system can act on: a language, a set of entities, an intent label, and a corrected token list.

Two of these classification steps use the same underlying machinery. Language detection classifies a query into one of the languages the system knows; intent classification classifies a query into one of the backends the system supports. Both are text classification problems and both can be solved with the same Naive Bayes model. This section develops that model once, then applies it to both problems.

Naive Bayes for text classification

Naive Bayes chooses the most probable class given the observed features by applying Bayes’ theorem and assuming that features are conditionally independent given the class. The derivation lives in the ML foundations appendix; here we state only the decision rule.

For a feature vector $\mathbf{x} = (x_1, \dots, x_M)$ and classes $C_1, \dots, C_K$:

Three things need to be learned from a labelled training corpus:

- **The prior $P(C_k)$:** fraction of training documents in class C_k . When the training data reflects real-world class frequencies, use it directly; when it is artificially balanced (equal examples per class) or when class balance is unknown, set a uniform prior $P(C_k) = 1/K$ and the priors drop out of the argmax.
- **The likelihood $P(x_j \mid C_k)$:** fraction of feature occurrences in class C_k that fall on token or feature x_j . This is a maximum-likelihood estimate over the training data.
- **A smoothing constant.** If x_j never appeared in the training data for class C_k , the maximum-likelihood estimate is zero, and one zero collapses the whole product to zero. Add-one (Laplace) smoothing avoids this by adding a small constant to every count.

The product in the decision rule is a direct consequence of the independence assumption: taking the features to be conditionally independent given the class turns the joint likelihood $P(\mathbf{x} \text{ mid } C_k)$ into the product $\prod_j P(x_j \text{ mid } C_k)$ of per-feature terms. This is what makes the estimation possible at all. In classical text classification the feature space has tens or hundreds of thousands of dimensions, so any particular full vector $\mathbf{x}$ almost never recurs in the training data, and $P(\mathbf{x} \text{ mid } C_k)$ could never be estimated directly. Each individual dimension x_j , by contrast, is observed many times across the corpus, so the per-feature factors $P(x_j \text{ mid } C_k)$ can be counted reliably. Independence trades a joint probability we cannot measure for a product of marginals we can.

The assumption is not literally true, since tokens in real text are not independent, but Naive Bayes still performs well on text because the errors it introduces tend to affect all classes similarly and rarely change which class wins the argmax. Its speed and small memory footprint are the reasons it stays in production stacks two decades after neural classifiers became viable.

Language detection as classification

Language detection uses character-based n-grams as features. For n from 1 to 5, count how many times each n-gram appears in the input text. This produces a bag-of-n-grams representation exactly like the bag-of-words representation used for topic classification, except that the tokens are short character strings instead of whole words.

Character n-grams work well as features because each language has characteristic short sequences that rarely appear in other languages:

- German trigrams: sch, cht, hen, und, der, die
- English trigrams: the, ing, and, ion, ent
- French trigrams: les, des, que, ent, oui
- Italian trigrams: che, non, per, ent

Some trigrams like ent appear in multiple languages but with different frequencies, which is enough for a Naive Bayes classifier to disambiguate.

Setting up the classifier:

- **Classes:** the target languages. lingua covers 70 languages; a domain-specific detector might cover only 5 or 10.
- **Features:** character n-grams from a per-language frequency profile built during training. Most of the discriminating signal sits in a modest set of frequent n-grams, so a frugal detector can store only those significant likelihoods and stay tiny. lingua in high accuracy mode keeps the full observed profile across five n-gram orders (unigram through fivegram): the English model holds about 380,000 n-grams (~ 3.3 MB), German about 470,000 (~ 4.2 MB). It trades size for robustness on short and ambiguous inputs rather than pruning to the discriminating core.
- **Prior:** uniform when the incoming text has unknown language distribution; empirical when the retrieval system knows its user base is 60% English, 30% German, 10% French.
- **Likelihood:** for the multinomial variant, $P(t_j \text{ mid } C_k) = n_{k,j} / \sum_l n_{k,l}$ where $n_{k,j}$ is the count of n-gram j in the training text for language k . With +1 smoothing, $P(t_j \text{ mid } C_k) = (n_{k,j} + 1) / (\sum_l n_{k,l} + V)$ where V is the vocabulary size.

The decision rule then takes the standard log-form:

$$\hat{C} = \arg \max_k \left(\log P(C_k) + \sum_j c_j \log P(t_j \text{ mid } C_k) \right) \quad (3.4)$$

where c_j is the observed count of n-gram j in the input.

For very short queries with only one or two trigrams that discriminate between languages, confidence drops. lingua reports its confidence as a normalized posterior, and the retrieval system can fall back to a default language (English, or the user's browser locale) when the top confidence is too close to the second-best.

```
from lingua import Language, LanguageDetectorBuilder

detector = LanguageDetectorBuilder.from_all_languages().build()
detector.detect_language_of("Bücher von Goethe")
# Language.GERMAN

# Restrict to expected languages and get confidence scores
euro = [Language.ENGLISH, Language.GERMAN, Language.FRENCH, Language.ITALIAN]
d2 = LanguageDetectorBuilder.from_languages(*euro).build()
d2.compute_language_confidence_values("Bücher von Goethe")
# GERMAN: 0.96, ENGLISH: 0.02, FRENCH: 0.01, ITALIAN: 0.01
```

langdetect is a lighter alternative: rule- and n-gram-based, 55 languages, ISO-code output.

```
from langdetect import detect
detect("Bücher von Goethe") # 'de'
detect("Livres de Goethe")  # 'fr'
detect("Books by Goethe")   # 'en'
```

Intent routing as classification

The same machinery routes the query to a backend. Once the language is known, an **intent classifier** decides which of the search system’s supported intents the query expresses:

- `book_search`: query looks for books (title, author, ISBN)
- `web_search`: general-purpose keyword search
- `image_search`: query looks for images
- `product_search`: query looks for shoppable products
- `people_search`: query looks for information about a person
- `news_search`: query has a temporal component or a current-events subject
- `map_search`: query looks for a location or directions
- `weather`, `calculator`, `definition`, `translation`: one-shot vertical intents

The training data is query logs annotated with the backend the user actually clicked into. Features are richer than for language detection because more of the pipeline’s output can be used:

- The token bag after stemming and stop-word removal
- POS-tag counts (how many nouns, verbs, WH-words)
- NER labels present in the query (has-PERSON, has-LOCATION, has-DATE, has-MONEY)
- Question-form markers (starts with a WH-word, ends with a question mark)
- Detected language (as a categorical feature)

With these features, “Bücher von Goethe” scores highly on `book_search` because it contains a book-domain plural noun (“Bücher”) and a PERSON entity (“Goethe”) in an author-attribution construction. “What to do in Basel?” scores highly on `map_search` and `web_search` because it contains a WH-word, a LOCATION entity, and no product or person entity. “Who is Albert Einstein?” scores highly on `people_search` because of the WH-word combined with a PERSON entity.

The classifier itself is exactly the same Naive Bayes model as for language detection, just with different features and different classes. In practice, a modern production stack fine-tunes a small neural classifier on top of the same features because it captures conjunctions (“has-PERSON AND WH-word AND is-question”) that Naive Bayes cannot express under the independence assumption. In a later chapter, we study the neural variants using LLM to make the routing decision (agentic RAG).

End-to-end walkthrough

Putting all five sections of the chapter together on the running query:

```
Input:          "Bücher von Goethe"

Tokenize:       ['Bücher', 'von', 'Goethe']
```

```

Normalize (case):  ['bücher', 'von', 'goethe']
Detect language:   German (confidence 0.96)
Stop words:        drop 'von' -> ['bücher', 'goethe']
Stem (Snowball DE): ['buch', 'goeth']
POS tag:           [(Bücher, NOUN), (Goethe, PROPN)]
NER:               [(Goethe, PERSON)]
Spell-check:       no correction needed
Intent classify:    book_search (features: language=DE, PERSON present,
                                plural NOUN of book-domain lemma)
Route:             library catalog with filters
                   language = "de"
                   author   = "Goethe"
                   content   = *

```

The library catalog answers a structured query with all Goethe titles it holds. No general-purpose keyword match against “Bücher von Goethe” was ever needed.

A second walkthrough on the other chapter-opener query. This one combined a WH-question form, tokens that would not appear in the answer, and a domain-vocabulary gap.

```

Input:             "Who won the F1 race on the weekend?"

Tokenize:          ['Who', 'won', 'the', 'F1', 'race', 'on',
                    'the', 'weekend', '?']
Normalize (case):   ['who', 'won', 'the', 'f1', 'race', 'on',
                    'the', 'weekend']
Detect language:    English
POS tag:            [(who, WH-PRON), (won, VERB), (the, DET),
                    (F1, PROPN), (race, NOUN), (on, ADP),
                    (the, DET), (weekend, NOUN)]
NER:               [(F1, EVENT), (the weekend, DATE)]
Selective stop-word filter: drop 'the', 'on'
                                keep 'who' as a classification feature
                                (WH-word signals question form)
Stem/lemma (English): ['win', 'F1', 'race', 'weekend']
Synonym expansion (domain): F1    -> {Formula 1, Grand Prix}
                             race  -> {Grand Prix, GP}
Temporal normalization: the weekend -> [Sat, Sun of the past weekend]
                             -> ['2026-08-08', '2026-08-09']
Intent classify:     news_search
                    features: WH-word + EVENT entity +
                               DATE entity + motorsport terms
                    features against 'question' subclass:
                               WH-word + won + who
Route:              news index with
                    query = "Formula 1" OR "Grand Prix" OR "GP"
                    date  = [2026-08-08, 2026-08-09]
                    sort   = date desc

```

The classical pipeline turned three tokens of question form and one date reference into a scoped keyword query the news index can answer: a Boolean OR over motorsport aliases, a date filter for the past weekend, and a sort by date. Each of the three gaps named in the chapter opener has a specific step above that addresses it. The missing “who” is absorbed into the classification features rather than searched as a keyword. The missing “weekend” becomes a temporal range that a race-results page dated in that range will match. The missing “race” is replaced by the domain-synonym expansion.

That is as far as this chapter takes the F1 example. What the news backend returns is a ranked list of race-result pages, not the two-word answer “Verstappen won” that the user actually requested. Two later chapters carry the example the rest of the way:

- The **semantic search** chapter shows how a dense-retrieval index closes the domain-vocabulary gap without a hand-maintained synonym list, catching the driver names, sponsor terms, and circuit aliases the classical stack never saw.
- The **retrieval-augmented generation** chapter runs a language model over the top-ranked page and returns the extracted name as the answer, turning a list of results into a direct answer.

The classical pipeline of this chapter is still there in both cases. It prepares the query and prunes the search space so the more expensive downstream steps only see the pages that matter. The router itself does not do the retrieval; it picks a backend and passes the structured query to it. Everything upstream in this chapter was building the structure that the router now dispatches on.

Where this sits in modern retrieval

Classical intent routing runs in every large search stack in 2025. Google Search, Amazon Search, and enterprise search products all have an intent-classification stage before the retriever. The classifier is rarely still Naive Bayes: production systems have moved to gradient-boosted trees or small transformers because the feature interactions matter and the classifier is on the query-latency critical path only in a few-millisecond budget. Naive Bayes remains the pedagogical baseline and a common first pass in cost-sensitive deployments (mobile, embedded).

Large language models can perform the whole pipeline in one prompt: extract entities, detect the language, classify the intent, and generate the structured backend query. This is a live area of production adoption, especially for open-ended assistant systems. The trade-offs against classical pipelines are latency (an LLM call is 100-1000 milliseconds versus sub-millisecond for the classical stack), cost (per-query LLM cost is measurable, per-query classical cost is not), and confidence quantification (Naive Bayes gives calibrated posteriors, LLMs do not). Production stacks in 2025 usually run the classical pipeline first, escalate to LLMs for the queries where the classical stack's confidence is low, and reserve LLMs for high-value verticals like agentic search and complex reformulation.

Two forward references pick up where this section leaves off:

- The text classification chapter covers deep-learning text classification with TextCNN and transformer-based classifiers, using sentiment analysis as its running example. The Naive Bayes classifier of this section is the baseline those methods are compared against.
- The semantic-search chapter covers embedding-based intent classification: instead of extracting features from the query and running Naive Bayes, embed the query into a vector space and compare it to prototype vectors for each intent. Modern hybrid systems combine both approaches.

The next page summarizes the chapter and points to the material that follows it in the book.

Summary

Method Comparison

Stemmers and lemmatizers side by side.

Method	Approach	Correctness	Speed	Multi-lingual	Needs dictionary
Porter	Rule-based suffix stripping (English)	pseudo-stem	fast	English only	no
Lancaster	Aggressive rule-based (English)	pseudo-stem, over-stems short words	fastest	English only	no
Snowball	Rule-based framework	pseudo-stem	fast	20+ languages	no
WordNet	Dictionary + POS lemmatizer	linguistic lemma	slow	English only (multilingual variants weaker)	yes
spaCy	Neural POS tagger + dictionary lemmatizer	linguistic lemma	medium	many languages	yes

Use a rule-based stemmer when performance dominates or when no dictionary is available for the language. Use a dictionary-based lemmatizer when accuracy on strongly-inflected languages matters (German case-inflected nouns, English strong verbs like “went -> go”) or when the downstream task needs linguistic base forms rather than arbitrary reductions. Snowball is the go-to compromise for multi-language corpora when only rule-based options are practical.

Key Takeaways

1. Classical retrieval fails in two directions: one concept has many surface forms on the document side (recall gap), and free-text queries carry structure that the retriever ignores (query understanding gap). Advanced text processing addresses both.
2. Tokenization is the first stage and the largest source of quality issues. A naive regex tokenizer breaks on possessives, numbers, abbreviations, and punctuation. Modern nltk and spaCy tokenizers handle these but still need retrieval-oriented cleanup.
3. Normalization decides which surface forms collapse to the same token. Case folding is nearly always applied. Unicode normalization is applied silently. Accent folding trades recall for precision and depends on the scenario.
4. Sentence segmentation is a small but essential step. Every RAG chunking strategy, every POS tagger, and every query-analysis pipeline depends on being able to find sentence boundaries reliably.
5. Stop-word removal is a size-versus-recall trade-off. Modern BM25-based systems handle stop words gracefully through IDF and can afford to keep them in the index. Legacy vector-space systems remove them aggressively. Phrase queries and titles like Stephen King’s “It” break under aggressive removal.
6. Stemming is a fast pseudo-linguistic reduction; lemmatization is a slower dictionary-based one that produces the true linguistic base form. Choose based on speed budget, language complexity, and whether downstream steps need real words.
7. Phrases and compounds sit on opposite sides of the word-boundary problem. Bi-grams like “New York” pack multiple tokens into one concept and need explicit phrase indexing; PMI and LHR are the classical selectors for which bi-grams are worth adding. Compounds like “Bücherregal” do the reverse, hiding multiple concepts in one token, and are split into their parts by log-frequency scoring, though only endocentric compounds split safely (a “Bücherregal” is a kind of “Regal”; a “Wolkenkratzer” is not a kind of “Kratzer”, so splitting it hurts precision).

8. Query understanding turns free text into structured features: language, POS tags, named entities, corrected spelling. Together they let the retriever route the query to the right backend and generate structured queries rather than doing bag-of-words matching.
9. A Naive Bayes classifier applied to hand-engineered features is enough to build both a language detector and an intent classifier. Modern production stacks use neural classifiers for accuracy but keep Naive Bayes as the baseline and as the fast path for cost-sensitive deployments.
10. Every classical technique in this chapter still runs in production 2025 search stacks. Dense retrieval subsumes some of them (synonym expansion) implicitly, but hybrid retrieval combines both branches and needs the classical stack on the lexical side.

Key Formulas

$$\text{pmi}(t_1, t_2) = \log_2 \frac{N \cdot \text{tf}(t_1, t_2)}{\text{tf}(t_1) \cdot \text{tf}(t_2)} \quad (3.5)$$

Pointwise Mutual Information: high when two tokens almost always appear together and rarely apart. Biased toward rare-word bi-grams; requires a minimum-frequency filter.

$$\log \lambda = \log \frac{L(H_1)}{L(H_2)} \quad (3.6)$$

Log Likelihood Ratio: log-ratio of the probability under independence to the probability under dependence. Robust on sparse data; ranks by $-2 \log \lambda$.

$$\text{score}(S) = \frac{1}{|S|} \sum_{p_i \in S} \log \frac{\text{tf}(p_i)}{N} \quad (3.7)$$

Compound Split Score: average log-frequency of the parts in split S . Choose the split with the highest score.

$$\hat{C} = \arg \max_k \left(\log P(C_k) + \sum_{j=1}^M \log P(x_j \text{ mid } C_k) \right) \quad (3.8)$$

Naive Bayes Maximum A Posteriori: pick the class that maximizes the log-prior plus the sum of log-likelihoods. Used for language detection over character n-grams and for intent classification over feature vectors combining tokens, POS tags, and NER labels.

Self-Check Questions

1. (Understand) The naive tokenizer produces ['Watson', 's'] from “Watson’s”. Under what retrieval scenario is this desirable, and under what scenario is it a problem? What information is permanently lost?
2. (Understand) Why does BM25 tolerate stop words that appear in the index better than a raw vector-space TF-IDF model does?
3. (Analyze) Given the *A Study in Scarlet* corpus where “said” occurs 207 times, “Holmes” 98 times, “Sherlock” 52 times, “Sherlock Holmes” 52 times, and “said Holmes” 12 times, compute PMI for the two bi-grams “Sherlock Holmes” and “said Holmes”. Explain why the ranking matches (or does not match) what a human would call “the more meaningful phrase”. Assume a corpus of $N = 43,200$ tokens.
4. (Analyze) Explain why aggressive accent folding hurts a legal-document retrieval system for German but helps a general-purpose web search engine over the same language.
5. (Analyze) A retrieval system indexes “Wolkenkratzer” only as itself (no compound splitting). A user queries “Wolke”. A different user queries “Kratzer”. Which query, if any, retrieves documents about skyscrapers, and why? Would splitting help both users equally? Would you split this specific compound?
6. (Evaluate) A production search team proposes replacing the entire classical query pipeline (tokenization, stemming, POS, NER, intent classification) with a single call to a large language model that outputs a structured query directly. State two reasons the team might keep the classical pipeline anyway, and one query type where the LLM call is likely to win.

7. (Apply) A user queries “Livres de Molière”. Walk through the end-to-end pipeline from section 5, listing the output of each step and the final routing decision. Assume the search stack supports French.

Further Reading

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- Kiss, T., & Strunk, J. (2006). **Unsupervised Multilingual Sentence Boundary Detection**. *Computational Linguistics*, 32(4), 485-525. [Free PDF](#). The Punkt algorithm, still the sentence-segmentation default in NLTK almost two decades later.
- Koehn, P., & Knight, K. (2003). **Empirical Methods for Compound Splitting**. *Proceedings of the 10th Conference of the European Chapter of the ACL*, 187-193. [Free PDF](#). The frequency-based compound-splitting method this chapter uses, evaluated for machine translation but applicable directly to retrieval.
- Minixhofer, B., Pfeiffer, J., & Vulić, I. (2023). **CompoundPiece: Evaluating and Improving Decomounding Performance of Language Models**. [arXiv:2305.14214](#). Shows that modern subword tokenizers (SentencePiece, BPE) still decompound German poorly, motivating why explicit decomponers remain in hybrid retrieval stacks.

Implementations and tools

- [NLTK](#) — Python toolkit for classical NLP; ships all the stemmers, tokenizers, POS taggers, and collocation finders used in this chapter.
- [spaCy](#) — Industrial-strength NLP library with neural taggers and lemmatizers for many languages.
- [WordNet](#) — Lexical database for English synonyms, hypernyms, and hyponyms.
- [Snowball](#) — Multi-language stemmer framework, source of the German and French stemmers used in section 2.
- [lingua-language-detector](#) — Character-n-gram Naive Bayes language detector covering 70+ languages.
- [Elasticsearch decompounder documentation](#) — Production compound-splitter configuration for German, Dutch, and other compounding languages.

4 - Index for Text Retrieval

By now we have most of the pieces of a text retrieval system. We can turn documents and queries into terms through tokenization and normalization, and we can say what a good result looks like using precision and recall. Chapter 1 gave us models that score how well a document matches a query, from Boolean matching to the vector space model and BM25. One practical question remains: how do we find the matching documents quickly when the collection holds millions of them? Scoring every document against every query, one after another, does not scale.

Fast retrieval is a balancing act between three competing goals: the quality of the ranked results, the latency of each query, and the storage the index consumes. The inverted index is the data structure that has held this balance for decades. It made full-text search practical on the modest hardware of the 1970s and 1980s, and it still sits at the core of today's search engines. Its defining property is that the work per query grows with the number of query terms, not with the size of the collection. Only recently have dense vector methods reached comparable speed at large scale, and even then they are often paired with an inverted index rather than replacing it. We return to dense retrieval in later chapters.

This chapter opens up the mechanics. We start from a small example and build the inverted file, then show how the same structure serves the retrieval models from Chapter 1. From there we add metadata filters and faceted search, and we look at how compression keeps the index small. We close with practical, low-cost ways to run production-grade search: full-text search inside a database you may already operate, the Lucene library, and the distributed engines built on top of it.

Inverted Files

Consider a tiny collection of three documents:

- D_1 : “The cat sat on the mat”
- D_2 : “The dog chased the cat”
- D_3 : “The cat and the dog played”

A query for “cat” should return D_1 , D_2 , and D_3 ; a query for “dog” should return D_2 and D_3 . With three documents we could simply read each one and check. With three million documents we cannot. We need a structure that takes us straight from a query term to the documents that contain it, without touching the rest of the collection. That structure is the inverted index, and this section explains why it makes query cost depend on the query rather than on the size of the collection.

Why scanning does not scale

Recall from [Classical Text Retrieval](#) that a document is represented as a sparse vector over the vocabulary. Assume a collection of N documents and a vocabulary of M terms. On average a document contains K distinct terms, with K much smaller than M , and a typical query contains about L terms, with L much smaller than K (five query terms is a common figure).

The most direct storage scheme reserves one entry for every term in every document: $N \cdot M$ entries in total. In the set-of-words model each entry is a single bit that records whether the term occurs; the bag-of-words model stores a term frequency instead. Almost all of these entries are zero, because each document uses only a small fraction K/M of the vocabulary. Storing the full matrix is therefore wasteful, and a sparse layout that keeps only the K non-zero entries per document reduces storage from $N \cdot M$ to $N \cdot K$ entries.

The sparse layout saves space, but it does not save time. To answer a query we still scan every document, and for each one we read terms that the query never asked about. Here we rely on a property of the classical models: Boolean retrieval, the vector space model, and BM25 all score a document from its query terms alone, under the assumption that terms contribute independently. Only the entries for the query terms affect the result, so almost everything we read is discarded. This is exactly the property the inverted index will exploit.

This property is what makes classical retrieval fast, and it is also its limit. It does not hold for dense retrieval, covered in [Semantic Search](#), where a document’s relevance depends on its whole embedding rather than on the presence of individual query terms. A dense system cannot restrict its attention to a handful of postings, which is why classical lexical retrieval remains orders of magnitude faster than dense retrieval and why the two are often combined.

Inverting the layout

Since only the query terms matter, we can organize storage around terms instead of documents. Rather than storing, for each document, the list of terms it contains, we store, for each term, the list of documents that contain it. This is the inverted index, also called the inverted file. Each term points to a **postings list**: the document identifiers in which the term appears. [Figure 4.1](#) shows the inverted index for the three-document collection above.

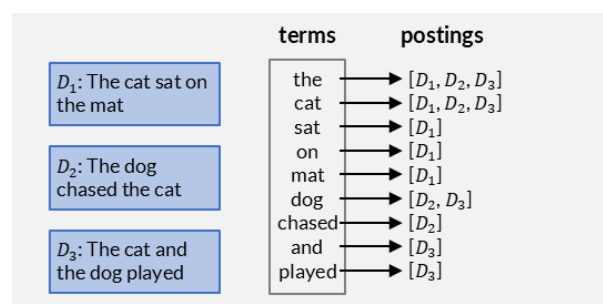


Figure 4.1: An inverted index over a three-document collection: each term in the vocabulary points to its postings list.

Inverting the layout does not change how much we store. There are still $N \cdot K$ term-document pairs. What changes is how much we read per query. A query touches only the postings lists of its query terms. If the $N \cdot K$ pairs are spread over M terms, an average postings list holds $N \cdot K/M$ entries, so a query of L terms reads about $N \cdot K \cdot L/M$ entries instead of all $N \cdot K$.

The read cost is proportional to $N \cdot K$, so inverted files scale linearly with the collection: double the documents and both the index and the average read roughly double. For small to large collections this is comfortable. The book and code indexes, under a gigabyte and around a gigabyte and a half, fit entirely in memory on one machine, so a query with a few keywords is answered almost instantly. The average-term assumption behind those reads often overstates the cost here, because someone searching a book catalog is usually after a specific title and types rare, discriminative words whose postings lists are shorter than average. Under practical conditions, inverted files work very well across this whole range.

The trouble is common words. Stop words and other high-frequency terms have postings lists far longer than average, and they are what makes large collections expensive. On the book and code collections they are a nuisance rather than a blocker: we can drop stop words from the query, searching “Lord of the Rings” as “lord” and “ring” while ignoring “of” and “the”, and lean on the rare, discriminative terms. At web scale the same query behaves differently. A stop word on half of 10 billion pages carries roughly 19 GB of postings on its own, and even after dropping it, “lord” and “ring” each occur in hundreds of millions of pages, some close to 1 GB of postings apiece. No amount of stop-word removal rescues a plain postings scan when the discriminative terms are themselves frequent.

The web index also no longer fits on one machine at more than 20 TB, so it lives mostly on disk and must be distributed. Inverted files degrade gracefully rather than suddenly, remaining practical across a wide range of sizes, but the combination of an enormous collection and unavoidably frequent terms is what eventually breaks the simple scheme. Keeping web-scale queries fast despite gigabyte postings lists is the job of the additional strategies we return to later: skip pointers and early termination, pruned and tiered indexes, caching, the sharding, and scaling out.

Query length is a cost lever in the same way, which matters for query expansion. Expanding a query, introduced in [Advanced Text Processing](#), adds related terms to raise recall, but every added term is another postings list to read and merge, and the extra terms add noise that can lower precision. On a large collection, where a single frequent term can already cost gigabytes, expansion has to be weighed against query cost, not against precision alone.

The structure of the index

An inverted index has three parts. The **vocabulary** (or dictionary) holds the M distinct terms and serves as the lookup key. Each term points to its **postings list**. A separate **document table** stores per-document metadata, such as a title, author, or URL, that the ranker and the result page need but that is not itself searched term by term.

The two tables below make this concrete, using book titles from the library collection we return to throughout the chapter. The document table records one row per document, keyed by an integer identifier:

ID	Title	Author	Year
1	Database Systems: The Complete Book	Garcia-Molina, Ullman, Widom	2008
2	Pattern Recognition and Machine Learning	Christopher Bishop	2006
3	Data Science from Scratch	Joel Grus	2015
4	Operating System Concepts	Silberschatz, Galvin, Gagne	2018
5	Artificial Intelligence: A Modern Approach	Russell, Norvig	2020
...	...	...	...
50	Frankenstein	Mary Shelley	1818

The inverted index maps each term in those titles to the sorted list of document identifiers where it occurs:

Term	Postings (document IDs)
computer	[9, 10, 12, 14]
design	[8, 9]
introduction	[7, 15]
software	[9, 13]
systems	[1, 14]
the	[1, 7, 8, 9, 13, 16, 19, 20, 26, 28, 31, 32, 35, 38, 43, 47]
...	...

The row for term the illustrates the earlier point about common words: even in a 50-title collection, the stop word “the” has a longer postings list than any content term.

For Boolean retrieval over the set-of-words model, the postings lists need only the document identifiers; term frequencies and document frequencies are not required. As documents are added, their identifiers are appended to the postings of the terms they contain. When documents arrive in order, each postings list stays sorted by increasing document identifier. That sorted order is not incidental: it is what makes the evaluation below efficient.

Boolean retrieval over postings

A Boolean query combines terms with AND, OR, and NOT. Because each postings list is a set of document identifiers, the operators map directly onto set operations:

- `expr1 AND expr2`: the intersection of the two postings sets.
- `expr1 OR expr2`: the union of the two postings sets.
- `expr1 AND NOT expr2`: the set difference, the documents of `expr1` that are not in `expr2`.

Nesting AND and OR combines these operations, and the same rules generalize to more than two operands. The NOT operator needs care. We never materialize `NOT expr2` on its own, because its complement can contain almost the whole collection. A pure negation, or an OR with a negated operand such as “cat OR NOT dog”, would force us to enumerate millions of identifiers. Such a query is also rarely what a user means: “cat OR NOT dog” selects every document except those that contain “dog” but not “cat”. We therefore allow NOT only inside an AND that has at least one non-negated operand, and we apply the negated parts last, as a subtraction from the candidates found so far.

In code this is remarkably direct: an inverted index is a dictionary from terms to postings sets, and the Boolean operators are the language’s own set operators. Taking an artificial example collection:

```
cat  = index['cat']      # [4, 5, 12, 13, 14, 15, 20, 22, 30, 34]
dog  = index['dog']      # [1, 3, 4, 6, 9, 10, 13, 21, 22, 23, 29, 30]
horse = index['horse']   # [6, 10, 11, 14]
bird  = index['bird']    # [2, 3, 8, 15, 26, 35, 36]

cat & dog      # cat AND dog      -> [4, 13, 22, 30]
horse | bird   # horse OR bird    -> [2, 3, 6, 8, 10, 11, 14, 15, 26, 35, 36]
cat - dog      # cat AND NOT dog  -> [5, 12, 14, 15, 20, 34]

# Queries nest freely, because each operation returns another set of IDs:
(cat & dog) | ((horse & cat) - bird) # (cat AND dog) OR (horse AND cat AND NOT
bird)                                # -> [4, 13, 14, 22, 30]
(cat | dog) & (horse | bird)         # (cat OR dog) AND (horse OR bird)
# -> [3, 6, 10, 14, 15]
(cat | dog) - (horse | bird)         # (cat OR dog) AND NOT (horse OR bird)
# -> [1, 4, 5, 9, 12, 13, 20, 21, 22, 23,
29, 30, 34]
```


The last two lines show the permitted use of NOT: negation applied to the result of an AND, never on its own. This set-based version is clear but reads every full postings list into memory. The next step scales it down.

Merging sorted postings as streams

Loading whole postings lists into memory does not scale to lists with millions of entries. Instead we read them as sorted streams and merge them, advancing one entry at a time, much like merging sorted lists. Suppose the postings are:

- cat: 1, 4, 10
- dog: 3, 4, 8, 10, 12

To evaluate “cat AND dog” we look at the head of each stream and always advance the smaller one. When both heads are equal, that document satisfies the AND, so we emit it and advance both streams.

cat head	dog head	comparison	action	output
1	3	$1 < 3$	advance cat	
4	3	$4 > 3$	advance dog	
4	4	equal	emit, advance both	4
10	8	$10 > 8$	advance dog	
10	10	equal	emit, advance both	10
exhausted	12	cat empty	stop	

The result is the sorted list 4, 10. The merge stops as soon as one stream is exhausted: once cat runs out, no later document can match, even though dog still has the entry 12.

The other operators follow the same pattern with a different emit rule. For “cat OR dog” we emit the smaller head at each step and advance the stream it came from, producing the union 1, 3, 4, 8, 10, 12. For “cat AND NOT dog” we emit a cat entry only when it is strictly smaller than the current dog head, producing 1. Because every operator consumes sorted streams and emits a sorted stream, the operators compose: the output of one merge feeds directly into the next.

The whole scheme depends on the postings being sorted, and we get that ordering for free as long as we only ever append documents with increasing identifiers: a new document’s identifier is larger than any already in a list, so it simply goes at the end. Deletions and updates would break this. Removing a document leaves a gap, and changing one alters its terms, so the tidy append-only order no longer holds without rewriting postings lists in place. Many systems avoid that cost by refusing in-place edits. Lucene, for example, treats its index as append-only: every new document is assigned the next identifier and is never modified afterwards, a deletion only flips a marker in a per-segment list of live documents so that the document is filtered out of results, and an update is a delete-by-term, which removes whatever document matches a chosen key term such as a unique identifier, followed by a fresh insert. Documents flagged as deleted are physically removed only later, in bulk, when segments are merged, which we revisit in [Practical Frameworks](#). This design has a valuable side effect. Because existing postings are effectively static, writers and readers barely need to coordinate: new documents are flushed into new segments while queries keep running against the existing ones, so the postings a reader is scanning never shift underneath it.

From matching to ranking

Boolean evaluation answers a yes-or-no question: does a document satisfy the query? It produces a candidate set, but no order within it. For anything beyond exact filtering we want the best documents first, which means scoring each candidate. That splits retrieval into two stages: a **retriever** that uses the inverted index to gather candidates, and a **ranker** that scores them. The next section shows how the ranked models from Chapter 1, the binary independence model, the vector space model, and BM25, all run over the same postings lists.

Ranked Retrieval over Inverted Files

Boolean retrieval returns an unordered candidate set. The ranked models from [Classical Text Retrieval](#), the binary independence model (BIR), the vector space model, and BM25, go further: they assign each document a score so we can return the best matches first. This section shows that all three run over the same inverted index. What changes is the content of the postings and the function that turns them into a score.

Retriever and ranker

Ranked retrieval separates into two stages, shown in [Figure 4.2](#). The **retriever** uses the inverted index to gather candidates, taking the union of the postings lists of the query terms, the same merge as a Boolean OR. The **ranker** then scores each candidate with the model's scoring function and keeps the top results. An optional filter can drop candidates that fail a metadata condition before ranking; we return to filtering in the next section.

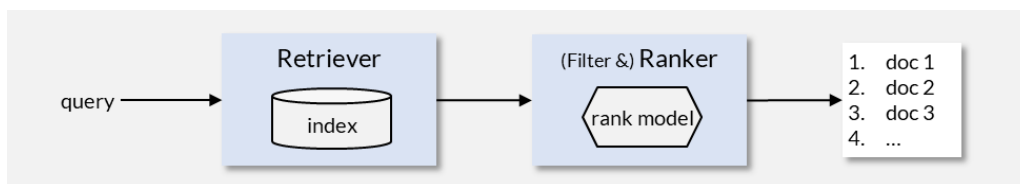


Figure 4.2: The two-stage pipeline: a retriever gathers candidates from the index, and a filter-and-ranker scores them into a ranked list.

The retriever only needs candidates with at least one query term, because a document with no query term scores zero under all three models. This is the same efficiency argument as before: we score a small candidate set, not the whole collection.

Two evaluation strategies

There are two natural orders in which to walk the postings and accumulate scores. We show both using the BIR model because its scoring function, a sum of per-term weights c_j , is additive and involves no term frequencies or document lengths, so the evaluation logic is fully visible without extra lookups.

Term-at-a-time (TAAT)

The most direct extension of the set-based Boolean evaluation from the previous section processes one query term fully before moving to the next. For each term we walk its postings list and add that term's c_j weight to a running score for the document. After all terms are processed, the score dictionary holds every candidate's final score, and we extract the top k .

```

def search_TAAT(query, k):
    query_vector = analyzer.set_of_words(query)
    # filter terms and obtain c_j-weights
    # returns a list of pairs (term_j, c_j)
    term_weights = query_weights(query_vector)

    scores = defaultdict(int)

    # iterate over terms and their postings
    for (term, weight) in term_weights:
        for doc_id in index[term]:
            # add this term's c_j to the document's running score
            scores[doc_id] += weight

    # avoid a full sort: use a heap to extract the top k
    topk = TopKList(k)
    for doc_id, score in scores.items():
        topk.add(doc_id, score)
    return topk
  
```

TAAT mirrors what we did with Boolean sets: read one term's postings, then the next, then combine. The difference is that instead of intersecting or uniting, we accumulate weights. Its simplicity comes at a cost: the scores dictionary can grow very large when a query contains common terms with long postings lists, and we cannot prune early because a document's score is incomplete until the last term is done.

Document-at-a-time (DAAT)

DAAT instead extends the sorted-stream merge from the end of the previous section. Rather than processing one term fully, it advances all query-term streams in parallel and produces one candidate at a time, in document-ID order. Each candidate receives its complete score the moment it appears, so we can feed it directly into a bounded top- k heap.

```
def search_DAAT(query, k):
    query_vector = analyzer.set_of_words(query)
    # filter terms and obtain c_j-weights
    term_weights = query_weights(query_vector)

    # get iterators for each term and fetch the first posting
    iters = [iter(index[term]) for (term, _) in term_weights]
    nexts = [next(it, None) for it in iters]

    topk = TopKList(k)

    while not all(e is None for e in nexts):
        # the smallest doc ID across all stream heads
        smallest = min(nexts, key=lambda x: x or math.inf)

        # score: sum c_j for every term whose head equals smallest
        score = 0
        for j in range(len(nexts)):
            if nexts[j] == smallest:
                score += term_weights[j][1]
        topk.add(smallest, score)

        # advance every stream whose head was the smallest
        for i, e in enumerate(nexts):
            if e == smallest:
                nexts[i] = next(iters[i], None)

    return topk
```

DAAT reads postings as streams and never builds a full score dictionary. Memory is bounded by the heap size k . Because the full score is known the moment a document is emitted, DAAT can also prune: once the heap is full, it can skip candidates whose maximum possible score cannot enter the top k (techniques such as WAND exploit this). The price is slightly more complex bookkeeping in the inner loop.

Comparison

Both strategies read the same postings and consider the same candidates, so their asymptotic cost is similar. TAAT is the clearer extension of the Boolean set approach and a fine choice for short queries. DAAT is generally preferred in production because of its bounded memory and its ability to prune.

The three models over postings

The models differ only in what the postings store and how a candidate is scored.

For **BIR**, the postings need only document identifiers. Each query term t_j carries a weight c_j derived from relevance feedback, and a document's score is the sum of the c_j over the query terms it contains. No term frequencies or document lengths are involved.

For the **vector space model**, the DAAT and TAAT patterns remain the same, but instead of summing a single c_j per query term, we sum the product $d_j \cdot q_j$ for each query term present in the document. Expand-

ing the TF-IDF weights, each term's contribution is $\text{tf}(D, t_j) \cdot \text{tf}(Q, t_j) \cdot \text{idf}(t_j)^2$. As a consequence, the postings for term t_j must store not only the document identifier but also the term frequency $\text{tf}(D, t_j)$ so the ranker can compute that product.

Using the inner product alone (the sum above) already gives a useful ranking that favors documents sharing many weighted terms with the query. The **cosine measure** refines this by normalizing both vectors:

$$\text{sim}_{\cos}(Q, D) = \frac{\sum_j d_j \cdot q_j}{\| \mathbf{d} \| \cdot \| \mathbf{q} \|}. \quad (4.1)$$

The query norm $\| \mathbf{q} \|$ is not a problem: we have the full query vector and can compute it once per query. The document norm $\| \mathbf{d} \|$ is the difficulty. During evaluation, whether DAAT or TAAT, we see only the subset of document components that overlap with the query terms, not the full document vector, so we cannot compute the norm from what we read. The solution is to store each document's length (or precomputed norm) in a separate table, indexed by document identifier. This adds one random lookup per candidate document.

BM25 faces the same requirement: its length-normalization factor uses the document length $| D |$ and the average document length avgdl , which likewise must be stored alongside the index. The cost is modest, a single integer or float per document, but it is a structural addition beyond the postings themselves.

For **BM25**, the postings again store term frequencies, and the ranker additionally needs the document length $| D |$ (the total number of term occurrences in D , not the vector norm $\| \mathbf{d} \|$), the average document length avgdl , and the parameters k and b .

A worked BM25 example

We index the titles of the 50-book library collection from [Performance Evaluation](#), applying the standard pipeline of tokenization, stop-word removal, and stemming. This gives $N = 50$ documents with an average title length of $\text{avgdl} = 2.72$ tokens. We run the query “database systems”, which the pipeline reduces to the stems “databas” and “system”, with $k = 1.2$ and $b = 0.75$.

Filtering, Facets, and Compression

Real queries rarely ask for text alone. A user searching a library wants “database systems” among books published after 2000, or filtered to the Computer Science shelf. And a production index must stay small enough to keep its hot data in memory. This section adds two capabilities to the inverted index: metadata filtering, including the faceted navigation built on top of it, and the compression that keeps postings compact.

Filtering on metadata

A predicate is a condition on document metadata, such as `year > 2000` or `genre = "Computer Science"`. There are three ways to combine such a predicate with a text query, differing in when and how the predicate is evaluated.

- **A priori filtering** evaluates the predicate first. We keep document metadata in a structure that answers the condition efficiently, a B-tree or an inverted list on the attribute, and look up the set of document identifiers that satisfy it. That set is passed into the retriever, which drops any candidate not in it during the postings merge. This is the best option when the predicate can be indexed, because it removes non-matching documents before scoring.
- **A posteriori filtering** evaluates the predicate last. When no index exists for the attribute, we rank first and check the predicate only on the documents we are about to return, pulling from the top- k heap in score order and skipping those that fail. For a loose predicate we test only the few documents we return plus a handful skipped. For a highly selective predicate we may have to walk deep into the heap, but that is still cheaper than testing every document in the collection.
- **Inline filtering** evaluates the predicate during the merge. We store the attribute in a stream aligned with the postings and ordered by document identifier, and advance it in lockstep with the postings streams, testing the condition as each candidate appears. This keeps filtering and scoring in a single pass.

In each case the retrieval algorithm itself is unchanged. Filtering slots into the same place where relevance feedback already removes documents: a candidate that fails the predicate is simply not scored or not returned.

Faceted search

Faceted search is filtering turned into navigation. Instead of a free-form predicate, the user picks from categorical attributes, a genre, a publication decade, a content type, and the system both filters the results and reports how many documents fall under each option. This is the a priori approach specialized to categories: the index keeps a separate postings list for each facet value, so filtering by a facet is an intersection with that list, and the facet counts are the list lengths.

Two refinements reduce the storage this adds. **Clustered indexing** groups documents that share facet values, so their postings lists overlap heavily and can be encoded relative to a shared reference list. **Hierarchical facet compression** exploits nested facets: a geographic hierarchy like Country then State then City stores only the differences between levels rather than the full path for every document.

Compressing postings lists

Compression is one of the highest-leverage optimizations in an inverted index. Cutting postings storage by a factor of four or more lets more of the index sit in memory, and reading fewer bytes from disk directly lowers query latency.

The key observation is that a postings list is a sorted sequence of document identifiers. Rather than storing each identifier, we store the gaps between consecutive identifiers, the **d-gaps**. In a list like 95673, 127088, ... we store the first identifier and then the gap 31415. Gaps are much smaller than absolute identifiers, especially for frequent terms whose postings are dense, and small integers compress well.

Variable-byte encoding

Variable-byte (VByte) encoding is the most widely used byte-aligned scheme. It uses the most significant bit of each byte as a continuation flag and the remaining seven bits for data. Every byte except the last has its high bit set to 1; the final byte has its high bit set to 0 to mark the end of the number.

Other schemes

VByte is simple and fast, but block-based schemes compress better. **PForDelta** (patched frame-of-reference delta) processes fixed blocks of, typically, 128 d-gaps. For each block it picks the smallest bit width b that fits about 90% of the values and stores those in b bits each; the roughly 10% that do not fit are recorded separately as exceptions. This packs the common case tightly while still handling outliers. Another byte-aligned family uses a two-bit length marker per value (one to four bytes), keeping data byte-aligned for fast decoding and reaching roughly 15 to 20 percent of the uncompressed size.

Metadata alongside postings

Modern indexes store more than document identifiers in a postings list, and the same compression applies to the extra data. **Term frequencies** are written as small integers after each identifier, enabling TF-IDF and BM25 scoring. Because BM25's saturation function flattens the contribution of high term frequencies, values above a modest ceiling barely affect the score. This means term frequencies can be capped and encoded in just a few bits (four bits suffice in practice) without noticeably hurting retrieval accuracy. **Length-normalization data**, such as document lengths and per-field statistics used by BM25, is precomputed at index time and kept in compressed form, trading a little storage for less work per query. Storing these values next to the postings is what lets the ranker of the previous section avoid a separate lookup for every candidate.

Practical Frameworks

Almost no application builds an inverted index from scratch. The mechanics of the previous sections, postings, merges, ranking, compression, are already implemented, tuned, and battle-tested in libraries and database engines. The engineering choice is which one to adopt, and that choice follows the quality, latency, and storage tradeoff from the chapter opener. This section covers two ready-to-use options: full-text search inside a relational database, and the Lucene library. The distributed engines built on Lucene are the subject of the next section.

Full-text search in your database

If an application already stores its documents in a relational database, the simplest option is often to search them there. PostgreSQL offers full-text search as a built-in feature, and SQLite provides a similar capability through its FTS5 module. There is no separate service to run and no index to keep synchronized with the source data.

PostgreSQL models text search with two data types. A `tsvector` is a document reduced to a sorted set of normalized lexemes, produced by `to_tsvector`, which tokenizes the text, removes stop words, and applies stemming through a configurable dictionary. A `tsquery` is a search expression with Boolean operators and phrase constraints, produced by `to_tsquery`. The match operator `@@` tests whether a `tsvector` satisfies a `tsquery`.

The speed comes from a GIN index, which stands for Generalized Inverted Index and is a genuine inverted index in the sense of this chapter. GIN stores a set of (key, posting list) pairs, where each key is a lexeme and its posting list holds the row identifiers of the documents that contain it. The lexemes themselves are organized in a B-tree so a lookup is fast, and posting lists are stored compressed; a lexeme with very many rows is promoted from a flat posting list to its own B-tree of row identifiers. A multi-word query resolves each lexeme to its posting list and intersects them, the same sorted-postings merge we saw in the first section. To keep inserts cheap, GIN can buffer new entries in a pending list and merge them into the main structure later, trading a little query overhead for faster writes.

Because a `tsvector` column and a GIN index live inside the database, full-text predicates combine naturally with ordinary SQL: a single query can match text, filter on other columns, join related tables, and run inside a transaction. The following example shows the full pattern, from schema to ranked search with a predicate:

```
-- 1. Define a table for movies
CREATE TABLE movies (
  id          SERIAL PRIMARY KEY,
  title       TEXT NOT NULL,
  overview    TEXT NOT NULL,
  rating      REAL,
  year        INT,
  tsv         tsvector
);

-- 2. Populate the tsvector column for full-text search
--   'A' = highest weight (title gets more influence)
--   'B' = lower weight (overview contributes less)
UPDATE movies
SET tsv = setweight(to_tsvector(title), 'A') ||
        setweight(to_tsvector(overview), 'B');

-- Create a GIN index for fast lookup
CREATE INDEX movies_tsv_idx ON movies USING GIN(tsv);

-- 3. Search with ranking and a predicate on year
SELECT id, title, year,
       ts_rank(tsv, to_tsquery('Star & Wars')) AS r
```

```

FROM movies
WHERE tsv @@ to_tsquery('Star & Wars')
      AND year < 2000
ORDER BY r DESC
LIMIT 10;

```

Step 2 builds the inverted index: `to_tsvector` tokenizes, normalizes, and stems the text, and `setweight` assigns one of four weight labels (A through D) so that `ts_rank` can weight matches differently by field; here title matches (weight A) count more than overview matches (weight B). Step 3 combines term matching (`@@`), a metadata predicate (`year < 2000`), and frequency-based ranking (`ts_rank`) in a single query. The database evaluates the GIN index lookup and the year filter together and chooses an efficient execution plan.

For collections up to a few million documents on a database you already operate, this is often all the search infrastructure an application needs. Conceptually, one could even build the inverted index by hand as an ordinary table of (term, document, frequency) rows with a B-tree index on the term column; the hands-on notebook does exactly this to make the structure concrete.

Lucene

Apache Lucene, created by Doug Cutting in 1999 and an Apache project since 2001, is the open-source Java library at the core of most production search platforms. It provides the full toolkit of this chapter: analysis, inverted indexing, term weighting, and ranking. [Figure 4.3](#) shows how its pieces fit together. The current major version is Lucene 10 (10.5 as of mid-2026). Major releases change APIs; in particular, Lucene 10 removed the convenience `searcher.doc()` method (use `reader.storedFields()` instead) and requires an explicit `Field.Store` argument when constructing numeric fields. The concepts below are stable across versions; the hands-on notebook uses the Lucene 10 API.

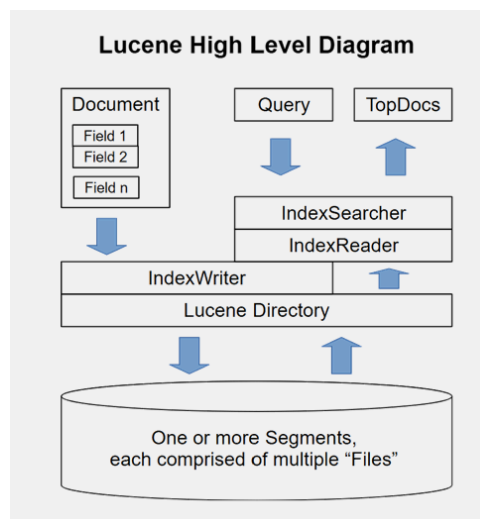


Figure 4.3: Lucene’s architecture: the write path (Document, IndexWriter, Directory, Segments) and the read path (Query, IndexSearcher, IndexReader, Directory, Segments, TopDocs) meet at the Directory storage abstraction.

Documents, fields, and analyzers

Lucene models a document as a set of named fields, and not every document needs every field. A field carries a name, a value, and a type that specifies whether the value is tokenized, what is written to the inverted index, and whether the original value is stored for retrieval. These three choices are independent: a long body field is typically tokenized and indexed for search but not stored, so the application fetches the original text from its own database. Field subclasses cover the common cases: a text field is tokenized and indexed with term frequencies and positions; string and numeric fields are indexed without tokenization for exact-match and range queries on metadata; a stored-only field is kept but not searchable.

Fields are indexed independently. If both a title field and a body field contain “house”, Lucene treats the two occurrences as distinct terms, internally prefixed as “title:house” and “body:house”. This is what lets a search weight a title match more heavily than a body match, or apply different normalization to each field.

Tokenization is the job of an analyzer, and the choice of analyzer must be identical for indexing and for querying. The standard analyzer lowercases and strips punctuation; the English analyzer additionally removes stop words and applies a Porter stemmer; custom analyzers can combine filters such as stemming, length limits, and case handling. Changing the analyzer means rebuilding the index.

```
// Comparing analyzer output for the same text
var text = "I think text's values' color goes here; WHAT happens ...";

// StandardAnalyzer: lowercase, strip punctuation, keep possessives
print_tokens(new StandardAnalyzer(), text);
// -> i think text's values color goes here what happens ...

// EnglishAnalyzer: + stop-word removal + Porter stemming
print_tokens(new EnglishAnalyzer(), text);
// -> think text valu color goe here what happen ...

// Custom stop words
var stopWords = new CharArraySet(Arrays.asList("i", "do"), false);
print_tokens(new EnglishAnalyzer(stopWords), text);
// -> think text valu color goe here what happen ...
```

Segments and updates

Lucene writes the index as a set of immutable **segments**. Each time an `IndexWriter` flushes, it creates a new segment, which reduces contention and protects against corruption. Because segments are immutable, Lucene never edits a document in place, and it has no primary key: internally a document is known only by a `docId` that can change when segments merge. Deletes and updates therefore identify documents by content. A **delete-by-term** marks every document whose chosen key field holds a given value, for example `id: "B-10432"`, by flipping a bit in the segment's live-documents list rather than erasing any data; a **delete-by-query** does the same for documents matching an arbitrary query. An update is a delete-by-term of the key value followed by an add of the new version, applied atomically. A configurable merge policy periodically combines smaller segments into larger ones, and only during a merge are the documents flagged as deleted physically dropped. Searching runs over all live segments in parallel and merges their results, which is the seed of the distributed scaling in the next section.

The following shows how documents are built and added to an index:

```
// Set up analyzer, directory, and writer
Analyzer analyzer = new EnglishAnalyzer();
Directory directory = FSDirectory.open(Paths.get("./index"));
IndexWriter writer = new IndexWriter(directory, new IndexWriterConfig(analyzer));

// Build a document from application data
Document doc = new Document();
doc.add(new TextField("title", "Star Wars", Field.Store.YES));
doc.add(new IntField("year", 1977, Field.Store.YES));
doc.add(new TextField("body", "A long time ago ...", Field.Store.NO));
writer.addDocument(doc);

// Close the writer to flush a new segment
writer.close();
```

Each `TextField` is tokenized and indexed; the `IntField` supports exact and range queries on metadata. Setting `Store.NO` on the body means its text is searchable but not retrievable from results, keeping the stored-fields footprint small.

Querying and scoring

A search uses an `IndexSearcher` over the index directory, usually with a query parser that turns user input into a query and applies the same analyzer used at indexing time. Results come back as a `TopDocs` object

holding the best-scoring documents and their scores. Beyond plain keyword queries, Lucene supports field-scoped terms with boosts (a title match weighted higher), range queries, wildcards, and fuzzy matching within an edit distance. Queries can also be built programmatically, combining clauses that must, should, or must-not match, or that only filter without affecting the score.

Lucene ranks with BM25 by default, computed per field so that the document frequency and length normalization use that field's statistics. Each query clause contributes a per-term score, and the document's final score is their sum:

$$\text{score}(D, Q) = \sum_i \text{score}_i$$

$$\text{score}_i = \text{boost}_i \cdot \text{idf}_i \cdot \frac{\text{tf}_i (k_1 + 1)}{\text{tf}_i + k_1 (1 - b + b \frac{dl}{\text{avgdl}})} \quad (4.2)$$

$$\text{idf}_i = \log \left(1 + \frac{N - n_i + 0.5}{n_i + 0.5} \right)$$

Here score_i is the contribution of one query term (or clause), and clauses marked as filters contribute zero.

This is the BM25 of the previous section with two practical additions: a per-clause boost factor, and the Lucene IDF variant with the extra 1 inside the logarithm that keeps the weight positive even for very common terms. A helpful diagnostic, `explain`, prints how each clause contributed to a document's score, which is invaluable when tuning boosts and field weights.

```
// Set up a searcher and a multi-field query parser
IndexSearcher searcher = new IndexSearcher(DirectoryReader.open(directory));
QueryParser parser = new MultiFieldQueryParser(
    new String[]{"title", "body"}, analyzer);

// Simple keyword query (searches title and body, ranked by BM25)
Query q = parser.parse("star wars");
TopDocs results = searcher.search(q, 10);

// Access stored fields (Lucene 10 API)
StoredFields storedFields = searcher.getIndexReader().storedFields();
for (ScoreDoc hit : results.scoreDocs) {
    Document d = storedFields.document(hit.doc);
    System.out.printf("%.2f %s (%s)\n", hit.score,
        d.get("title"), d.get("year"));
}

// Programmatic query with filter, boost, and explain
Query query = new BooleanQuery.Builder()
    .add(IntField.newRangeQuery("year", 1950, 2000), Occur.FILTER)
    .add(new TermQuery(new Term("title", "shawshank")), Occur.MUST)
    .add(new BoostQuery(
        new TermQuery(new Term("body", "hope")), 1.5f), Occur.SHOULD)
    .build();

TopDocs top = searcher.search(query, 10);
// Score breakdown for the top hit
System.out.println(searcher.explain(query, top.scoreDocs[0].doc));
```

The `explain` output decomposes the score into each clause's contribution, showing the IDF, TF, field length, boost, and normalization parameters. For the query above, it might look like this:

```
7.4038 = sum of:
  0.0 = match on required clause, product of:
    0.0 = # clause
    1.0 = year:[1950 TO 2000]           // FILTER: must match, no score
```

```

1.0 = year:[1990 TO 2000]           // SHOULD: contributes 1.0
3.0980 = weight(title:shawshank in 0) [BM25Similarity], result of:
  3.0980 = score(freq=1.0), computed as boost * idf * tf from:
    6.5023 = idf, computed as  $\log(1 + (N - n + 0.5) / (n + 0.5))$  from:
      1 = n, number of documents containing term
      999 = N, total number of documents with field
    0.4765 = tf, computed as  $\text{freq} / (\text{freq} + k_1 * (1 - b + b * dl / \text{avgdl}))$  from:
      1.0 = freq, occurrences of term within document
      1.2 = k1, term saturation parameter
      0.75 = b, length normalization parameter
      2.0 = dl, length of field
      2.2533 = avgdl, average length of field
  3.3057 = weight(body:hope in 0) [BM25Similarity], result of:
    3.3057 = score(freq=1.0), computed as boost * idf * tf from:
      1.5 = boost                       // the BoostQuery factor
      4.7687 = idf, computed as  $\log(1 + (N - n + 0.5) / (n + 0.5))$  from:
        8 = n, number of documents containing term
        1000 = N, total number of documents with field
      0.4621 = tf, computed as  $\text{freq} / (\text{freq} + k_1 * (1 - b + b * dl / \text{avgdl}))$  from:
        1.0 = freq, occurrences of term within document
        1.2 = k1, term saturation parameter
        0.75 = b, length normalization parameter
        8.0 = dl, length of field
        8.335 = avgdl, average length of field

```

Every quantity from the BM25 formula is visible: the IDF with its $\log(1 + \dots)$ variant, the saturating TF with k_1 , b , dl , and $avgdl$, and the boost factor. Filter clauses contribute zero to the score but must still be satisfied. This makes it possible to diagnose why a document ranks where it does and to tune field boosts accordingly.

Higher-level pipelines

Above these engines sit frameworks that orchestrate retrieval as one step in a larger pipeline, combining keyword search with dense retrieval, rerankers, and language-model generation. Haystack is a widely used example. Because these frameworks belong to the retrieval-augmented generation setting, we cover them in [Retrieval-Augmented Generation](#) rather than here.

Scaling Out

A single Lucene index scales a long way. It handles up to about 2.1 billion documents, and its segment structure lets one query run in parallel across segments. But a single node has limits. Once the index grows past roughly 20 to 40 GB, query latency is bound by memory and disk throughput, and a single machine cannot serve hundreds of concurrent queries. Beyond that point we distribute the index across many machines. This section works up from parallelism inside one node to a globally distributed deployment, using the three engines built on Lucene: Apache Solr, Elasticsearch, and OpenSearch.

The distributed engines

All three engines wrap Lucene and add clustering, horizontal scaling, and near-real-time updates. **Solr** offers a schema-driven, configuration-centric model with strong support for faceting and multilingual text, and is a common choice for enterprise and site search. **Elasticsearch** exposes a schema-flexible REST API and is heavily used for log and security analytics, where together with Logstash and Kibana it forms the ELK stack, though it is equally a capable document search engine. **OpenSearch** began in 2021 as an Apache 2.0 fork of Elasticsearch, created after Elastic changed the Elasticsearch license, and is led by a community with Amazon's backing while staying largely compatible with Elasticsearch.

Elasticsearch and OpenSearch matter beyond keyword search: both have added dense vector (approximate nearest-neighbor) search, so the same cluster can serve lexical and semantic retrieval together. We return to that hybrid setup when we cover vector search in a later chapter.

Parallelism within a node

The first level of scaling needs no extra machines. Lucene can split an index into several segments, and a merge policy controls how many there are. Figure 4.4 shows the pattern: suppose we keep 15 equal segments on a server with 16 virtual CPUs. A coordinator thread parses the query and hands each segment to a searcher thread on one of the remaining 15 CPUs. Each thread searches its segment, and the coordinator merges the partial results before replying. This is a scatter-gather, or map-reduce, pattern: map the query across segments, reduce the partial results into one ranked list.

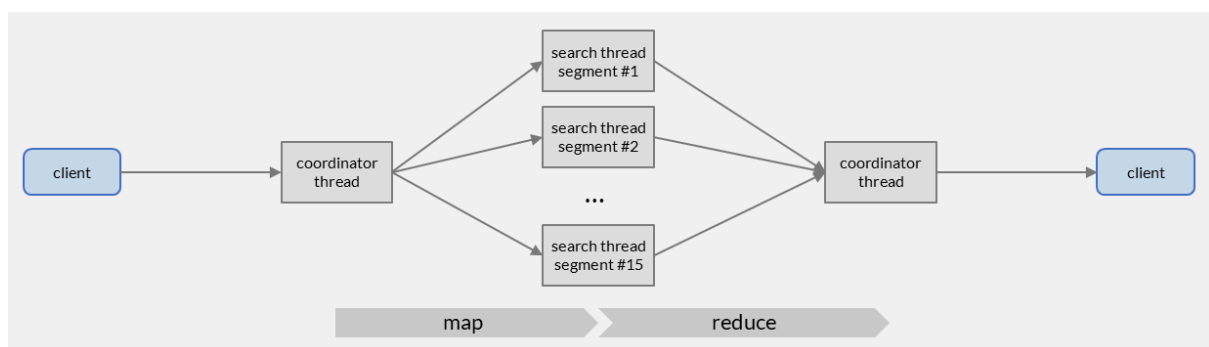


Figure 4.4: Segment-level parallelism: a coordinator thread scatters a query across per-segment searcher threads and gathers their partial results.

The speedup is real but bounded. If 99% of the work is the segment search, Amdahl's law gives roughly a 13-fold speedup over a single thread. But latency still grows with the index size, only one query runs at a time at full width, and pushing to ever-larger machines produces a monolithic server with no high availability. To go further we need more nodes, not more cores.

Sharding across nodes

Sharding distributes the collection across many smaller worker nodes. Each **shard** holds a distinct subset of the documents as its own Lucene index, with its own segments and term statistics. Search now uses two levels of map-reduce, shown in Figure 4.5: the query is scattered across shards, and within each shard across its segments, then reduced back up.

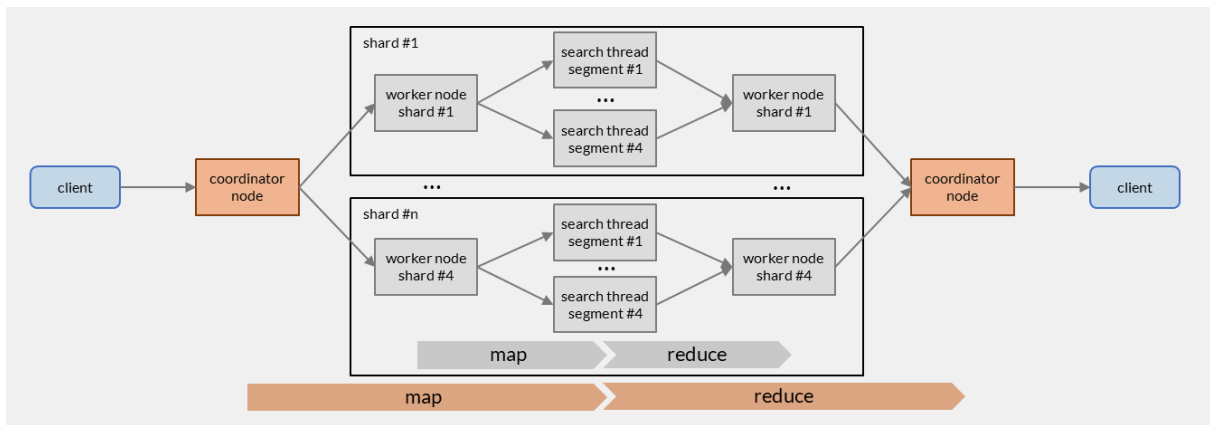


Figure 4.5: Two-level map-reduce: a coordinator fans the query out across shards, and each shard fans it out across its segments before results merge bottom-up.

Because each shard keeps its own term statistics, the same document could score slightly differently depending on which shard holds it. Search is not an exact operation, so small score differences across shards are acceptable as long as they do not distort the overall ranking. When a document is added, a coordinator node assigns it to a shard, either by a policy such as round-robin or by hashing a chosen prefix of the document identifier so that related documents land on the same shard. The shard count is usually fixed, because redistributing documents is expensive. With n shards, the system can handle up to n times the single-node document limit (about 2.1 billion documents), so adding shards is what lets a collection grow beyond what one machine supports. Each shard runs on a modest worker node, for example a container in a Kubernetes cluster, rather than requiring one ever-larger server.

Replication and availability

Sharding spreads data out but does not by itself protect against node failure. For that we replicate each shard and place the replicas in different availability zones, avoiding the same server, rack, or data center. Figure 4.6 shows a deployment of four shards, each with one leader and one replica, arranged so that a shard's leader and replica sit in different zones. Within a shard, the leader node indexes new documents and propagates the changes to its replicas; the engines replicate at the storage level so every replica returns identical results.

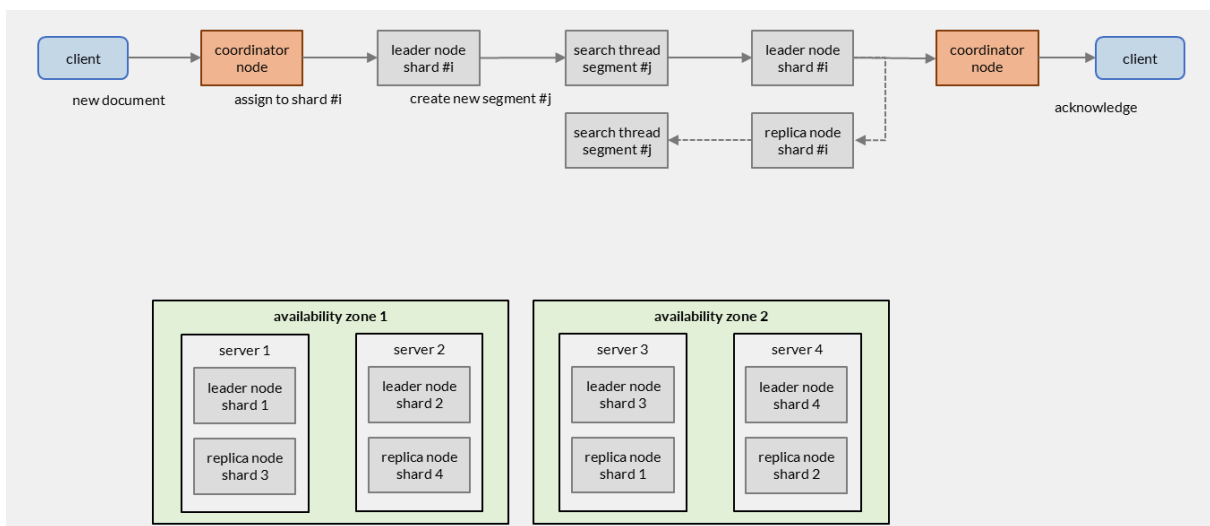


Figure 4.6: Leader and replica shards placed across two availability zones: new documents are indexed by the leader and propagated to the replica, and each shard's replica sits in a different zone from its leader.

Replicas do more than protect against failure. Each leader can also act as a coordinator, routing each shard's part of a query to any replica or leader that holds it. Adding replicas therefore adds concurrent-query capacity roughly in proportion to their number, up to the capacity of the coordinating nodes. Scaling

concurrency this way needs more physical servers, but they can be modest ones hosting small container nodes.

Distributing across regions

The highest level of distribution places whole clusters in several geographic regions near the users they serve. Figure 4.7 shows two regions, one for Europe and one for Asia. Writes go to a primary region and are replicated across regions; a DNS service with a geoproximity policy resolves the application hostname to the entry point of the nearest region, with the other region as a fallback. A client is routed to its closest cluster, cutting the round-trip latency that a distant single region would impose.

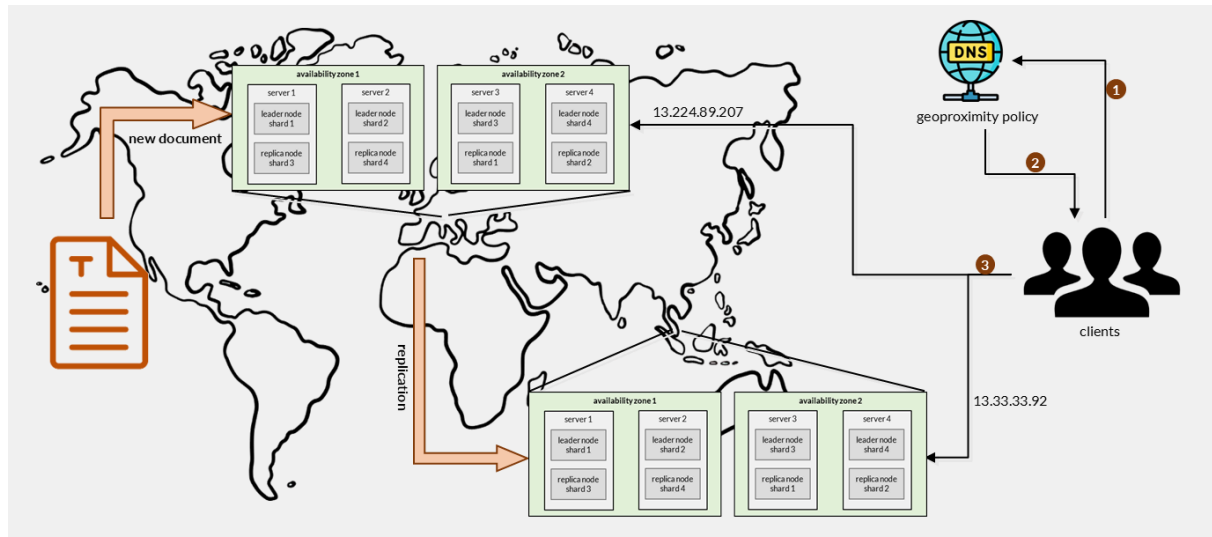


Figure 4.7: Geo-distributed clusters: writes replicate from a primary region to another, and DNS geoproximity routing sends each client to its nearest cluster.

Regional distribution does not speed up a single query. With inter-region latencies of 100 to 200 ms, splitting one search across regions would cost more than it saves, so each query runs within one region. What regional distribution buys is more total concurrent capacity, higher availability, and lower latency for clients near a region. Without it, clients far from a single cluster would see every query pay the long round trip.

Summary

Approach Comparison

The chapter surveyed three ways to run inverted-file search in practice. They sit at different points on the quality, latency, and storage tradeoff.

Property	In-database FTS	Lucene library	Distributed engines
Examples	PostgreSQL, SQLite FTS5	Apache Lucene	Solr, Elasticsearch, OpenSearch
Scale	up to a few million documents	one node, up to ~2.1B docs or 20-40 GB before latency-bound	10s of billions, sharded and replicated
Default ranking	frequency and proximity, no BM25	BM25	BM25, plus vector search
Vector search	via pgvector extension	Lucene 9+ (HNSW)	native, with hybrid scoring
Extra operations	none, it is your database	embed in a JVM application	run and operate a cluster
Best fit	an app that already has a database	custom search inside one application	high query volume, high availability, hybrid retrieval

Each tier builds on the same core: postings, merges, and BM25 scoring are present everywhere. What grows from one tier to the next is the surrounding machinery for scale and availability.

Key Takeaways

1. Scoring every document per query does not scale. The inverted index maps each term to a postings list, so query cost grows with the number of query terms, not with the size of the collection.
2. Retrieval splits into a retriever that gathers candidates by merging query-term postings and a ranker that scores them. Boolean, BIR, vector space, and BM25 all run over the same index; only the postings content and scoring function differ.
3. Document-at-a-time evaluation keeps memory bounded and allows pruning; term-at-a-time is simpler but accumulates a potentially large score table and cannot prune early.
4. Metadata predicates attach through a priori, a posteriori, or inline filtering, and faceted search is the a priori case specialized to categories. Compression via d-gaps and variable-byte encoding keeps postings small so more of the index stays in memory.
5. Sharding scales data volume and per-query latency; replication scales concurrent-query capacity and availability. A single query is not split across regions, so geo-distribution buys capacity, availability, and proximity, not lower per-query latency.

Key Formulas

Self-Check Questions

1. (Understand) Explain why an inverted index reads only a fraction L/M of the data a full scan would touch, and what L and M stand for.
2. (Understand) Given the postings cat = 1, 4, 10 and dog = 3, 4, 8, 10, 12, evaluate “cat AND dog”, “cat OR dog”, and “cat AND NOT dog” by streaming merge.
3. (Analyze) When would you prefer term-at-a-time over document-at-a-time evaluation, and when would you choose a posteriori filtering over a priori filtering?
4. (Analyze) Why can the same document receive slightly different scores on different shards, and why is that usually acceptable?
5. (Evaluate) An application with two million documents already stored in PostgreSQL needs search with good relevance ranking. Weigh in-database full-text search against adding a Lucene-based engine, using the quality, latency, and cost tradeoff.

Further Reading

- Manning, C. D., Raghavan, P., & Schütze, H. (2008). **Introduction to Information Retrieval**. Cambridge University Press. [Free HTML](#). The chapters on index construction and index compression cover this chapter's core structures in depth.
- Zobel, J., & Moffat, A. (2006). **Inverted Files for Text Search Engines**. *ACM Computing Surveys*, 38(2). The canonical survey of inverted-file representation, construction, and query evaluation, including compression and query-time optimizations.
- Robertson, S., & Zaragoza, H. (2009). **The Probabilistic Relevance Framework: BM25 and Beyond**. *Foundations and Trends in Information Retrieval*, 3(4), 333-389. [PDF](#). Derives the BM25 ranker used throughout the chapter, including the multi-field BM25F variant behind Lucene's per-field scoring.
- Büttcher, S., Clarke, C. L. A., & Cormack, G. V. (2010). **Information Retrieval: Implementing and Evaluating Search Engines**. MIT Press. An implementation-focused treatment of indexing, compression, and query processing.

Tools and documentation

- Apache Lucene: lucene.apache.org
- Apache Solr: solr.apache.org
- Elasticsearch: elastic.co
- OpenSearch: opensearch.org
- PostgreSQL Full Text Search: postgresql.org
- Zhou, Z. (2019). **Lucene IndexWriter: An In-Depth Introduction**. *Alibaba Cloud Blog*. [Article](#). Walks through IndexWriter internals: the concurrency model, segment creation, delete-by-term, flush, commit, and merge lifecycle.

5 - Semantic Search

Classical retrieval matches keywords; semantic search matches meaning. This chapter develops the full progression from matrix factorization methods that first revealed latent structure in text, through the neural revolution of word embeddings and transformers, to the production-ready retrieval pipelines deployed in modern search systems. By the end, you will understand how to design a system that retrieves documents about “cardiac conditions” when a user searches for “heart disease”, even when no terms overlap.

Introduction

A user searching for “heart disease” expects to find documents about myocardial infarction, coronary artery conditions, and cardiology treatment, even when those documents share no keywords with the query. Classical retrieval models based on BM25 or the vector space model treat terms as independent symbols: they can match “heart” to “heart” but not to “cardiac”. Stemming and lemmatization help at the surface level, and hand-curated ontologies can map synonyms, but these approaches are labor-intensive, language-specific, and brittle when facing the full diversity of natural language expression.

Semantic search solves this problem by representing text as dense vectors in a continuous space where geometric proximity encodes meaning. Two words that never co-occur in a query-document pair can still be recognized as related if their embeddings point in similar directions. This shift from discrete term matching to continuous meaning matching is the central theme of this chapter.

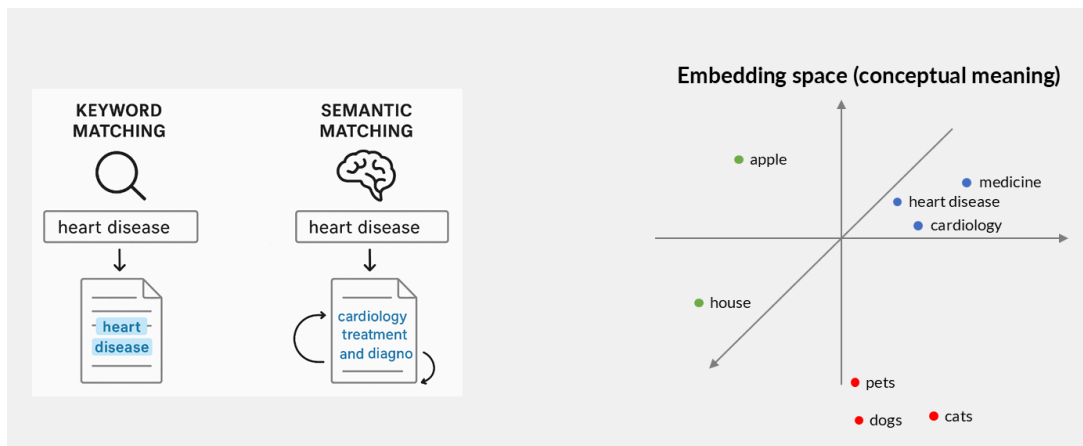


Figure 5.1: Keyword matching operates on exact surface forms (left). Semantic matching maps terms into a continuous embedding space where conceptually related terms cluster together (right), enabling retrieval even when query and document share no words.

The Evolution of Semantic Representations

The path from keyword retrieval to modern semantic search spans three decades of research, each generation addressing limitations of its predecessor:

1. **Latent Semantic Indexing** (Deerwester et al., 1988) applied Singular Value Decomposition to the term-document matrix, discovering latent topics that link related terms. LSI proved that dimensionality reduction could capture semantic relationships, but its computational cost and inability to use inverted indices limited practical adoption.
2. **Static word embeddings** (Mikolov et al., 2013; Pennington et al., 2014) learned dense vector representations from local context windows. Word2Vec and GloVe showed that simple neural architectures trained on large corpora produce vectors where semantic similarity corresponds to geometric proximity. However, each word receives a single vector regardless of context: “bank” has the same representation whether it refers to a financial institution or a riverbank.
3. **Contextual embeddings and transformers** (Vaswani et al., 2017; Devlin et al., 2019; Reimers and Gurevych, 2019) introduced attention mechanisms that produce different representations for the same word depending on its surrounding context. Sentence-BERT restructured transformers into efficient bi-encoders that generate high-quality sentence embeddings suitable for large-scale retrieval.
4. **Modern retrieval architectures** (Khattab and Zaharia, 2020; Kusupati et al., 2022) introduced late interaction models like ColBERT that balance quality and efficiency, Matryoshka embeddings that enable multi-resolution search, and hybrid pipelines that combine sparse and dense retrieval for robustness.

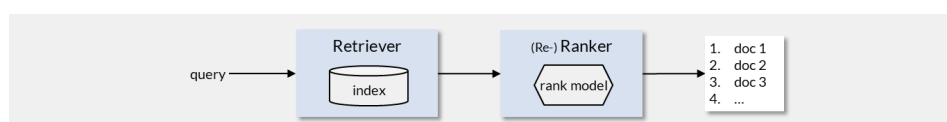


Figure 5.2: The modern retrieval pipeline: a fast retriever selects candidate documents from the index, then a re-ranker applies deeper semantic analysis to produce the final ranked list.

How This Chapter Is Organized

This chapter covers the full progression from early matrix methods to production-ready semantic search systems:

- **Latent Semantic Indexing** introduces dimensionality reduction via SVD and shows how LSI discovers latent topics that connect related terms.
- **Word Embeddings** presents Word2Vec, GloVe, and subword models that learn distributional word representations from context.
- **Transformer Embeddings** explains the attention mechanism, BERT, bi-encoders (SBERT), cross-encoders, and late interaction models (ColBERT).
- **Building Semantic Search** covers practical pipeline design: embedding model selection, document chunking, hybrid retrieval, and efficiency techniques.

The next chapter (**Vector Search**) addresses the indexing problem: how to efficiently search through millions of embedding vectors using approximate nearest neighbor algorithms.

Latent Semantic Indexing

In classical retrieval, documents are sparse high-dimensional vectors with TF-IDF weights, and similarity depends on shared terms. This term independence assumption limits recall: a document about “automobile maintenance” will not match a query for “car repair” unless we explicitly expand the query with synonyms. Stemming and lemmatization reduce some mismatches, but they are language-specific, require manual effort, and cannot capture domain-specific relationships. A search for “tokens” in a text processing context should also retrieve paragraphs discussing “terms” or “words”, but no stemmer will make that connection.

Latent Semantic Indexing (LSI) addresses this by projecting the high-dimensional term-document space into a compact, lower-dimensional representation where semantically related terms are brought closer together. The key insight is that terms appearing in similar document contexts are likely related, even if they never co-occur in the same document. By reducing dimensionality, LSI forces the model to merge these co-occurrence patterns into latent topics.

Singular Value Decomposition

LSI builds on the Singular Value Decomposition (SVD), which factorizes any $m \times n$ matrix into three components. Given a term-document matrix A of rank r , the SVD produces:

$$A = USV^T \quad (5.1)$$

where U is an $m \times r$ orthonormal matrix (columns represent terms in the latent space), S is an $r \times r$ diagonal matrix of singular values in decreasing order, and V is an $n \times r$ orthonormal matrix (columns represent documents in the latent space).

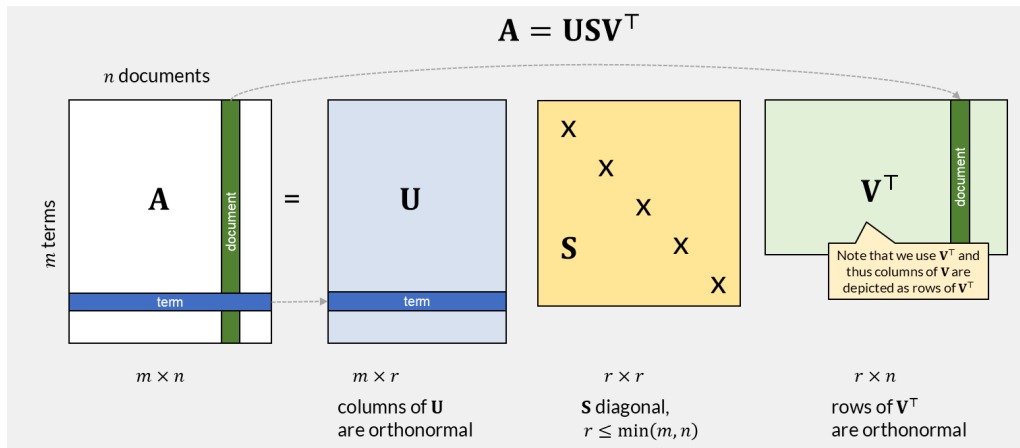


Figure 5.3: SVD decomposes the $m \times n$ term-document matrix A into three factors: U (term representations), S (singular values capturing topic importance), and V^T (document representations).

The singular values on the diagonal of S decrease rapidly in magnitude. By retaining only the k largest singular values and their corresponding columns in U and V , we obtain the best rank- k approximation of A under the Frobenius norm:

$$A_k = U_k S_k V_k^T \quad (5.2)$$

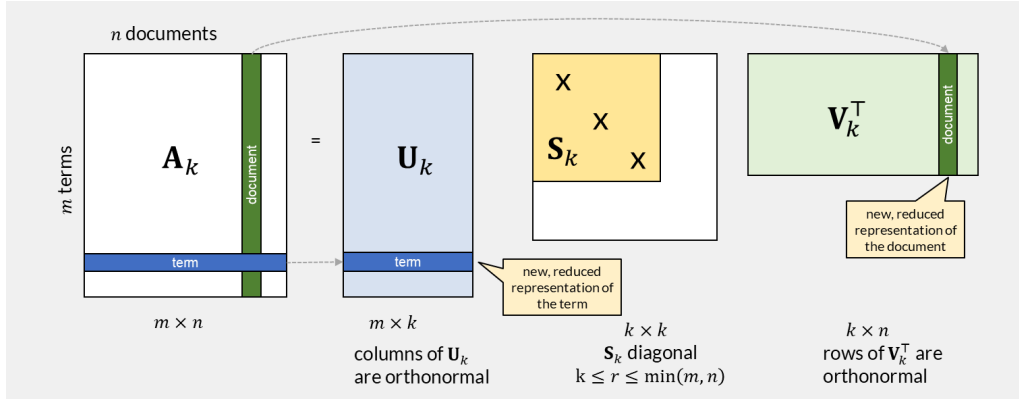


Figure 5.4: Truncated SVD retains only k dimensions. The reduced matrices U_k , S_k , V_k^T provide compact representations: columns of V_k are k -dimensional document vectors, and columns of U_k are k -dimensional term vectors.

The k dimensions no longer correspond to individual terms. Each dimension represents a latent topic, a pattern of term co-occurrence across documents. We can interpret a topic by examining the terms with the largest weights in the corresponding column of U_k .

Application in Text Retrieval

In text retrieval, A is typically the TF-IDF weighted term-document matrix with m vocabulary terms and n documents. After computing the truncated SVD:

- **Document representations:** columns of V_k (equivalently, rows of V_k^T) give k -dimensional vectors for each document.
- **Term representations:** columns of U_k give k -dimensional vectors for each term.
- **Topic interpretation:** the i -th column of U_k describes which terms contribute to the i -th latent topic.

Folding in new documents and queries

When a new document d arrives (represented as a term vector in the original m -dimensional space), we project it into the k -dimensional latent space without recomputing the full SVD:

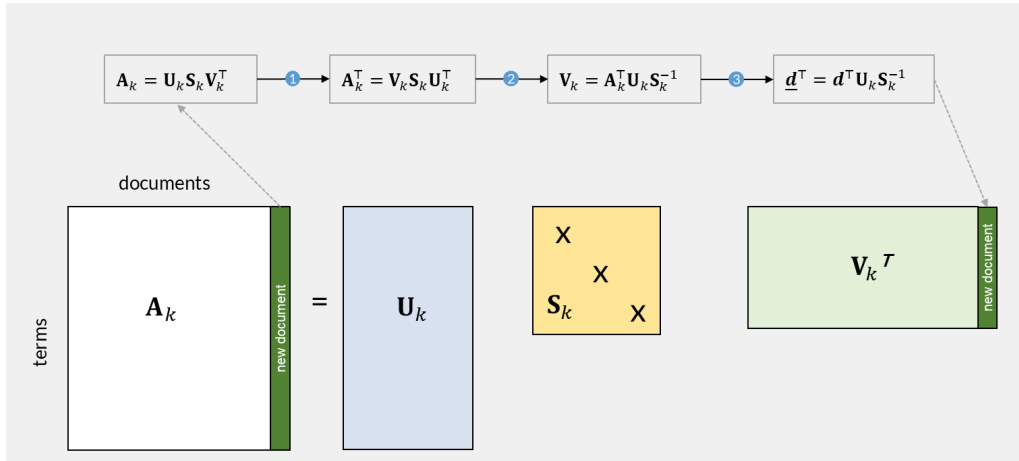


Figure 5.5: Derivation of the folding-in formula in three steps: transpose the truncated SVD equation, multiply by the pseudo-inverse, and isolate a single document vector.

Queries are treated as short documents. The query vector q (with non-zero entries for query terms) is projected identically:

$$\bar{q}^T = q^T U_k S_k^{-1} \quad (5.3)$$

Retrieval then computes cosine similarity between the projected query and all projected documents:

$$\text{sim}_{\cos}(Q, D_i) = \frac{\bar{q} \cdot \bar{d}_i}{|\bar{q}| \cdot |\bar{d}_i|} \quad (5.4)$$

Worked Example

Consider a collection of 9 documents split into two groups: five about human-computer interfaces (c1-c5) and four about graph algorithms (m1-m4):

ID	Title
c1	Human machine interface for Lab ABC computer applications
c2	A survey of user opinion of computer system response time
c3	The EPS user interface management system
c4	System and human system engineering testing of EPS
c5	Relation of user-perceived response time to error measurement
m1	The generation of random, binary, unordered trees
m2	The intersection graph of paths in trees
m3	Graph minors IV: Widths of trees and well-quasi-ordering
m4	Graph minors: A survey

A search for “human computer interaction” should retrieve c1-c5, but not every document contains those exact terms. Traditional retrieval requires explicit query expansion.

After constructing the term-document matrix (using term frequencies, excluding stop words and terms appearing only once), we compute the SVD and truncate to $k = 2$ topics.

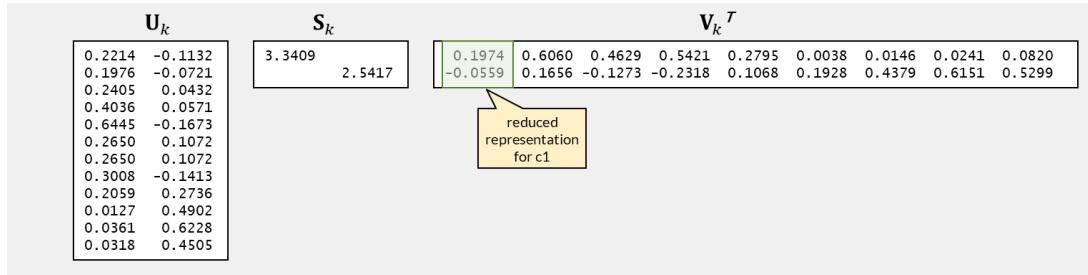


Figure 5.6: Truncated SVD with $k = 2$: the matrices U_k (12 terms $\times$ 2 topics), S_k (2 $\times$ 2), and V_k^T (2 $\times$ 9 documents). Each document is now a 2D vector.

Projecting the query “human computer interaction” (only “human” and “computer” are in the vocabulary) into the 2D topic space yields $\bar{q} = (0.138, -0.028)$ (after normalizing sign conventions).

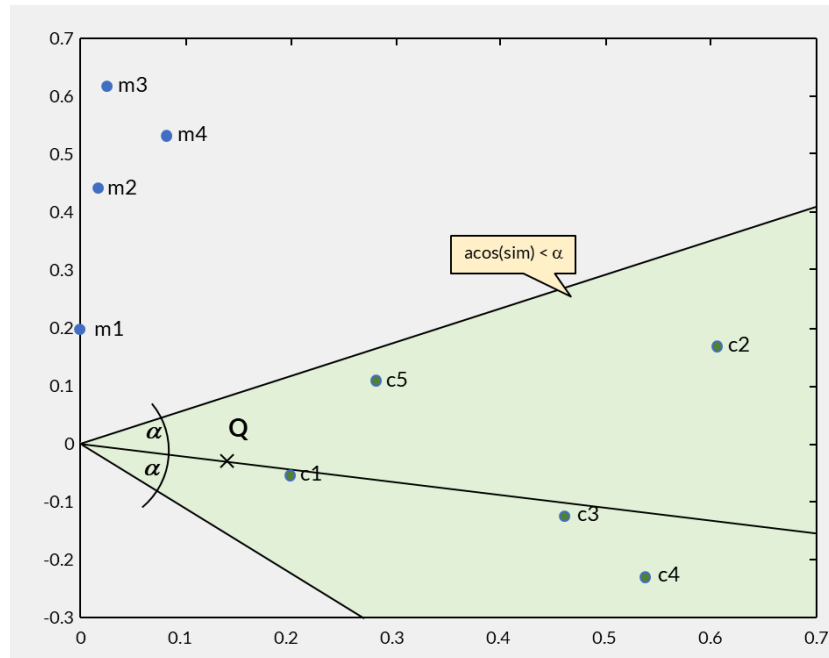


Figure 5.7: Documents and query in the 2D latent space. Topic 1 (horizontal) aligns with the interface documents c1-c5, while topic 2 (vertical) aligns with the graph documents m1-m4. The shaded angular cone shows all documents within angle α of the query vector.

The query vector points toward the c-documents. Using cosine similarity, documents rank as: c3 (0.997) > c1 (0.997) > c4 (0.979) > c2 (0.895) > c5 (0.846), with all c-documents retrieved before any m-document. Notably, c3 (“The EPS user interface management system”) ranks highest despite containing neither “human” nor “computer”. The SVD has recognized that its terms (“EPS”, “user”, “interface”, “system”) co-occur with query-related terms in other documents, connecting them through the latent topic structure.

The two topics can be interpreted from U_k :

- **Topic 1:** $0.64 \cdot \text{system} + 0.40 \cdot \text{user} + 0.30 \cdot \text{eps} + 0.27 \cdot \text{time} + 0.27 \cdot \text{response} + 0.24 \cdot \text{computer}$
- **Topic 2:** $0.62 \cdot \text{graph} + 0.49 \cdot \text{trees} + 0.45 \cdot \text{minors} + 0.27 \cdot \text{survey}$

Limitations of LSI

LSI demonstrated that latent semantic structure improves retrieval, but several limitations prevented widespread adoption:

1. **No inverted index acceleration:** projected document vectors are dense, so every query requires comparison against all documents. The pruning strategies that make inverted indices fast (early termination, skip pointers) do not apply.
2. **Corpus-specific topics:** the learned mapping depends on both terms and documents in the collection. An LSI model trained on IT articles performs poorly on biomedical text. Unlike modern embeddings, LSI representations do not transfer across domains.
3. **Static vocabulary:** new terms can only be approximated through the existing topic structure. Two new terms in the same document receive identical representations, since both depend solely on that document’s projection into V_k .
4. **No sub-word awareness:** LSI operates at the word level and cannot generalize to morphological variants or misspellings not already in the vocabulary.
5. **Linear assumption:** SVD captures linear relationships between terms and documents. Non-linear semantic patterns (metaphor, irony, context-dependent meaning) are beyond its reach.

Despite these limitations, LSI established the foundational principle that underpins all modern semantic search: by reducing dimensionality, we can discover latent structure that connects terms sharing similar usage patterns, even when they never co-occur directly. The methods that follow in this chapter, from Word2Vec to transformers, all build on this insight while addressing LSI’s specific shortcomings.

Word Embeddings

Word embeddings map each word in a vocabulary to a dense vector of d dimensions (typically 100-300), where words appearing in similar contexts receive similar vectors. Unlike LSI, which derives representations from a global term-document matrix, word embeddings learn from local context windows and scale efficiently to billions of training tokens. This section covers the foundational models: Word2Vec, GloVe, and fastText.

From LSI to Embeddings

While both LSI and word embeddings produce dense vector representations, they differ in fundamental ways:

Aspect	LSI	Word Embeddings
Unit of representation	Documents (and terms as by-product)	Words (documents via aggregation)
Context signal	Global co-occurrence across entire documents	Local context windows (typically 5-10 words)
Training	SVD on term-document matrix	Neural network with gradient descent
Vocabulary	Fixed; retraining needed for new terms	Sub-word variants possible (fastText)
Transferability	Corpus-specific; poor cross-domain	Language-level; transfers across collections
Scalability	$O(mnk)$ for SVD; expensive for large corpora	$O(n \cdot d \cdot w)$; scales to web corpora

The key conceptual shift is the **distributional hypothesis**: a word's meaning is defined by the company it keeps (Firth, 1957). Word2Vec operationalizes this by training a neural network to predict either context from a center word (Skip-Gram) or a center word from its context (CBOW).

Word2Vec

Word2Vec, introduced by Mikolov et al. in 2013, maps words to a d -dimensional vector space using self-supervised learning on raw text. No labeled data is required. It has two architectures:

Skip-Gram

The Skip-Gram model operates within a context window of size $2m + 1$ around a center word and learns to predict context words from the center word. For the phrase “the dog chases a cat” with center word “chases” and $m = 2$:

$$P(\text{the, dog, a, cat} \mid \text{chases}) = P(\text{the} \mid \text{chases}) \cdot P(\text{dog} \mid \text{chases}) \cdot P(\text{a} \mid \text{chases}) \cdot P(\text{cat} \mid \text{chases}) \quad (5.5)$$

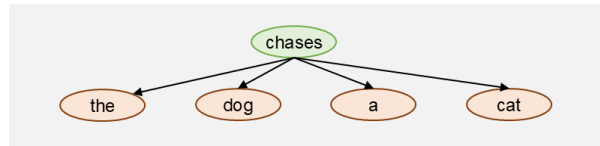


Figure 5.8: Dependency tree for “the dog chases a cat”. In the Skip-Gram model, the center word “chases” predicts each surrounding word independently.

Each word w_i has two d -dimensional vectors: $\mathbf{v}_i$ when used as a center word, and $\mathbf{u}_i$ when used as a context word. The conditional probability uses a softmax over the entire vocabulary $\mathbb{T}$:

$$P(w_s \mid w_c) = \frac{\exp(\mathbf{u}_s^\top \mathbf{v}_c)}{\sum_{i \in \mathbb{T}} \exp(\mathbf{u}_i^\top \mathbf{v}_c)} \quad (5.6)$$

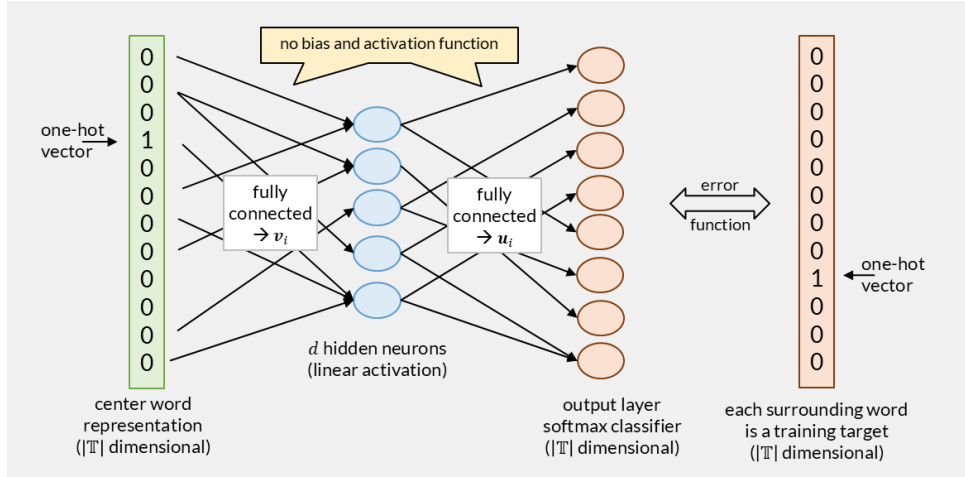


Figure 5.9: Skip-Gram as a neural network: a one-hot input (center word) is projected through a weight matrix (yielding v_i), passed through d hidden neurons with linear activation, then projected to vocabulary size (yielding u_i), and compared to the target context word via softmax and cross-entropy loss.

CBOW

The Continuous Bag of Words (CBOW) model reverses the prediction direction: it predicts the center word from the average of its context word vectors:

$$P(w_c \text{ mid } w_{c-m}, \dots, w_{c+m}) = \frac{\exp(\frac{1}{2m} u_c^\top (v_{c-m} + \dots + v_{c+m}))}{\sum_{i \in \mathcal{T}} \exp(\frac{1}{2m} u_i^\top (v_{c-m} + \dots + v_{c+m}))} \quad (5.7)$$

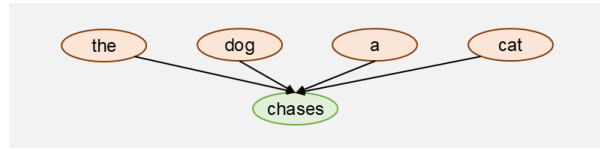


Figure 5.10: In CBOW, all context words point to the center word “chases”, which serves as the prediction target.

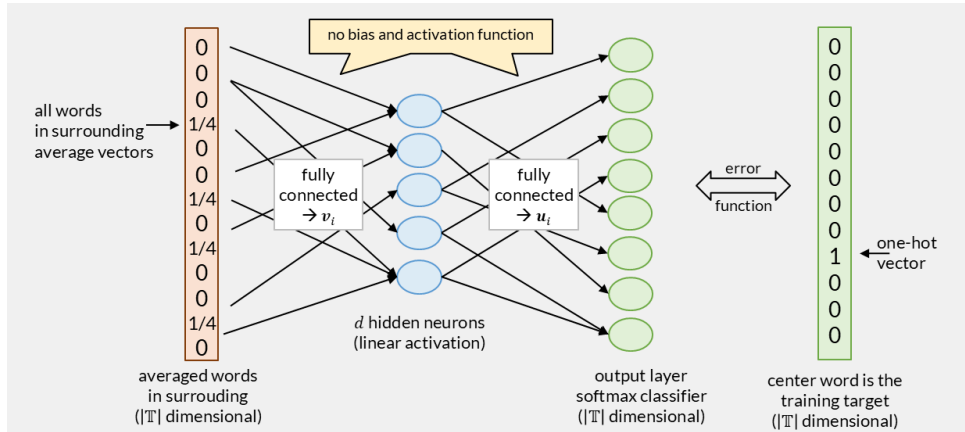


Figure 5.11: CBOW architecture: context word vectors are averaged into a single $|T|$ -dimensional input, passed through a hidden layer of d neurons, then a softmax classifier predicts the center word.

CBOW produces one training sample per window (context around center word), while Skip-Gram produces $2m$ samples (center word around each context word). In practice, Skip-Gram performs better on rare words because it generates more training signal per occurrence, while CBOW trains faster and works well for frequent words.

Training

A PyTorch implementation uses the Embedding layer to avoid explicit one-hot vectors:

```
class SkipGramModel(nn.Module):
    def __init__(self, vocab_size: int, embed_size: int):
        super().__init__()
        self.embeddings = nn.Embedding(vocab_size, embed_size)
        self.linear = nn.Linear(embed_size, vocab_size)

    def forward(self, inputs):
        x = self.embeddings(inputs)
        x = self.linear(x)
        return x
```

Training creates pairs from context windows across the corpus. Sub-sampling frequent words (reducing pairs with stop words to ~1% of their count) improves quality and speed.

GloVe and Subword Models

GloVe

GloVe (Pennington et al., 2014) also learns word vectors from co-occurrence, but uses a global co-occurrence matrix rather than local windows. Let X_{ij} count how often word w_j appears in the context of w_i . GloVe's insight is that the *ratio* of co-occurrence probabilities P_{ik}/P_{jk} distinguishes related from unrelated word pairs better than raw counts.

The model optimizes a weighted least-squares objective:

fastText

fastText (Bojanowski et al., 2017) improves Word2Vec by representing each word as the sum of its sub-word (character n-gram) vectors. For the center word “chases” with 3-grams:

$$\mathbf{v}_{\text{chases}} = \sum_{g \in \mathcal{Z}} \mathbf{z}_g = \mathbf{z}_{<\text{ch}} + \mathbf{z}_{\text{cha}} + \mathbf{z}_{\text{has}} + \mathbf{z}_{\text{ase}} + \mathbf{z}_{\text{ses}} + \mathbf{z}_{\text{es} >} \quad (5.8)$$

This sub-word approach has two advantages:

1. **Morphological generalization:** “chasing”, “chased”, and “chases” share sub-words, so they receive related vectors without explicit stemming.
2. **Handling unknown words:** any new word can be represented by combining its sub-word vectors, even if the word itself was never seen during training.

fastText also introduces **negative sampling** to replace the expensive softmax over the full vocabulary. Instead of normalizing across all $|\mathbb{T}|$ terms, it contrasts the true context pair against m randomly sampled negative words, dramatically reducing computation.

Sub-word lengths can be fixed (3-6 characters) or variable using BPE or WordPiece algorithms, trading vocabulary size against representation granularity.

Emergent Properties and Limitations

Word embeddings trained on large corpora exhibit remarkable emergent structure:

- **Semantic clustering:** words with related meanings form clusters in the embedding space.
- **Analogies:** vector arithmetic captures relational patterns. The classic example: $\mathbf{v}_{\text{Paris}} - \mathbf{v}_{\text{France}} + \mathbf{v}_{\text{Germany}} \approx \mathbf{v}_{\text{Berlin}}$.
- **Odd-one-out detection:** the word least similar to a group can be identified by distance from the group centroid.

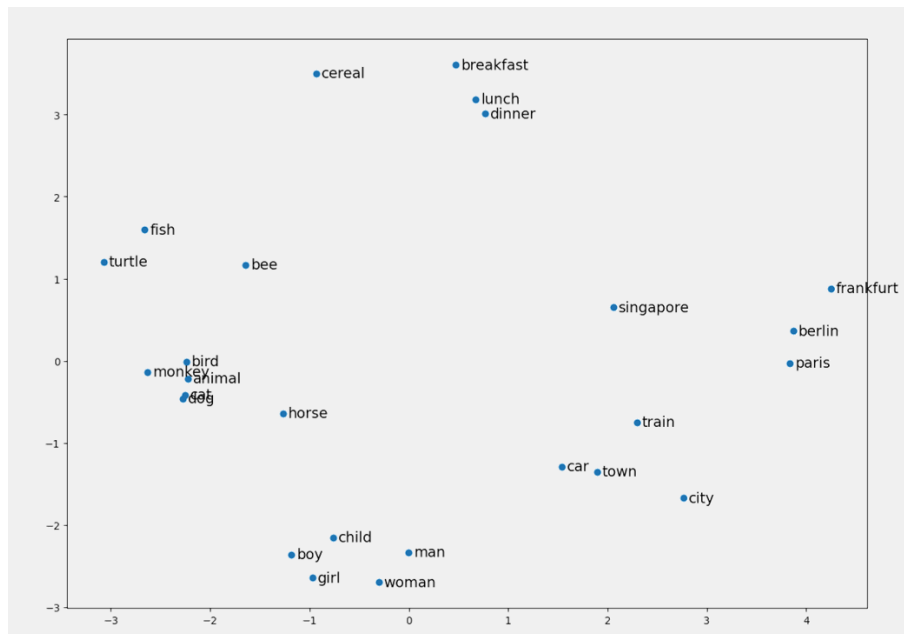


Figure 5.12: PCA projection of GloVe embeddings for 25 words. Semantic clusters emerge: animals (bottom-left), meals (top), cities (right), and people (bottom-center) occupy distinct regions despite no explicit category supervision.

Despite these strengths, static embeddings have a fundamental limitation: **each word receives a single vector regardless of context**. The word "bank" gets the same embedding whether it means a financial institution or a riverbank. In the sentence "I went to the bank on Park Street; I sat and watched the boats go by", humans resolve the ambiguity instantly, but Word2Vec cannot.

This polysemy problem, combined with the lack of word-order sensitivity, motivated the development of contextual embeddings using transformer architectures, covered in the next section.

Transformer Embeddings

Static word embeddings assign a single vector to each word regardless of context. This fails for polysemous words and ignores the compositional meaning that emerges from word combinations. The transformer architecture (Vaswani et al., 2017) solved this by introducing self-attention: a mechanism that lets each token's representation be influenced by every other token in the sequence. This section traces the path from the attention mechanism through BERT to the modern retrieval architectures: bi-encoders, cross-encoders, and late interaction models.

The Attention Mechanism

The core idea of self-attention is that a word's meaning in context depends on its relationships with all other words in the sequence. In "The bank approved the loan", the meaning of "bank" is determined by its relationship with "approved" and "loan". Self-attention computes these relationships explicitly.

For a sequence of n token embeddings, self-attention transforms each token by three steps:

1. **Project** each token embedding x_i into three vectors: a query q_i , a key k_i , and a value v_i using learned weight matrices W_Q, W_K, W_V :

$$q_i = W_Q x_i, \quad k_i = W_K x_i, \quad v_i = W_V x_i \quad (5.9)$$

2. **Score** how much each token should attend to every other token via scaled dot products between queries and keys:

$$\alpha_{ij} = \text{softmax} \left(\frac{q_i^\top k_j}{\sqrt{d_k}} \right) \quad (5.10)$$

3. **Aggregate** the values weighted by attention scores to produce the new representation:

$$z_i = \sum_{j=1}^n \alpha_{ij} v_j \quad (5.11)$$

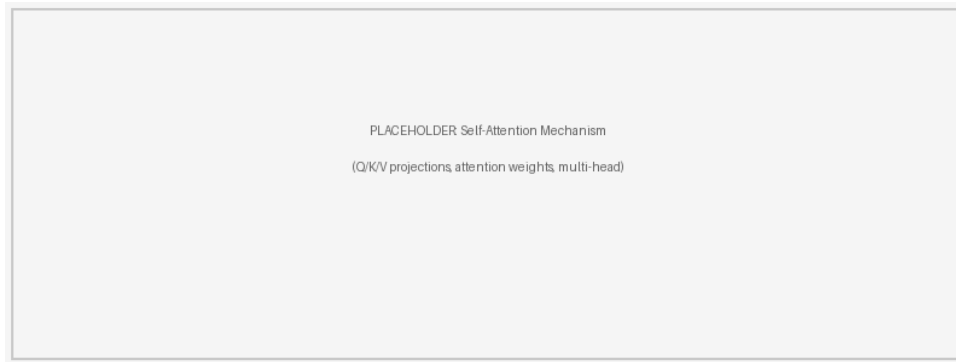


Figure 5.13: *[PLACEHOLDER]* Self-attention mechanism: each token produces a query, key, and value vector. Attention weights (computed from query-key dot products) determine how much each token's value contributes to the output representation. Multi-head attention runs h parallel attention operations with different projections, then concatenates results.

Multi-head attention runs h parallel attention operations with different projection matrices, allowing the model to attend to different types of relationships simultaneously (syntactic, semantic, positional). The outputs are concatenated and projected:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W_O \quad (5.12)$$

A transformer encoder stacks multiple layers of multi-head self-attention followed by feed-forward networks. BERT-Base uses 12 layers with 12 attention heads per layer; BERT-Large uses 24 layers with 16 heads.

BERT: Bidirectional Context

BERT (Devlin et al., 2019) applies the transformer encoder to produce contextualized token representations. Unlike GPT, which processes text left-to-right (suitable for generation), BERT attends to context from both directions simultaneously, making it better suited for understanding and representation tasks.

Input construction

BERT processes input through three summed embeddings:

1. **Token embeddings:** WordPiece sub-word units (vocabulary of $\sim 30,000$ tokens). The word “playing” becomes [“play”, “##ing”].
2. **Positional encodings:** learned position vectors that preserve sequence order.
3. **Segment encodings:** markers distinguishing sentence A from sentence B when processing sentence pairs.

Two special tokens frame the input: [CLS] at the start (whose final hidden state serves as a sequence-level representation) and [SEP] separating segments. Sequences are padded to 512 tokens maximum.

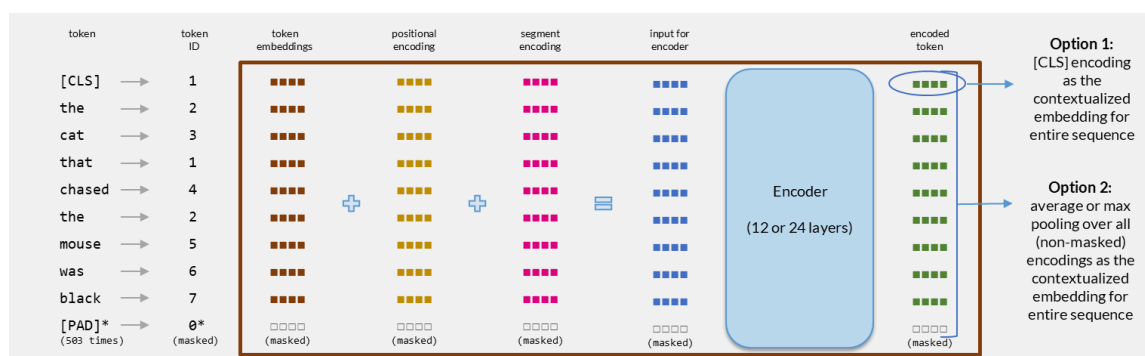


Figure 5.14: BERT input construction for “the cat that chased the mouse was black”. Token embeddings, positional encodings, and segment encodings are summed to form the encoder input. The 12/24-layer encoder produces contextualized output vectors. Option 1: use the [CLS] token encoding as the sequence embedding. Option 2: pool (average or max) over all non-masked token encodings.

Pre-training

BERT is pre-trained on two self-supervised tasks:

- **Masked Language Modeling (MLM):** randomly mask 15% of input tokens and predict them from context.
- **Next Sentence Prediction (NSP):** given two sentences, predict whether the second follows the first in the original text.

After pre-training on large corpora (Wikipedia + BookCorpus), BERT’s encoder produces 768-dimensional (Base) or 1024-dimensional (Large) contextualized vectors for each token. The same word receives different representations depending on its surrounding context.

From BERT to sentence embeddings

Using raw BERT output for retrieval was initially disappointing. The [CLS] token, while designed for classification, produces poor sentence embeddings without task-specific fine-tuning. Mean pooling over token outputs performs marginally better but still underperforms even simple Word2Vec averaging on semantic similarity benchmarks. The breakthrough came with Sentence-BERT.

Bi-Encoders: Sentence-BERT

Sentence-BERT (Reimers and Gurevych, 2019) restructures BERT into a **bi-encoder** (also called dual-encoder or siamese network): two sentences are encoded independently by the same transformer, then compared via cosine similarity.

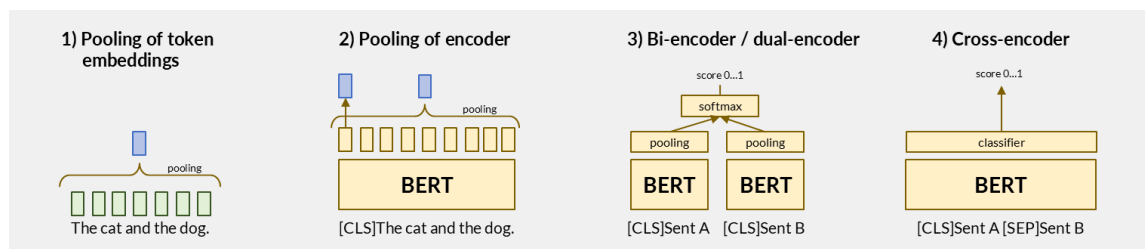


Figure 5.15: Four architectures for sentence similarity, progressing from simple to powerful: (1) pooling static word embeddings, (2) pooling BERT token outputs, (3) bi-encoder with independently encoded sentences compared via learned similarity, (4) cross-encoder that jointly processes both sentences for maximum accuracy.

Architecture and training

The bi-encoder produces fixed-size sentence embeddings:

1. Pass a sentence through BERT (or any transformer encoder).
2. Apply mean pooling over the non-masked token outputs to produce a single d -dimensional vector.
3. Normalize the vector to unit length.

Training uses similarity-based losses on labeled sentence pairs:

- **Contrastive loss:** pulls similar pairs close, pushes dissimilar pairs apart with a margin.
- **Triplet loss:** given an anchor, a positive, and a negative, ensures $\text{sim}(a, p) > \text{sim}(a, n) + \epsilon$.
- **Multiple negatives ranking loss (MNRL):** treats all other examples in the batch as negatives, maximizing efficiency of each training step.

Hard negative mining (selecting negatives that are challenging but incorrect) is critical for training high-quality bi-encoders.

Inference efficiency

The bi-encoder's key advantage for retrieval: documents are encoded once during indexing. At query time, only the query needs encoding. Comparison is a dot product (equivalent to cosine similarity for normalized vectors), enabling sub-millisecond ranking over millions of pre-computed embeddings.

Modern bi-encoder variants

Since SBERT, the field has advanced significantly:

- **Instruction-tuned embeddings:** models like E5-Mistral, GTE-Qwen2, and Nomic Embed accept a task instruction prefix (e.g., "Represent this document for retrieval:") that adapts the embedding to the downstream task without retraining.
- **Decoder-based encoders:** Qwen3-Embedding and similar models build on decoder transformers (using the final [E0S] token representation instead of [CLS]), achieving state-of-the-art quality at the cost of larger models (0.6B to 8B parameters).
- **Extended context:** modern embedding models support 8k-32k token inputs, enabling direct encoding of long documents without chunking.

Cross-Encoders and Reranking

A cross-encoder processes the query and document together as a single input sequence, separated by [SEP]:

[CLS] query [SEP] document [SEP]

The [CLS] token's final hidden state passes through a classification layer to produce a relevance score between 0 and 1. Because all tokens from both query and document attend to each other through every transformer layer, the cross-encoder captures fine-grained token-level interactions that bi-encoders miss.

Quality vs efficiency tradeoff

Property	Bi-Encoder	Cross-Encoder
Encoding	Query and document separately	Query and document jointly

Property	Bi-Encoder	Cross-Encoder
Pre-computation	Documents encoded at index time	Every pair must be processed at query time
Complexity per query	$O(1)$ per document (dot product)	$O(n \cdot L)$ per document (L = sequence length)
Semantic depth	Good (independent representations)	Excellent (full token-level interaction)
Typical use	First-stage retrieval	Reranking top- k candidates

Two-stage pipeline

Cross-encoders are too expensive for exhaustive search over large collections. The standard deployment pattern is:

1. **Retrieve**: a bi-encoder (or BM25) selects the top- k candidates (typically $k = 100 - 1000$).
2. **Rerank**: a cross-encoder scores each candidate with full token interaction and reorders the list.

This hybrid approach achieves near cross-encoder quality at near bi-encoder speed.

Late Interaction: ColBERT

ColBERT (Khattab and Zaharia, 2020) occupies the middle ground between bi-encoders and cross-encoders by computing interaction at the token level while still allowing document pre-computation.

Architecture

Unlike bi-encoders (which compress each text into a single vector) or cross-encoders (which process both texts jointly), ColBERT:

1. **Encodes** query and document independently through a transformer, retaining all token-level embeddings (not pooling to a single vector).
2. **Interacts** via late interaction: each query token computes maximum similarity against all document tokens.
3. **Scores** by summing these maximum similarities across query tokens:

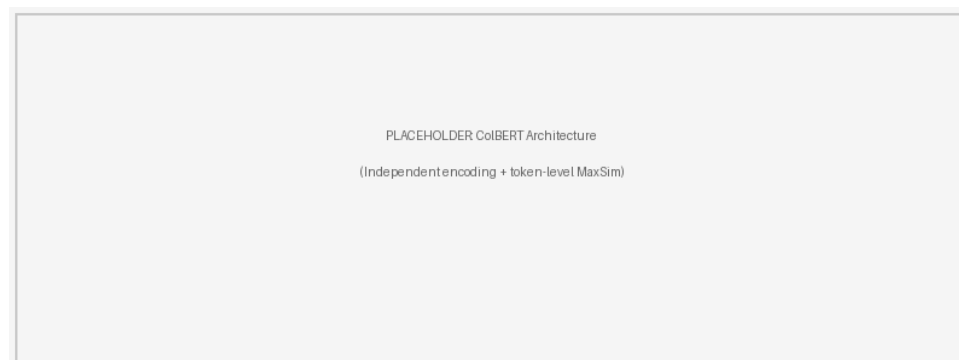


Figure 5.16: [PLACEHOLDER] ColBERT architecture: query and document are encoded independently (allowing document pre-computation), but scoring uses token-level MaxSim interaction rather than a single-vector dot product.

Why late interaction works

The MaxSim operation captures important phenomena that single-vector similarity misses:

- **Exact term matching**: a query token “Python” will have very high similarity with the document token “Python”, directly contributing to the score.
- **Semantic matching**: a query token “automobile” will have high similarity with the document token “car” through their embedding proximity.
- **Partial matching**: not every query token needs to match strongly; the sum allows documents to score well by matching the most important query aspects.

ColBERT v2 and efficiency

ColBERT v2 (Santhanam et al., 2022) addresses the storage cost of retaining per-token embeddings for every document through **residual compression**: token embeddings are quantized by storing only the residual from the nearest centroid, reducing storage by 6-10x while maintaining quality.

The PLAID engine further accelerates retrieval by using centroid interaction for candidate generation before computing full MaxSim only on the top candidates.

Architecture comparison

Architecture	Document representation	Query-time cost	Quality
Bi-encoder	1 vector per document	1 dot product per doc	Good
ColBERT	n vectors per document	\$	q
Cross-encoder	None (must re-encode)	Full transformer pass per doc	Excellent

ColBERT achieves 95-98% of cross-encoder quality with 100-1000x faster query processing, making it practical for direct retrieval (not just reranking) over moderately large collections.

Building Semantic Search

The previous sections presented the models that produce semantic representations. This section addresses the engineering question: how do we assemble these components into a working retrieval system? A production semantic search pipeline must choose an embedding model, decide how to chunk documents, select a retrieval architecture, and manage efficiency at scale.

Embedding Model Selection

The Massive Text Embedding Benchmark (MTEB) provides a standardized evaluation framework for comparing embedding models across multiple tasks: retrieval, semantic textual similarity, classification, clustering, pair classification, and reranking. As of mid-2025, MTEB evaluates models on 56+ datasets across these tasks, with retrieval-specific metrics (nDCG@10) being most relevant for search applications.

Key selection criteria

When choosing an embedding model for a retrieval system, consider:

Criterion	Range	Tradeoff
Embedding dimension	256-4096	Higher dimensions capture more nuance but increase storage and computation
Model size	33M-8B parameters	Larger models produce better embeddings but require more compute for encoding
Context window	512-32k tokens	Longer windows encode more context per chunk but increase encoding time
Multilingual support	1-100+ languages	Multilingual models sacrifice some single-language quality for breadth
Instruction support	Yes/No	Instruction-tuned models adapt to specific tasks via prefixes

Instruction-tuned embeddings

Modern embedding models accept task-specific instructions that adapt the representation without retraining:

```
# For encoding documents
instruction = "Represent this document for retrieval: "
embedding = model.encode(instruction + document_text)

# For encoding queries
instruction = "Represent this query for retrieving relevant documents: "
embedding = model.encode(instruction + query_text)
```

This asymmetric encoding (different instructions for queries vs documents) improves retrieval quality by 3-5% on MTEB benchmarks compared to symmetric encoding.

Model landscape (2025)

Representative models spanning the quality-efficiency spectrum:

Model	Parameters	Dimensions	Context	MTEB Retrieval (nDCG@10)
all-MiniLM-L6-v2	33M	384	512	~49
nomic-embed-text-v1.5	137M	768	8192	~55
jina-embeddings-v3	570M	1024	8192	~58
GTE-Qwen2-7B	7B	3584	32k	~61
Qwen3-Embedding-8B	8B	4096	32k	~63

Note

Benchmark scores evolve rapidly. The specific numbers above reflect mid-2025 results. Always consult the current MTEB leaderboard (<https://huggingface.co/spaces/mteb/leaderboard>) when making production decisions.

Document Chunking and Encoding

Embedding models have finite context windows, and retrieval granularity affects both precision and recall. Documents must be split into chunks that balance:

- **Semantic coherence:** each chunk should represent a complete thought or topic.
- **Size constraints:** chunks must fit within the model's context window.
- **Retrieval granularity:** smaller chunks enable precise matching but may lose context; larger chunks provide more context but dilute the embedding with irrelevant content.

Chunking strategies

Fixed-size chunking splits text at regular token intervals (e.g., every 256 or 512 tokens) with overlap between adjacent chunks (typically 10-20%). Simple to implement but may split sentences or paragraphs mid-thought.

Semantic chunking uses structural cues (paragraph boundaries, section headers, sentence boundaries) to create naturally coherent chunks. More complex but produces chunks aligned with the document's information structure.

Recursive chunking starts with large structural splits (sections), then recursively subdivides any chunk exceeding the target size by progressively finer boundaries (paragraphs, sentences, fixed size).

Encoding pipeline

A typical encoding pipeline for a document collection:

1. **Preprocessing:** extract text, normalize whitespace, optionally remove boilerplate.
2. **Chunking:** split into overlapping or non-overlapping segments.
3. **Metadata attachment:** each chunk retains a reference to its source document, position, and any relevant metadata (title, section heading, date).
4. **Batch encoding:** the embedding model encodes chunks in batches on GPU.
5. **Normalization:** L2-normalize all vectors for dot-product retrieval.
6. **Indexing:** store vectors in a vector database or approximate nearest neighbor index (covered in Chapter 6).

Retrieval Architectures

Three dominant patterns structure modern semantic search pipelines:

Option A: BM25 + semantic reranker

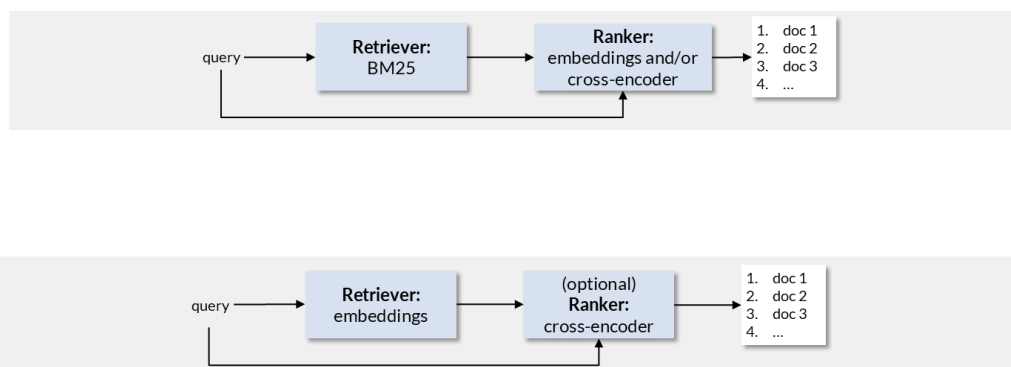


Figure 5.17: Two retrieval pipeline architectures. Top: BM25 retrieves candidates, then an embedding model or cross-encoder reranks them. Bottom: dense embedding retrieval with optional cross-encoder reranking.

BM25 first-stage retrieves the top- k candidates (typically $k = 100 - 1000$) using the inverted index. A **semantic reranker** (cross-encoder or bi-encoder) then reorders these candidates by semantic similarity. This architecture leverages existing infrastructure (inverted indices are well-understood and fast) while adding semantic depth.

Strengths: fast, handles exact-match queries well, low infrastructure cost. Weakness: BM25 recall ceiling limits semantic reranking; if the relevant document is not in the top- k candidates, no reranker can recover it.

Option B: Dense retrieval + optional reranker

A bi-encoder retrieves the top- k documents directly from a vector index using approximate nearest neighbor search. An optional cross-encoder reranks the results for higher precision.

Strengths: captures semantic matches that BM25 misses entirely. Weakness: poor on exact-match queries (proper nouns, codes, IDs); requires vector index infrastructure.

Option C: Hybrid retrieval (sparse + dense)

The most robust modern approach fuses both BM25 and dense retrieval results:

1. BM25 retrieves top- k_1 candidates (handles exact matches, rare terms).
2. Dense retrieval retrieves top- k_2 candidates (handles semantic matches).
3. **Reciprocal Rank Fusion (RRF)** or linear score combination merges both result lists.
4. A cross-encoder reranks the merged top- k for final ordering.

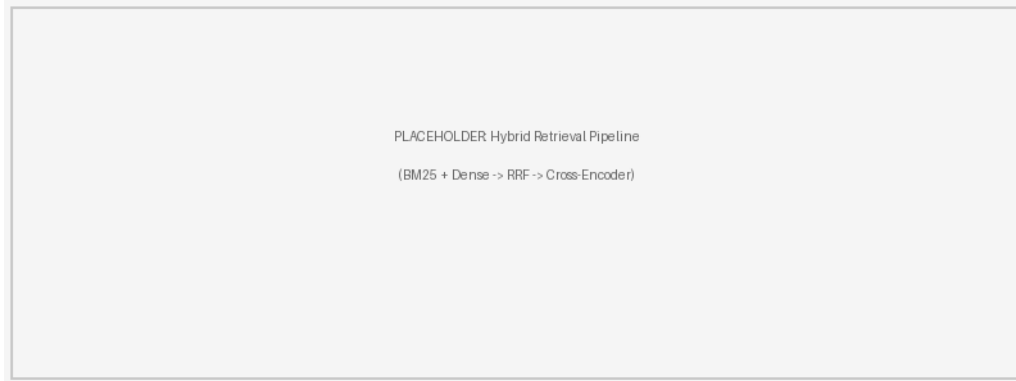


Figure 5.18: [PLACEHOLDER] Hybrid retrieval: BM25 and dense retrieval run in parallel, results are fused via Reciprocal Rank Fusion, then reranked by a cross-encoder. This captures both lexical and semantic matches.

RRF assigns each document a fused score based on its rank in each result list:

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)} \quad (5.13)$$

where R is the set of result lists and k is a constant (typically 60) that dampens the influence of high ranks.

Efficiency at Scale

Matryoshka Representation Learning (MRL)

Traditional embeddings use a fixed number of dimensions. Matryoshka embeddings (Kusupati et al., 2022) train the model so that truncated prefixes of the embedding remain useful: the first 64 dimensions capture coarse semantics, the first 128 add detail, and the full vector provides maximum precision.

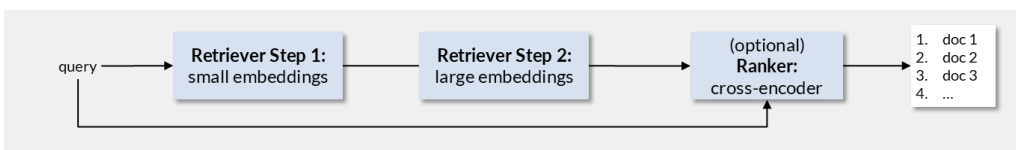


Figure 5.19: Multi-stage retrieval with Matryoshka embeddings: a fast initial retrieval using small (truncated) embeddings narrows candidates, then full-dimension embeddings provide precise reranking. A single model serves both stages.

The training objective adds loss terms at multiple truncation levels:

$$\mathcal{L}_{\text{MRL}} = \sum_{d \in \mathcal{D}} \mathcal{L}_d(\mathbf{x}_{[:d]}, \mathbf{y}_{[:d]}) \quad (5.14)$$

where $\mathcal{D}$ is a set of target dimensions (e.g., {64, 128, 256, 512, 1024}) and $\mathbf{x}_{[:d]}$ denotes the first d components of the embedding.

This enables **multi-stage retrieval within a single model**:

1. Use truncated embeddings (e.g., 128 dimensions) for fast coarse retrieval over the full collection.
2. Use full embeddings (e.g., 1024 dimensions) to rerank the top candidates with higher precision.

Binary and scalar quantization

Storage and comparison costs can be further reduced through quantization:

- **Binary quantization**: replace each float dimension with a single bit (positive arrow.r 1, negative arrow.r 0). Reduces storage by 32x and enables fast Hamming distance computation. Quality loss is typically 5-10% on retrieval benchmarks.
- **Scalar quantization** (int8): map each float to an 8-bit integer. Reduces storage by 4x with minimal quality loss (typically < 2%).

Quantization is most effective as a first-stage filter: use quantized embeddings for coarse candidate retrieval, then rescore with full-precision embeddings.

Practical Considerations

Asymmetric retrieval

Queries are typically short (5-20 tokens) while documents may be long (hundreds to thousands of tokens). This asymmetry means:

- Query embeddings may not capture full intent in limited tokens.
- Document embeddings must compress extensive content into a fixed vector.
- Instruction-tuned models address this with different prefixes for queries vs documents.

Domain adaptation

General-purpose embedding models may underperform on specialized domains (legal, medical, scientific) where vocabulary and language patterns differ from training data. Options for adaptation:

- **Fine-tuning**: train on domain-specific query-document pairs (requires labeled data).
- **Synthetic data**: use LLMs to generate query-document pairs from the target corpus.
- **Domain-specific models**: some domains have specialized embedding models (e.g., PubMedBERT for biomedical text).

Embedding freshness

Unlike inverted indices where adding a document requires only updating postings lists, embedding models may drift: a model trained on 2024 data may produce suboptimal representations for 2026 content discussing new concepts. Re-encoding the full collection with an updated model requires significant compute but ensures consistent representation quality.

Connection to vector search

Dense embeddings create the vectors; the next chapter ([Vector Search](#)) addresses how to efficiently index and search these vectors at scale using approximate nearest neighbor algorithms (HNSW, IVF, product quantization).

Summary

Model Comparison

This chapter presented a progression of methods for representing text semantically. Each generation addressed specific limitations of its predecessors while introducing new tradeoffs:

Method	Representation	Context	Transfer	Key Limitation
LSI (1988)	Dense document vectors via SVD	Global (entire corpus)	Corpus-specific	No inverted index; expensive SVD
Word2Vec (2013)	Dense word vectors	Local window (5-10 words)	Language-level	One vector per word (no polysemy)
GloVe (2014)	Dense word vectors	Global co-occurrence ratios	Language-level	One vector per word (no polysemy)
fastText (2017)	Sub-word vectors	Local window + sub-words	Handles OOV words	Still no contextual meaning
BERT (2019)	Contextualized token vectors	Full sequence (512 tokens)	Task-specific fine-tuning	Poor sentence embeddings without SBERT
Bi-encoder / SBERT (2019)	Single sentence/document vector	Full sequence	Pre-trained, adaptable	No token-level interaction at query time
Cross-encoder (2019)	Relevance score (no vector)	Joint query+document	Task-specific	Too expensive for first-stage retrieval
ColBERT (2020)	Per-token vectors	Full sequence	Pre-trained	Higher storage than bi-encoders
Matryoshka (2022)	Multi-resolution vectors	Full sequence	Pre-trained	Training complexity

The field has converged on a practical consensus: bi-encoders for efficient first-stage retrieval, optionally combined with BM25 for hybrid coverage, followed by cross-encoder reranking for precision-critical applications. ColBERT offers a middle path when single-vector bi-encoders lack sufficient quality but cross-encoder latency is unacceptable.

Key Takeaways

1. **The vocabulary mismatch problem** is the fundamental motivation for semantic search: related terms that share no surface forms cannot be matched by keyword methods alone.
2. **Dimensionality reduction discovers latent structure**: whether via SVD (LSI) or neural training (embeddings), mapping text to a compact space forces the model to group co-occurring patterns into latent topics or semantic directions.
3. **Context transforms meaning**: the same word receives different representations depending on surrounding words in transformer models, solving the polysemy problem that plagued static embeddings.
4. **Bi-encoders enable scalable retrieval**: encoding documents once and comparing via dot product achieves sub-millisecond search over millions of documents. Normalization converts cosine similarity to dot product.
5. **Cross-encoders provide precision**: joint encoding of query and document captures fine-grained token interactions, producing the highest quality relevance scores at the cost of per-pair computation.

6. **Late interaction (ColBERT)** preserves token-level detail while keeping documents pre-computable, achieving 95-98% of cross-encoder quality with manageable latency.
7. **Hybrid retrieval is robust:** combining BM25 (exact match, rare terms) with dense retrieval (semantic matching) via Reciprocal Rank Fusion covers failure modes of either method alone.
8. **Model selection is empirical:** MTEB benchmarks provide standardized comparison, but production performance depends on domain, language, and latency constraints. No single model dominates all use cases.

Key Formulas

LSI document projection:

$$\bar{\mathbf{d}}^\top = \mathbf{d}^\top \mathbf{U}_k \mathbf{S}_k^{-1} \quad (5.15)$$

Maps a term-frequency vector into the k -dimensional latent topic space.

Skip-Gram probability:

$$P(w_s \text{ mid } w_c) = \frac{\exp(\mathbf{u}_s^\top \mathbf{v}_c)}{\sum_{i \in \mathbb{T}} \exp(\mathbf{u}_i^\top \mathbf{v}_c)} \quad (5.16)$$

Probability of context word w_s given center word w_c , parameterized by embedding vectors.

Scaled dot-product attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (5.17)$$

Core operation of the transformer: queries select relevant keys, whose associated values are aggregated.

ColBERT MaxSim scoring:

$$\text{score}(q, d) = \sum_{i=1}^{|q|} \max_{j=1}^{|d|} \mathbf{q}_i^\top \mathbf{d}_j \quad (5.18)$$

Each query token's best document-token match is summed for the final relevance score.

Self-Check Questions

1. **(Understand)** Why does truncating the SVD to k dimensions bring semantically related terms closer together, even if they never co-occur in the same document?
2. **(Understand)** Explain why Word2Vec assigns the same embedding to “bank” regardless of whether it appears in a financial or geographic context. What architectural change in transformers solves this?
3. **(Analyze)** A bi-encoder encodes a 500-word document into a single 768-dimensional vector. What information is necessarily lost compared to the per-token representations? Give a concrete example of a query that might fail because of this compression.
4. **(Analyze)** Compare the computational cost of ranking 10,000 documents using: (a) a bi-encoder, (b) a cross-encoder, (c) ColBERT with MaxSim. Express each in terms of the number of transformer forward passes required.
5. **(Evaluate)** Your system uses BM25 retrieval followed by cross-encoder reranking with $k = 100$. A user reports that relevant documents for the query “ML model deployment best practices” are not appearing in results, despite existing in the collection. Diagnose the most likely failure point and propose a fix.
6. **(Evaluate)** You are designing a semantic search system for a multilingual legal corpus (5 million documents, 12 languages). Justify your choice of embedding model, chunking strategy, and retrieval architecture.
7. **(Apply)** Given Matryoshka embeddings with dimensions in {64, 128, 256, 768}, design a two-stage retrieval pipeline. Specify which dimension you use at each stage and why.

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